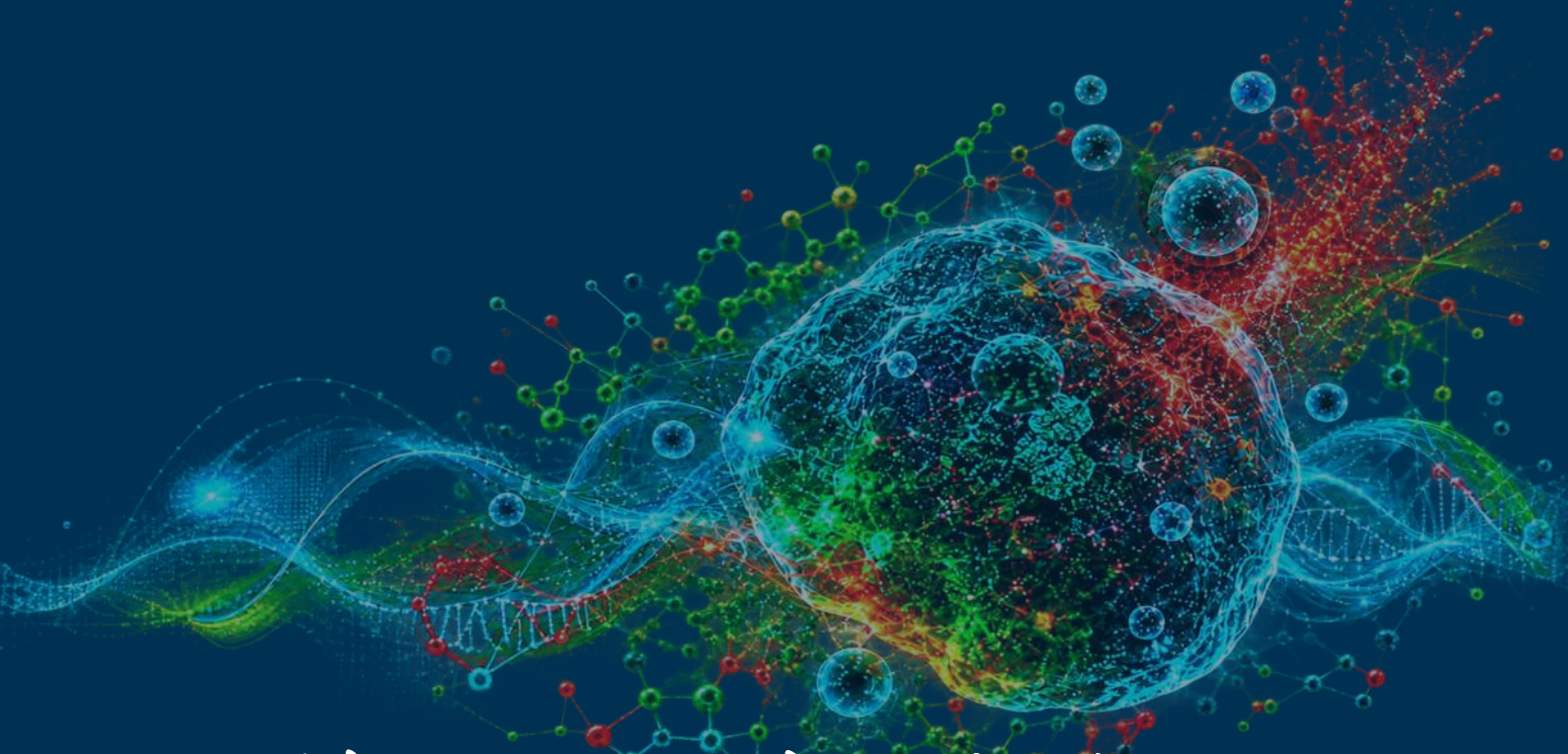


第八屆多體學及精準醫學聯合會議

The 8th Multiomics and Precision Medicine Joint Conference

會議手冊 / Program Book

NOVEMBER 14–15, 2026

國立成功大學成杏校區醫學院

College of Medicine, National Cheng Kung University

Tainan, Taiwan

主辦單位 | Organized by

Taiwan Proteomics Society

Taiwan Society for Extracellular Vesicles

補助單位 | Supported by

衛生福利部 Ministry of Health and Welfare

NSTC 國家科學及技術委員會 National Science and Technology Council

國家實驗研究院
國家高速網路與計算中心
National Center for High-performance Computing

GP1

國家生醫數位資料與分析運算雲端服務平台

服務簡介

本設施提供生醫專屬的**高效能運算與儲存平台**，整合 CPU/GPU 平行運算、高速檔案系統及異地備份。我們支援基因體二級分析加速（DRAGEN/Parabricks）、QIAGEN IPA 資料庫及 AlphaFold 結構預測。對國科會、學術界計畫提供折扣。協助生科與精準醫療產業，減輕計算資源部署的負擔。

生醫專用計算量能

節點種類	節點數量	核心數	記憶體大小
CPU節點	18	128	1000G
GPU節點(H200)	2	8GPU	1600G
6TB大記憶體節點	2	128	6000G

服務項目及收費

服務代號	服務項目	國科會計畫收費	一般學術界收費	產業界收費	服務用途及獨特性
GP1-1	CPU計算節點	0.1/核心小時	0.2/核心小時	0.37/核心小時	基因體註解、蛋白質結構與電腦模擬
GP1-2	CPU大記憶體節點	0.7/核心小時	1.4/核心小時	2.59/核心小時	基因體組裝分析、大資料統計分析
GP1-3	GPU計算節點	30/GPU小時	60/GPU小時	150/GPU小時	基因體鹼基判定與蛋白質結構計算
GP1-4	巨量資料儲存(VirtualStor Scaler/Aspera)	1440/TB/year	4,200/TB/year	6,240/TB/year	物件儲存與高速上傳服務
GP1-7	IPA資料庫	470/時	670/時	暫不提供	IPA生物資訊資料庫及分析工具
GP1-9	基因體分析硬體加速服務	5.2/1Gbases	5.2/1Gbases	7.8/1Gbases	提供DRAGEN基因體分析硬體加速服務
GP1-10	客製化服務	依服務內容議價			客製化分析流程設計與開發
GP1-11	生醫功能性增強模型建置服務	依服務內容議價			AI輔助生醫軟體分析服務
GP1-12	生醫代理人服務	依服務內容議價			MadCowork 代理人服務
NCHC-5	高速計算儲存(GPFS)	250/100GB/年	1,260/100GB/年	5,040/100GB/年	提供高速計算空間，滿足專案計畫執行之需求

主持人：林沿妊 組長/副研究員
執行機關：國家高速網路與計算中心
Email：yjlin@niar.org.tw

聯絡人：黃介瑋 助理研究員
電話：03-5776085 #376
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網址：http://lions.nchc.org.tw/ngs.jsp

國家科學及技術委員會 National Science and Technology Council

籌備委員會致參與人員

2026 多體學與精準醫學聯合會議(Multiomics and Precision Medicine Joint Conference, MOPM)的籌辦，在台灣蛋白體學會、台灣生物資訊與系統生物學會、台灣基因醫學暨生物標記學會、台灣演化與計算生物學會、台灣精準醫學學會、及台灣胞外體學會的共同努力下，重回舉辦首屆會議的台南市來舉行，令人感到別具意義。本次年會由台灣蛋白體學會主辦，並由五大學會協助辦理。本屆會議不另設主題，試想回歸首屆多體學聯合會議的初心，試想回歸當初投身臨床或基礎研究的出發心，試想回歸最先從大學選系、研究所選擇題目的初衷。

本次聯合會議仍是緊扣精準醫療，以多種體學的研究技術探究與各種疾病相關連的生物醫學機制，與發展可促進人類全體健康的研究題材。在人工智慧的協助下，以人機協作方式發展應用與解決方案，相信未來多體學研究配合大數據資料庫和人工智慧，個人化醫療指日可待。2026MOPM 非常榮幸能邀請到多位國際級講者前來分享其研究成果，包括中研院副院長陳玉如院士，其專長為蛋白質體學與磷酸化後轉譯分析，曾任中研院化學所所長、HUPO 理事長、台灣化學學會會長、台灣質譜學會理事長與台灣蛋白體學會理事長。來自 Stanford University 的 Newman 副教授，其專長於開發單細胞基因體學與生物資訊學工具，致力於解析腫瘤微環境與細胞異質性。第三位為日本京都大學 CiRA Foundation 前研發中心副主任 Shinsuke Yoshida 博士，其專長為幹細胞研究和應用；另外也邀請到南韓蛋白體學會(KHUPO)的 Hyun Joo An 博士與 Je-Hyun Baek 博士。Hyun Joo An 博士現任韓國忠南國立大學分析科學與科技研究所教授，亦是 Glycometrix, Inc.公司的共同創辦人。Je-Hyun Baek 博士現為韓國 Seegene Medical Foundation R&D Center for Clinical Mass Spectrometry 的研究主管，其主要研究涵蓋層析分析、質譜儀器、蛋白體學研究與 protein biomarkers screening 等領域。

台灣近年來基礎研究逐步落實，多體學的研究推動精準醫療，因此整合從上游的基因體至中游蛋白體學以及下游的代謝體學資料，其大數據資料庫以 AI 分析逐漸成為生醫研究及臨床轉譯的重要發展方向。2026MOPM 廣邀多位在近年備受關注的空間蛋白質體學、空間生物資訊、基因體學、生醫資訊學、精準醫療及外泌體等領域專精的教授、相關從業人員與精準醫療公司的執行長，在不同 sessions 中分享最新的研究成果，盼望與會人員能踴躍參與並加入討論與互動，彼此激發出更多的研究靈感、具創意的技術與更多的合作機會。期望給予 2026 MOPM 與會人員暨關心台灣多體學發展的眾人，收穫滿滿的學術盛宴。

大會誠摯歡迎您共襄盛舉，再次感謝大家的參與

陳健生　台灣蛋白體學會 理事長
梁世欣　台灣蛋白體學會 秘書長
李宗夷　台灣生物資訊與系統生物學會 理事長
朱彥煒　台灣生物資訊與系統生物學會 副理事長
唐筠　台灣生物資訊與系統生物學會 秘書長
蔡婉琪　台灣基因醫學暨生物標記學會 理事長
莊智弘　台灣基因醫學暨生物標記學會 秘書長
廖本揚　台灣演化與計算生物學會 理事長
張筱涵　台灣演化與計算生物學會 秘書長
王智弘　台灣精準醫學學會 理事長
陳相成　台灣精準醫學學會 副理事長
周涵怡　台灣胞外體學會 理事長

2026 年 11 月

主辦單位的話

台灣蛋白體學會謹代表 2026MOPM 大會誠摯歡迎各位參與「2026MOPM 第八屆多體學與精準醫學聯合會議」。承 MOPM 輪流舉辦，本屆會議於 2026 年 11 月 14 日至 11 月 15 日假國立成功大學醫學院的場地舉辦，重回最初首屆舉辦的場地。在此需感謝學校商借場地，特別感謝衛生福利部科技發展組、國家科學及技術委員會、國立成功大學醫學院，以及各大學會邀請的廠商們鼎力襄助經費支持，使團隊得以順利籌辦本次大會。本次大會特別邀請中研院副院長陳玉如院士、Stanford University 的 Newman 副教授與日本京都大學 CiRA Foundation 前研發中心副主任 Shinsuke Yoshida 博士為大會講者，分享其優異的研究成果。會議期間共匯聚 36 位國內外學者專家參與分組演講，分享多體學相關的研究技術、成果應用及人工智慧於生醫領域的發展。大會期間，多體學相關從業者也將透過設攤、科儀新知和午餐論壇會報等活動，向與會人員展示多體學設備的開發、多體學技術的發展與現今和未來的可能應用。籌備團隊成員們盼望大會兩天的精彩研究分享能夠激盪出更多的研究火花，開拓與會人員的視野，除了給予新進研究生們更大的信心之外，能進一步促成學者們之間的交流。在此，謹向參與年會的同學們暨所有支持本會議的講者們、廠商們及各大學會的支持表達衷心且誠摯的謝意。各學會的理事長、祕書長與祕書，在百忙之外仍需要聯絡講員、安排講員資料的蒐集，海報投稿與報名資料的彙整及追蹤，鼓勵年會的參與報名。在此要特別感謝我的實驗室團體成員們，在祕書婉綺的分工授權之下，逐步完成年會的籌備。要感謝蛋白體學會的理事張權發老師與他的學生們、理事于松桓老師與他的學生們、祕書長梁世欣老師與他的學生們、廖寶琦老師與他的學生們、吳謂勝老師與他的學生們，若沒有他們努力付出與籌劃這一切，本次大會的工作是無法順利完成。最後，籌備委員會的成員們再次誠摯邀請您共襄盛舉，在氣候宜人的 11 月份，一同享受台南滿滿的人情味與本次大會精心安排的學術盛宴。

國立成功大學食安所所長　　台灣蛋白體學會理事長 陳健生

高雄醫學大學生物科技學系教授 台灣蛋白體學會祕書長 梁世欣

2026 年 11 月





Multiomics and Precision Medicine Joint Conference
2026 多體學及精準醫學聯合會議

主辦單位

台灣蛋白體學會
TPS

台灣生物資訊與系統生物學會
TBSB

台灣基因醫學暨生物標記學會
TGMBS

台灣演化與計算生物學會
TSECB

台灣精準醫學學會
TPMS

台灣胞外體學會
TSEV

國立成功大學
NCKU

國立成功大學醫學院
College of Medicine
NCKU

補助單位

衛生福利部
Ministry of Health and Welfare

國家科學及技術委員會
National Science and Technology Council

交通資訊

自行開車（國道路線）

- ❑ 南下：
  沿國道一號南下 → 下大灣交流道右轉 → 沿復興路(西向)往台南市區直走 → 接小東路 → 直走即可抵達本院。【自國道三號南下者，轉國道 8 號（西向），可接國道一號（南向）】

- ❑ 北上：
  沿國道一號北上 → 下仁德交流道左轉 → 沿中山路(西向)往台南市區直走 → 接東門路 → 中華東路右轉(北向) → 小東路口左轉(西向)，直走即可抵達本院。 【自國道三號北上者，轉 86 號快速道路（西向），可接國道一號（北向）】

搭乘火車

- ❑ 於台南站下車後，自後站出口走大學路 → 勝利路左轉 → 遇小東路右轉約 100 公尺，左側即為醫學院成杏校區入口。

搭乘高鐵

- ❑ 搭乘台灣高鐵抵台南站者，可轉搭台鐵電車抵達台南火車站，自後站出口走大學 → 遇勝利路左轉 → 遇小東路右轉約 100 公尺，左側即為醫學院成杏校區入口。

成大醫院停車位即時查詢網頁
(周邊停車場及市府路邊車格即時資訊)

成功大學校本部周邊停車場及路邊停車位置說明：

- 本校收費停車場:

(A)力行校區: 附設醫院第二大樓地下室(約 400 停車格), 入口位於小東路上.
(B)成功校區: 工科系館地下室(約 150 停車格), 入口位於長榮路上.
(C)勝利校區: 成大會館地下室(約 200 停車格), 入口位於大學路上.

- 免費停車場: 前鋒路力行校區對面(約 40 停車格).

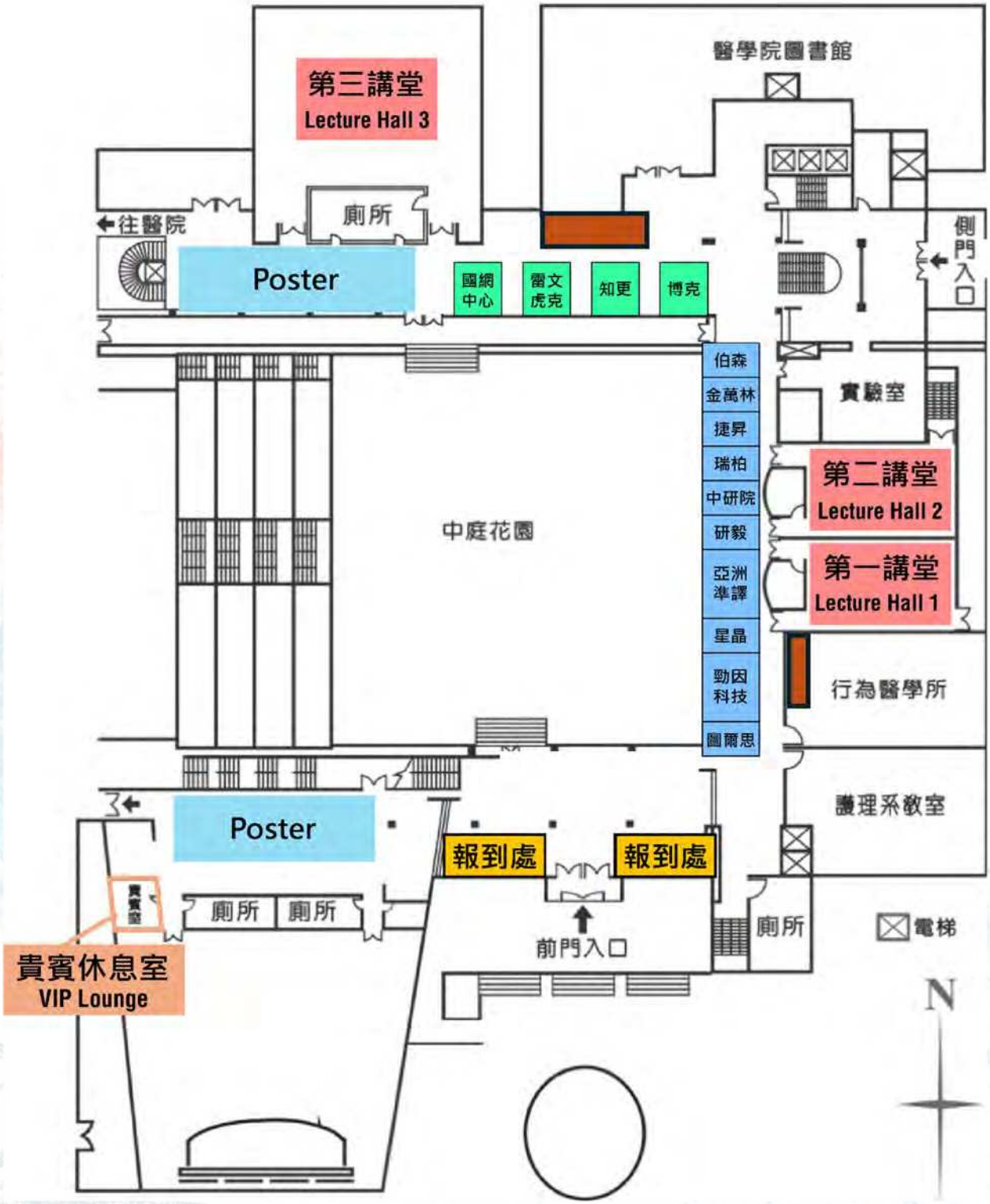
成功大學校外周邊汽車停車場及路邊停車位置圖



A 收費停車場(約400格)
B 收費停車場(約150格)
C 收費停車場(約200格)
D 收費停車場(約160格)

— 路邊停車格
測速照相機
闖紅燈照相機

會場平面圖



圖爾思生物科技股份有限公司
亞洲準譯股份有限公司
瑞柏生物科技股份有限公司
伯森生物科技股份有限公司
雷文虎克生物技術股份有限公司

勁因科技有限公司
研毅企業有限公司
捷昇生物科技有限公司
博克科技股份有限公司
國家實驗研究院國家高速網路與計算中心

星晶科技有限公司
中央研究院 臺灣人體生物資料庫
金萬林企業股份有限公司
知更科技有限公司



講員與主持人休息室
位於二樓 **R208** 室

Lunch Seminar

- Lunch Seminar 將於會場 3 樓教室舉行，歡迎踴躍參與。
- 午餐餐盒同步於 3 樓發放，敬請與會者屆時前往領取。
- 各場次 Seminar 結束後，參與者可持集章卡至指定地點蓋章。



11/14 *12:50~13:25*

R303 NCHC 國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing

R304 APG 亞洲準譯 ASIA PATHOGENOMICS

11/15 *12:20~12:55*

R301 J&H 博克科技股份有限公司 J&H TECHNOLOGY CO., LTD.

R302 PharmiGENE 世基生物醫學股份有限公司

Technology Showcase

11/14 *14:25-14:35* **第三講堂**

主題：**Multi-Omics Tools Decipher Human Microbiome Biomarkers and Therapeutic Potentials**

講者：**周士軒** FAS 領域應用科學家
兼任副教授/國立中興大學 生命科學院

Banquet

濃園滿漢餐廳 Nong Yuan Man Han Restaurant

701 台南市東區長榮路二段 248 號

晚宴訂於 11/14（六）18:30 開席

受邀貴賓敬請於 17:50 前往會場大門集合，搭乘接駁專車。

接駁車 Shuttle Bus					
18:00	成功大學醫學院 College of Medicine, NCKU	→	（18:05 中途停靠） 太子文旅 Academy Hotel - Chengda Branch	→	濃園滿漢餐廳 Restaurant
21:00	濃園滿漢餐廳 Restaurant	→	太子文旅 Academy Hotel - Chengda Branch	→	（成功大學醫學院） （College of Medicine, NCKU）

自行開車貴賓，請將車輛停放於餐廳對面的長榮女中校園內。



【Day 1】2026 年 11 月 14 日 星期六 【Day 1】Saturday, November 14, 2026			
8:00-9:00	註冊報到（成大醫學院穿堂） Registration (College of Medicine Lobby)		
9:00-9:20	第三講堂 Lecture Hall 3		
	Opening Remarks 主持人：陳健生 理事長/台灣蛋白體學會 所長/國立成功大學 食品安全衛生暨風險管理研究所 Moderator: President Chien-Sheng Chen / Taiwan Proteomics Society Director / Department of Food Safety/Hygiene and Risk Management, National Cheng Kung University		
9:20-9:30	Group Photo		
	第三講堂 Lecture Hall 3	第一講堂 Lecture Hall 1	第二講堂 Lecture Hall 2
	Session 1：AI 與精準醫療研究 AI and Precision Medicine Research 主持人：廖寶琦 特聘教授 國立成功大學 工業衛生學科暨環境醫學研究所 Moderator: Distinguished Prof. Pao-Chi Liao Department of Environmental and Occupational Health, National Cheng Kung University Organized by TPS	Session 2：AI 於生物資訊學之應用 AI for Bioinformatics 主持人：張資昊 教授 台北醫學大學 醫學資訊研究所 Moderator: Prof. Tzu-Hao Chang Graduate Institute of Biomedical Informatics, Taipei Medical University Organized by TBSB	Session 3：精準醫學檢測與多體學判讀 Precision Medicine Testing and Multi-Omics Interpretation 主持人：王智弘 總院長 臺北市立聯合醫院 理事長/臺灣精準醫學學會 Moderator: Dr. Chih-Hung Wang Superintendent / Taipei City Hospital President / Taiwan Precision Medicine Society Organized by TPMS
9:30-10:05	Talk 1-1 主題：From Multi-Omics to Multimodal Deep Learning: Advancing AI-Driven Precision Medicine 講者：陳璿宇 教授 研究員/中央研究院 統計科學研究所 Speaker: Dr. Hsuan-Yu Chen Research Fellow / Institute of Statistical Science, Academia Sinica	Talk 2-1 主題：Evolutionary Learning in Bioinformatics: From Signature Discovery to Intelligent Systems 講者：何信瑩 副院長 國立陽明交通大學 工程生物科學學院 Speaker: Prof. Shinn-Ying Ho Vice Dean / College of Engineering Bioscience, National Yang Ming Chiao Tung University	Talk 3-1 主題：隱藏的診斷線索：全外顯子定序的臨床經驗與啟發 Topic: Hidden in Plain Sight: Uncovering Diagnostic Clues in Whole-Exome Sequencing 講者：陳燕麟 主任 三軍總醫院 精準醫學暨基因體中心 Speaker: Dr. Yen-Lin Chen Director / Center for Precision Medicine and Genomics, Tri-Service General Hospital
10:05-10:40	Talk 1-2 Topic: What Gene Editing Cannot Predict: Glycan Remodeling in Xenogeneic Tissues and Blood Speaker: Prof. Hyun Joo An Graduate School of Analytical Science and Technology, Chungnam National University Director / Asia-Pacific Glycomics Reference Site, Chungnam National University	Talk 2-2 主題：超越平衡基準：具類別不平衡感知的血腦屏障穿透胜肽預測 Topic: Beyond Balanced Benchmarks: Imbalance-Aware Prediction of Blood-Brain Barrier Penetrating Peptides 講者：陳鯨太 助理教授 臺北市立大學 資訊科學系 Speaker: Assistant Prof. Ching-Tai Chen Department of Computer Science, University of Taipei	Talk 3-2 主題：AI 在臨床醫療數據分析的應用 講者：張廷彰 教授級顧問醫師 林口長庚紀念醫院 婦產部 Speaker: Dr. Ting-Chang Chang Professor-Level Consulting Physician / Department of Obstetrics and Gynecology, Chang Gung Memorial Hospital, Linkou Branch
10:40-11:15	Talk 1-3 Topic: Beyond Genomic Liquid Biopsy: Tumor-Proximal BALF Extracellular Vesicle Multi-Proteomics for Precision Lung Cancer Care Speaker：Dr. Je-Hyun Baek Research Director / R&D Center for Clinical Mass Spectrometry, Seegene Medical Foundation	Talk 2-3 主題：長讀長定序於精準醫學之應用：從腫瘤變異分相到感染症診斷 Topic: Long-Read Sequencing for Precision Medicine: From Tumor Variant Phasing to Infectious Disease Diagnosis 講者：黃耀廷 系主任 教授/國立中正大學 資訊工程學系 Speaker: Prof. Yao-Ting Huang Department Chair / Institute of Computer Science and Information Engineering, National Chung Cheng University	Talk 3-3 主題：Precision Medicine in Hematology Practice 講者：陳昱光 主任 三軍總醫院 再生醫療中心 Speaker: Dr. Yu-Guang Chen Director / Center for Cell Therapy and Regenerative Medicine, Tri-Service General Hospital

<table>
<tr><td>11:15-11:45</td><td colspan="3">Refreshment Break</td></tr>
<tr><td rowspan="2">11:45-12:25</td><td colspan="3">第三講堂
Lecture Hall 3</td></tr>
<tr><td colspan="2">Keynote Talk 1
主題：癌症蛋白質基因體學：從分子發現到精準腫瘤醫學
Topic：Cancer Proteogenomics: From Molecular Discovery to Precision Oncology
講者：陳玉如 副院長
中央研究院
Speaker: Prof. Yu-Ju Chen
Vice President / Academia Sinica</td><td>主持人：蔡少正 校長/國立中正大學
講座教授/國立成功大學 生理學科暨研究所
Moderator: Chair Prof. Shaw-Jenq Sean Tsai
Department of Physiology,
National Cheng Kung University
President / National Chung Cheng University</td></tr>
<tr><td>12:25-13:40</td><td colspan="3">Lunch & Poster Session</td></tr>
<tr><td rowspan="2">Lunch Seminar
12:50-13:25</td><td>Room 303 (3F)
國家實驗研究院國家高速網路與計算中心
National Center for High-performance Computing, NARLabs</td><td colspan="2">Room 304 (3F)
亞洲準譯股份有限公司
Asia Pathogenomics Co., Ltd.</td></tr>
<tr><td>主題：HPC x Agentic AI for Science：打造新世代智慧生醫科研平台
講者：林沿妊 博士
大數據服務平台組 組長/國家高速網路與計算中心 副研究員</td><td colspan="2">主題：From Variant to Gene Expression, Cell by Cell: Single-Cell RNA + DNA Co-Sequencing
講者：Dr. Leo Chan
Head of APAC Distribution and Global FAS Program,
Atrandi Biosciences</td></tr>
<tr><td rowspan="2">13:40-14:25</td><td colspan="3">第三講堂
Lecture Hall 3</td></tr>
<tr><td colspan="2">Keynote Talk 2
Topic：Rethinking Cell Therapy Development: Toward a Learning Cell Therapy System
講者：吉田信介 博士
前研發中心副主任/日本京都大學 CiRA Foundation
Speaker: Dr. Shinsuke Yoshida
Former Assistant Head / R&D Center, CiRA Foundation, Kyoto, Japan</td><td>主持人：王智弘 理事長/台灣精準醫學學會
總院長/臺北市立聯合醫院
Moderator: President Chih-Hung Wang
Taiwan Precision Medicine Society
Superintendent / Taipei City Hospital</td></tr>
<tr><td>14:25-14:35</td><td colspan="3">Technology Showcase
圖爾思生物科技股份有限公司 BIOTOOLS Co., Ltd
Topic：Multi-Omics Tools Decipher Human Microbiome Biomarkers and Therapeutic Potentials
講者：周士軒 博士 FAS 領域應用科學家 兼任副教授/國立中興大學 生命科學院
Speaker: Dr. Shih-Hsuan Chou
Associate Prof. Rank Specialist / College of Life Sciences, National Chung Hsing University</td></tr>
<tr><td>14:35-14:50</td><td colspan="3">Refreshment Break</td></tr>
<tr><td></td><td>第三講堂
Lecture Hall 3</td><td>第一講堂
Lecture Hall 1</td><td>第二講堂
Lecture Hall 2</td></tr>
<tr><td></td><td>Session 4：演化基因體學
Evolutionary Genomics
主持人：許家郎 研究員
國立臺灣大學 附設醫院醫學研究部
Moderator: Dr. Chia-Lang Hsu
Research Fellow / Department of Medical Research, National Taiwan University Hospital

Organized by TSECB</td><td>Session 5：癌症微環境之多體學剖析：從細胞訊號到精準治療
Multi-Omics Dissection of the Tumor Microenvironment: From Cell Signaling to Precision Therapy
主持人：莊智弘 教授
高雄醫學大學 醫學檢驗生物技術學系
Moderator: Prof. Chih-Hung Chuang
Department of Medical Laboratory Science and Biotechnology, Kaohsiung Medical University

Organized by TGMBS</td><td>Session 6：外泌體與人類疾病
Exosomes in Human Diseases
主持人：沈湯龍 教授
國立臺灣大學 植物病理與微生物學系
Moderator: Prof. Tang-Long Shen
Department of Plant Pathology and Microbiology, National Taiwan University

Organized by TSEV</td></tr>
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14:50-15:25	Talk 4-1 **主題：When Spatial Meets Long Reads: Unveiling Novel Genes and Alternative Splicing in Maize Embryonic Leaves** **講者：呂美曄 研究技師** 中央研究院 生物多樣性研究中心 **Speaker: Dr. Mei-Yeh Jade Lu** Senior Research Specialist / Biodiversity Research Center, Academia Sinica	Talk 5-1 **主題：標靶 KIF2C 以克服紫杉醇抗藥性：微管解聚酶如何解讀微管蛋白密碼** **Topic: Targeting KIF2C to Overcome Paclitaxel Resistance: How a Microtubule Depolymerase Reads the Tubulin Code** **講者：王慧菁 所長** 國立清華大學 分子與細胞生物研究所 **Speaker: Prof. Lily Hui-Ching Wang** Director / Institute of Molecular and Cellular Biology, National Tsing Hua University	Talk 6-1 **主題：Macrophage-Tumor Crosstalk Remodels Extracellular Vesicles to Drive Pancreatic Cancer Progression** **講者：王竹安 助理教授** 國立成功大學 基礎醫學研究所 **Speaker: Assistant Prof. Chu-An Wang** Institute of Basic Medical Sciences, National Cheng Kung University
15:25-16:00	Talk 4-2 **主題：羽毛形態轉變的多樣化及其分子調控機制** **Topic: Diversification of Feather Transitions and Their Underlying Molecular Control** **講者：陳志冠 助理教授** 國立東華大學 生化暨分子醫學科學系 **Speaker: Assistant Prof. Chih-Kuan Chen** Department of Biochemistry and Molecular Medicine, National Dong Hwa University	Talk 5-2 **主題：From tumor dedifferentiation to systemic wasting: Roles of NSUN2-mediated m^5C modification and macrophage-derived CHI3L1 in pancreatic cancer** **講者：黃柏憲 副教授** 國立成功大學 生物化學暨分子生物學研究所 **Speaker: Dr. Po-Hsien Huang** Associate Prof. / Department of Biochemistry and Molecular Biology, National Cheng Kung University	Talk 6-2 **主題：工程化外泌體於腦部疾病應用之開發** **Topic: Therapeutic Applications of Engineered Extracellular Vesicles in Brain Disorders** **講者：謝明佑 教授** 中國醫藥大學 生物醫學工程學系 **Speaker: Prof. Ming-You Shie** Department of Biomedical Engineering, China Medical University
16:00-16:35	Talk 4-3 **主題：The origin and spread of adzuki bean across Asia** **講者：李承叡 教授** 國立臺灣大學 生態學與演化生物學研究所 **Speaker: Prof. Cheng-Ruei Lee** Institute of Ecology and Evolutionary Biology, National Taiwan University	Talk 5-3 **主題：從多體學角度解析 Tirzepatide 治療代謝功能障礙相關脂肪性肝病之分子機制** **Topic: Unraveling the Molecular Mechanisms of Tirzepatide in MASLD: A Multi-Omics Perspective** **講者：李佳陽 教授** 高雄醫學大學 醫學研究所 副研發長/高雄醫學大學 研究發展處 **Speaker: Prof. Chia-Yang Li** Graduate Institute of Medicine, Kaohsiung Medical University Associate Vice President / Office of Research and Development, Kaohsiung Medical University	Talk 6-3 **主題：AI 工程化細胞外囊泡：退化性關節炎疾病修飾治療的新前沿** **Topic: AI-Engineered Extracellular Vesicles: A New Frontier for Disease-Modifying Osteoarthritis Therapy Extracellular Vesicles** **講者：廖秀蓉 研究員** 亞東醫院 醫學研究部 助理教授/國立陽明交通大學 生物藥學研究所 **Speaker: Dr. Hsiu-Jung Liao** Research Fellow / Department of Medical Research, Far Eastern Memorial Hospital Assistant Prof. / Institute of Biopharmaceutical Sciences, National Yang Ming Chiao Tung University
16:35-17:30	TSECB General Assembly	TGMBS General Assembly	TSEV (Board of Directors Meeting & General Assembly)
18:30-21:00	**Invited Banquet**（濃園滿漢餐廳）		

【Day 2】2026 年 11 月 15 日 星期日 【Day 2】Sunday, November 15, 2026			
8:00-9:00	**註冊報到**（成大醫學院穿堂） **Registration** (College of Medicine Lobby)		
9:00-9:45	**第三講堂** **Lecture Hall 3**		
	Keynote Talk 3 Topic：Decoding cellular plasticity and spatial phenotypes in cancer Speaker: Associate Prof. Aaron Newman Department of Biomedical Data Science, Stanford University	**主持人：劉宗霖 副院長** 國立成功大學 生物科學與科技學院 系主任/國立成功大學 生物科技與產業科學系 **Moderator: Prof. Tsung-Lin Liu** Vice Dean / College of Bioscience and Biotechnology, National Cheng Kung University Chairman / Department of Biotechnology and Bioindustry Sciences, National Cheng Kung University	
9:45-10:20	Refreshment Break		
	第三講堂 **Lecture Hall 3**	**第一講堂** **Lecture Hall 1**	**第二講堂** **Lecture Hall 2**
	Session 7：系統生物學 **Systems Biology** **主持人：吳謂勝 教授** 國立成功大學電機工程學系 **Moderator: Prof. Wei-Sheng Wu** Department of Electrical Engineering, National Cheng Kung University Organized by TSECB	**Session 8：代謝體學新紀元：從精密質譜分析到精準醫學實踐** **A New Era of Metabolomics: From High-Precision Mass Spectrometry Analysis to Precision Medicine Practice** **主持人：黃友利 副校長** 高雄醫學大學 **Moderator: Prof. Yeou-Lih Huang** Senior Vice President / Kaohsiung Medical University Organized by TGMBS	**Session 9：植物與微生物胞外囊泡** **Plant and Microbial EVs** **主持人：周涵怡 副教授** 國立臺灣大學 口腔生物科學研究所 **Moderator: Associate Prof. Han-Yi E. Chou** Graduate Institute of Oral Biology, National Taiwan University Organized by TSEV
10:20-10:55	Talk 7-1 **主題：Nucleolar Proteins Coordinate Telomere Maintenance in Telomerase-Positive and ALT Cancer Cells** **講者：林之儀 助理教授** 國立中興大學 生命科學系 **Speaker: Dr. Chih-Yi Gabriela Lin** Assistant Prof. / Department of Life Sciences, National Chung Hsing University	Talk 8-1 **主題：Generalizability of machine-learning–identified metabolites for breast cancer diagnosis across MS platforms and medical centers** **講者：徐丞志 執行長** 創辦人/雷文虎克生物技術 副教授/國立臺灣大學 化學系暨研究所 **Speaker: Dr. Cheng-Chih (Richard) Hsu** Associate Prof. / Department of Chemistry, National Taiwan University Founder / Leeuwenhoek Biotechnology	Talk 9-1 **主題：Extracellular Vesicle-Mediated Host-Microbe Crosstalk** **講者：徐瑋萱 副教授** 國立成功大學食品安全衛生暨風險管理研究所 **Speaker: Associate Prof. Wei-Hsuan Hsu** Department of Food Safety/Hygiene and Risk Management, National Cheng Kung University
10:55-11:30	Talk 7-2 **主題：利用 *E. coli* 研究單一細胞中的 ATP 與生長動態** **Topic: Studying ATP and growth dynamics at the single-cell level in *E. coli*** **講者：林暐翔 助研究員** 中央研究院 分子生物研究所 **Speaker: Dr. Wei-Hsiang Lin** Assistant Research Fellow / Institute of Molecular Biology, Academia Sinica	Talk 8-2 **主題：Beyond LDL Cholesterol: Lysophosphatidylcholine as Cargo and Driver of Pathogenic Electronegative Lipoproteins** **講者：詹華蓁 助理教授** 義守大學 醫學檢驗技術學系 **Speaker: Assistant Prof. Hua-Chen Chan** Department of Medical Laboratory Science, I-Shou University	Talk 9-2 **主題：植物來源奈米囊泡於腸道發炎與氧化壓力之調控作用及潛在應用** **Topic: Regulatory Effects and Potential Applications of Plant-Derived Nanovesicles in Intestinal Inflammation and Oxidative Stress** **講者：潘敏雄 副院長** 國立臺灣大學 生物資源暨農學院 **Speaker: Prof. Min-Hsiung Pan** College of Bioresources and Agriculture, National Taiwan University

11:30-12:05	Talk 7-3 **主題：Beyond Transcription: Novel Mechanisms of HIF-1α in Small RNA Regulation, the Exoswitch** **講者：李婕寧 助理教授** 國立成功大學 細胞生物與解剖學研究所 **Speaker: Dr. Jie-Ning Li** Assistant Prof. / Department of Cell Biology and Anatomy, National Cheng Kung University	Talk 8-3 **主題：以多體學整合分析探討口腔癌前病變惡化進程相關之分子路徑** **Topic: Integrated Multi-Omics Analysis of Molecular Pathways Driving Malignant Transformation in Oral Potentially Malignant Disorders** **講者：朱俐潔 副教授** 長庚大學 生物醫學研究所 **Speaker: Associate Prof. Lichieh Julie Chu** Graduate Institute of Biomedical Sciences, Chang Gung University	Talk 9-3 **主題：Beyond the Cell: Extracellular Vesicles in Viroid-Associated Systemic Pathogenesis** **講者：沈湯龍 教授** 國立臺灣大學 植物病理與微生物學系 **Speaker: Prof. Tang-Long Shen** Department of Plant Pathology and Microbiology, National Taiwan University
12:05-14:05	**Lunch & Poster Session**		
Lunch Seminar 12:20-12:55	**Room 301** (3F) **博克科技股份有限公司** **J&H Technology Co., Ltd.** **主題：從多體學到空間生物學：解碼單細胞層級的組織微環境** **Topic: From Multi-omics to Spatial Biology: Decoding the Tissue Microenvironment at Single-Cell Resolution** **講者：盧幸岑 產品經理**		**Room 302** (3F) **世基生物醫學股份有限公司** **Pharmigene Inc.** **主題：從轉錄體走向調控機制：以單細胞多體學解碼細胞調控** **Topic: Beyond the Transcriptome: Decoding Gene Regulation with Single-Cell Multiomics** **講者：Steiner Chen**
	第三講堂 **Lecture Hall 3**	**第一講堂** **Lecture Hall 1**	**第二講堂** **Lecture Hall 2**
13:10-13:50	TPS General Assembly	TBSB Board of Directors Meeting & General Assembly	TPMS Board of Directors Meeting
	Session 10：質譜多體學大數據生醫分析 **Mass Spectrometry-Based Multi-Omics Big Data Analysis in Biomedicine** **主持人：王家義 副研發長** 國立成功大學 研究發展處 **Moderator: Prof. Chia-Yih Wang** Associate Vice President / Office of Research and Development, National Cheng Kung University **Organized by TPS**	**Session 11：多體學於轉譯醫學之應用** **Multi-Omics in Translational Medicine** **主持人：林峻宇 副教授** 國立陽明交通大學 生物資訊及系統生物研究所 **Moderator: Associate Prof. Chun-Yu Lin** Institute of Bioinformatics and Systems Biology, National Yang Ming Chiao Tung University **Organized by TBSB**	**Session 12：精準醫學臨床應用與未來治療策略** **Clinical Applications of Precision Medicine and Future Therapeutic Strategies** **主持人：陳相成 處長** 國防部 軍醫局醫務管理處 **Moderator: Prof. Hsiang-Cheng Chen** Director / Department of Medical Management, Medical Affairs Bureau, Ministry of National Defense **Organized by TPMS**
14:05-14:40	Talk 10-1 **主題：ANALomics: An Integrated Computational Toolkit for Multi-Omics Analysis Across Genomics, Transcriptomics, and Multi-Layered Proteomics** **講者：于松桓 助理教授** 國立中山大學 精準醫學研究所 **Speaker: Assistant Prof. Sung-Huan Yu** Institute of Precision Medicine, National Sun Yat-sen University	Talk 11-1 **主題：Decoding Multi-layer *Cis*-regulatory Module Functional Mechanisms with Deep Learning** **講者：楊子賢 副教授** 國立成功大學 生物醫學工程學系 **Speaker: Associate Prof. Tzu-Hsien Yang** Department of Biomedical Engineering, National Cheng Kung University	Talk 12-1 **主題： 量子感測精準檢驗神經退化性疾病血液生物標誌物：臨床案例分享** **講者：楊謝樂 董事長** 磁量生技股份有限公司 **Speaker: Dr. Shieh-Yueh Yang** General Manager / MAGQU Co., Ltd.

14:40-15:15	Talk 10-2 **主題：Machine Learning and Deep Learning of MALDI-TOF Mass Spectrometry Data for Predicting Vancomycin Resistance in Enterococci** **講者：陳志杰 副教授** 國立中山大學 醫學科技研究所 **Speaker: Dr. Chih-Chieh Chen** Associate Prof. / Institute of Medical Science and Technology, National Sun Yat-sen University	Talk 11-2 **主題：從單細胞空間微環境到癌前阻斷：空間多體學解析肺腺癌演化與 TIM-3 預防標靶** **Topic: Deciphering Single-Cell Spatial Architecture in Lung Adenocarcinoma: Cell-Cell Interactions and TIM-3 Precancer Interception via Spatial Multiomics** **講者：李建融 助理教授** 長庚大學 生物醫學研究所 **Speaker: Dr. Jian-Rong Li** Assistant Prof. / Graduate Institute of Biomedical Sciences, Chang Gung University	Talk 12-2 **主題：循環腫瘤細胞於精準腫瘤學之應用：計數、表型分析、微量殘留病灶及臨床決策支持** **Topic: Circulating Tumor Cells in Precision Oncology: Enumeration, Phenotyping, Minimal Residual Disease, and Clinical Decision Support** **講者：周輝政 醫師** 臺安醫院 婦產科 **Speaker: Dr. Hei-Jen Jou** Visiting Staff / Department of Obstetrics and Gynecology, Taiwan Adventist Hospital
15:15-15:50	Talk 10-3 **主題：An AI-powered Multimodal Platform for Pancancer Kinase Activity Profiling and Drug Response Prediction** **講者：李宗夷 理事長** 台灣生物資訊與系統生物學會 所長/國立陽明交通大學 生物資訊及系統生物研究所 **Speaker: Prof. Tzong-Yi Lee** President / Taiwan Bioinformatics and Systems Biology Society Director / Institute of Bioinformatics and Systems Biology, National Yang Ming Chiao Tung University	Talk 11-3 **主題：AI-driven multi-omics integration for drug response prediction and clinical outcomes in precision oncology** **講者：周緻雅 博士** 助理研究員/國立陽明交通大學 生物資訊中心 **Speaker: Dr. Chi-Nga Chow** Research Assistant / Center for Bioinformatics, National Yang Ming Chiao Tung University	Talk 12-3 **主題：Toward Personalized Medicine in Pulmonary Arterial Hypertension: Deciphering the TLR7/MyD88 Axis to Guide Phenotype-Specific Immunomodulation** **講者：葉富強 醫師** 主治醫師/三軍總醫院 風濕免疫過敏科 **Speaker: Dr. Fu-Chiang Yeh** Visiting Staff / Division of Rheumatology, Immunology and Allergy, Tri-Service General Hospital
15:50-16:30	**第三講堂** **Lecture Hall 3** Award Ceremony **主持人：陳健生 理事長**/台灣蛋白體學會 所長/國立成功大學 食品安全衛生暨風險管理研究所 **Moderator: President Chien-Sheng Chen** / Taiwan Proteomics Society Director / Department of Food Safety/Hygiene and Risk Management, National Cheng Kung University		
16:30-16:40	Closing Remarks & Handover Ceremony		

Keynote

Keynote 1 | 主持人

蔡少正 校長/國立中正大學

講座教授/國立成功大學 生理學科暨研究所

Moderator

Chair Prof. Shaw-Jenq Sean Tsai

Department of Physiology, National Cheng Kung University
President / National Chung Cheng University

Keynote 2 | 主持人

王智弘 理事長/台灣精準醫學學會

總院長/臺北市立聯合醫院

Moderator

President Chih-Hung Wang

Taiwan Precision Medicine Society
Superintendent / Taipei City Hospital

Keynote 3 | 主持人

劉宗霖 副院長

國立成功大學 生物科學與科技學院
系主任/國立成功大學 生物科技與產業科學系

Moderator

Prof. Tsung-Lin Liu

Vice Dean / College of Bioscience and Biotechnology,
National Cheng Kung University
Chairman / Department of Biotechnology and Bioindustry Sciences,
National Cheng Kung University

陳玉如 副院長
Prof. Yu-Ju Chen



中央研究院
中央研究院院士/中央研究院 化學研究所
Vice President / Academia Sinica
Academician / Institute of Chemistry, Academia Sinica
E-mail: yujuchen@as.edu.tw

Education and Training

- 1997 Ph.D., Department of Chemistry, Iowa State University, U.S.A.
- 1992 B.S., Department of Chemistry, National Taiwan University, Taiwan

Professional Experience

- 2020–Present Distinguished Research Fellow, Institute of Chemistry, Academia Sinica
- 2020–Present Associate Editor, Analytical Chemistry, American Chemical Society
- 1999–2019 Assistant/Associate/Full Research Fellow, Institute of Chemistry, Academia Sinica
- 2023–2025 President, Chemical Society of Taiwan
- 2021–2022 President, Human Proteome Organization (HUPO)
- 2013–2021 Director, Institute of Chemistry, Academia Sinica, Taiwan
- 2012–2015 President, Taiwan Society for Mass Spectrometry
- 2008–2011 President, Taiwan Proteomics Society

Research Interests

Chen group is interested in developing advanced mass spectrometry-based proteomics strategies towards comprehensive profiling of human proteome, especially the druggable membrane sub-proteome and its post-translational modification to decipher complex disease mechanisms. She launched the Taiwan Cancer Moonshot Program for proteogenomic mapping across major cancers, revealing etiology and progression mechanism for non-smoking lung adenocarcinoma (*Cell*, 2020; *Cancer Cell*, 2025), gastric cancer (*Gut*, 2026), and luminal breast cancer (*Molecular Cancer*, 2026). Her patented technology was translated into a start-up company for early cancer diagnosis.

Major Honors and Awards

- 2025 Academic Award, 69th Session, Ministry of Education, Taiwan
- 2025 15th Outstanding Breast Cancer Research Award (TBCF)
- 2025 Distinguished Achievement in Proteomic Sciences Award, Human Proteome Organization (HUPO)
- 2025 IUPAC Distinguished Women in Chemistry or Chemical Engineering Award
- 2025 Taiwan Bio-development Foundation (TBF) Chair in Biotechnology
- 2023 Tung-Ho Outstanding Research Award, THS Foundation
- 2023 Outstanding Research Award, National Science Council
- 2023 16th Taiwan Outstanding Female Scientist Award (L'ORÉAL Taiwan)

- 2021 National Innovation Award (Gastric cancer diagnosis by nanoprobe-based affinity mass spectrometry, technology transfer to JUN ZHI Biomedical Co)
- 2020 Medal of Taiwan Society for Mass Spectrometry
- 2013 Outstanding Research Award, Ministry of Science and Technology

Cancer Proteogenomics: From Molecular Discovery to Precision Oncology

Yu-Ju Chen
Academia Sinica

Genomic profiling has transformed cancer classification and treatment, yet tumors with similar genomic alterations or clinical stages can follow markedly different courses. Cancer proteogenomics integrates genomic, proteomic, phosphoproteomic, and clinical data to connect molecular alterations with functional changes that shape disease progression and treatment response. Drawing on the Taiwan Cancer Moonshot program and international collaborations, this talk will illustrate how proteogenomics addresses distinct clinical questions in lung, breast, and gastric cancers. In lung adenocarcinoma, we validated a “late-like” proteomic phenotype in a subset of clinically early-stage tumors and linked carcinogen-associated mutational signatures to proteomic programs that vary by sex and oncogenic driver (*Cancer Cell*, 2025). In breast cancer, we proposed a proteogenomic pipeline refining molecular subtype and corresponding potential therapeutic vulnerabilities (*Mol. Cancer*, 2026). In gastric cancer, our integrative proteogenomic analysis revealed a complex interplay among the gastric microbiome, environmental carcinogen signatures, and host proteomic and immune profiles, identifying molecular subtypes with distinct clinical outcomes and potential therapeutic vulnerabilities (*Gut*, 2026). These examples show how molecular discoveries can inform risk stratification and guide the search for therapeutic targets. Emerging spatial and single-cell proteomics can further resolve where signaling pathways are active within tumors and how different cell populations respond to treatment.

吉田信介 博士
Dr. Shinsuke Yoshida



前研發中心副主任/日本京都大學 CiRA Foundation
傑出研究員/台灣精準醫學學會
Former Assistant Head / R&D Center, CiRA Foundation, Japan
Distinguished Researcher / Taiwan Precision Medicine Society (TPMS)

Education

- 2021 Master of Business Administration, Doshisha University, Kyoto, Japan
- 2010 Bachelor of Dentistry, Osaka University, Osaka, Japan
- 2000 Ph.D. in Earth and Planetary Physics, The University of Tokyo, Tokyo, Japan

Experience

- 2026-, Distinguished Researcher, Taiwan Precision Medicine Society (TPMS)
- 2020-2026, CiRA Foundation, Kyoto, Japan
  - 2021-2026, Assistant Head, R&D center
  - 2020, QA manager, Facility for iPS Cell Therapy (FiT)
- 2012-2020, Center for iPS Cell Research and Application (CiRA), Kyoto University, Kyoto, Japan
  - 2013-2020, QA manager(2017-2020) / QC manager(2014-2017) / Researcher, FiT
- 2010-2011, Dental Resident, Oral and Maxillofacial Surgery, Kyoto University Hospital, Kyoto, Japan
- 2000-2006, Researcher, ISAS/JAXA, Kanagawa, Japan

Honors and Awards

「細胞及癌症研究菁英獎」(PPSSC2023)

Research Interests

Regenerative medicine
Stem cell engineering
Regulatory science

Selected Publications

1. Shinsuke Yoshida; Tomoaki M. Kato; Yoshiko Sato; Masafumi Umekage; Tomoko Ichisaka; Masayoshi Tsukahara; Naoko Takasu; Shinya Yamanaka. A clinical-grade HLA haplobank of human induced pluripotent stem cells matching approximately 40% of the Japanese population. Med 2023, 4(1), 51-66.

Rethinking Cell Therapy Development: Toward a Learning Cell Therapy System

Yoshida Shinsuke[1*]
[1] *Taiwan Precision Medicine Society (TPMS)*

This year marks the 20th anniversary of the discovery of induced pluripotent stem (iPS) cells, and two iPS cell-derived products have now received regulatory approval in Japan. More than a dozen clinical studies and trials involving iPS cell-derived cells have been conducted in Japan. To date, no adverse events specifically associated with iPS cell-derived products have been reported, and further products are expected to emerge.

Although both products received conditional and time-limited approval, they represent the first regulatory approvals in the world for iPS cell-derived products. Furthermore, the inclusion of cell transplantation therapies using these products in Japan's national health insurance coverage marks a significant step toward their wider clinical adoption.

In this presentation, I will first review the current status of therapeutic development using iPS cells, covering not only the recently approved products but also other ongoing initiatives, including examples of how omics data are being used to advance these efforts.

I will then discuss how we can promote the sustainable development and broader adoption of cell therapies, including those derived from iPS cells. This discussion will address emerging challenges across manufacturing, clinical development, and healthcare delivery, including potential drug lag and drug loss in Japan. Drawing on the concept of continuous improvement in Good Manufacturing Practice (GMP), I will propose the idea of a "Learning Cell Therapy System": a framework in which knowledge and data generated throughout research and development, manufacturing, and clinical practice are continuously fed back into and shared across these domains to improve future therapies and their implementation.

Associate Prof. Aaron Newman



副教授/美國史丹佛大學 生物醫學數據科學系
Associate Professor / Department of Biomedical Data Science,
Stanford University

Education

PhD: Biomolecular Science and Engineering Program, University of California, Santa Barbara (2010).

Academic Appointments

- Associate Professor, Department of Biomedical Data Science, Stanford University
- Investigator, Chan Zuckerberg Biohub (San Francisco)
- Member, Institute for Stem Cell Biology and Regenerative Medicine
- Member, Stanford Cancer Institute
- Member, Bio-X
- Member, Stanford Cancer Biology and Immunology Programs

Honors & Awards

- AACR Trailblazer Cancer Research Grant for Mid-Career Investigators, American Association for Cancer Research (2026-2029)
- 3rd Annual Daisy Ho Memorial Award, Department of Experimental Therapeutics, MD Anderson Cancer Center (2023)
- CZ Biohub Investigator, Chan Zuckerberg Biohub (2022 - 2027)
- Baxter Faculty Scholar Award, Donald E. and Delia B. Baxter Foundation (2020)
- Interdisciplinary Initiatives Seed Grants Program Award, Stanford Bio-X (2018 - 2020)
- Stem Cell Research Award, Stinehart/Reed Foundation (2018 - 2020)
- K99/R00 Pathway to Independence Award, NIH/NCI (2015 - 2020)
- Visionary Postdoctoral Fellowship, Dept. of Defense (2012 - 2015)
- NIH T32 Cancer Biology Training Grant, Stanford University (2012)
- Siebel Fellow, The Siebel Stem Cell Institute (Stanford/UC Berkeley) (2011 - 2015)
- Graduate Dissertation Fellowship Award, UCSB (2009)

Research Interests

Bioinformatics and Computational Biology
Cancer Genomics
Digital Cytometry

Selected Publications

1. Non-invasive profiling of the tumour microenvironment with spatial ecotypes
Nature Zhang W Brown EL Usmani A Earland N Kang M Olelewe C Viswanathan A Chauhan PS Steen CB Jeon HS Avagyan S Alahi I Semenkovich NP Schwab JC Sachs CM Qaium F Harris PK Cai Q Gentles AJ Knight J Graham RP Bacchiocchi A Lucas PC Fields RC Sznol M Halaban R Chen DY Chaudhuri AA Newman AM 2026. https://doi.org/10.1038/s41586-026-10452-4.

2. Improved reconstruction of single-cell developmental potential with CytoTRACE 2
Nature Methods Kang M Gulati GS Brown EL Qi Z Avagyan S Armenteros JJA Gleyzer R Zhang W Steen CB D'Silva JP Schwab J Clarke MF Chaudhuri AA Newman AM 2025. https://doi.org/10.1038/s41592-025-02857-2.
3. Profiling cell identity and tissue architecture with single-cell and spatial transcriptomics
Nature Reviews Molecular Cell Biology Gulati GS D'Silva JP Liu Y Wang L Newman AM 2024. https://doi.org/10.1038/s41580-024-00768-2.
4. High-resolution alignment of single-cell and spatial transcriptomes with CytoSPACE
Nature Biotechnology Vahid MR Brown EL Steen CB Zhang W Jeon HS Kang M Gentles AJ Newman AM 2023. https://doi.org/10.1038/s41587-023-01697-9.
5. T cell characteristics associated with toxicity to immune checkpoint blockade in patients with melanoma
Nature Medicine Lozano AX Chaudhuri AA Nene A Bacchiocchi A Earland N Vesely MD Usmani A Turner BE Steen CB Luca BA Badri T Gulati GS Vahid MR Kahemeneh F Harris PK Chen DY Dhodapkar K Sznol M Halaban R Newman AM 2022. https://doi.org/10.1038/s41591-021-01623-z.

Decoding cellular plasticity and spatial phenotypes in cancer

Aaron Newman
Department of Biomedical Data Science, Stanford University

Multicellular ecosystems—arising from immune, stromal, and/or malignant cells—are powerfully associated with cancer pathogenesis and response to therapy but are difficult to identify and profile clinically. Within these ecosystems, cancer cells exhibit varying degrees of developmental plasticity, a phenotype that is strongly linked to poor outcomes but challenging to characterize. In this talk, I will introduce complementary machine learning frameworks to address these challenges. The first, called CytoTRACE 2, leverages interpretable AI trained on millions of data points to infer developmental potency from single-cell expression data, enabling new insights into cancer cell plasticity. The second, termed Spatial EcoTyper, facilitates discovery of multicellular spatial ecosystems, called spatial ecotypes, from single-cell spatial expression data, and enables their multimodal recovery from spatial, single-cell, and bulk expression data. A third method, Liquid EcoTyper, leverages interpretable AI for non-invasive interrogation of spatial cellular ecosystems from plasma cell-free DNA. Together, these frameworks provide a unified basis for deciphering and tracking of developmental potency and tissue biogeography in cancer and other pathological conditions.

Reference:

1. Kang M, Gulati GS, Brown EL, Qi Z, Avagyan S, Armenteros JJA, Gleyzer R, Zhang W, Steen CB, D'Silva JP, Schwab J, Clarke MF, Chaudhuri AA, and AM Newman. Improved reconstruction of single-cell developmental potential with CytoTRACE 2. *Nature Methods*, 2025. 22:2258-2263.
2. Zhang W, Brown EL, Usmani A, Earland N, Kang M, Olelewe C, Viswanathan A, Chauhan PS, Steen CB, Jeon HS, Avagyan S, Alahi I, Semenkovich NP, Schwab JC, Sachs CM, Qaium F, Harris PK, Cai Q, Gentles AJ, Knight J, Graham RP, Bacchiocchi A, Lucas PC, Fields RC, Sznol M, Halaban R, Chen DY, Chaudhuri AA, Newman AM. Non-invasive profiling of the tumour microenvironment with spatial ecotypes. *Nature*, 2026. 654:1076-1087.



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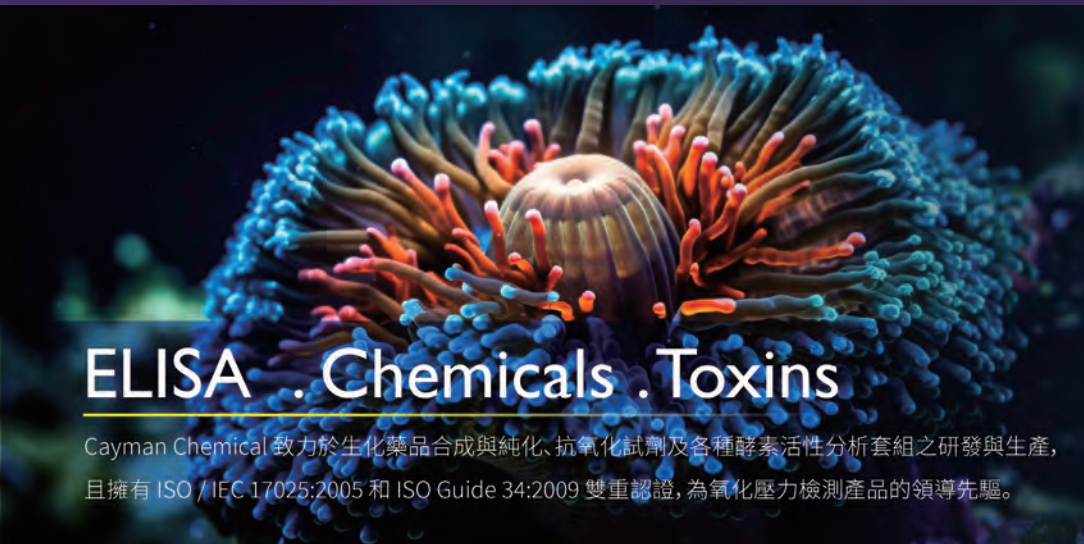
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Session 1：AI 與精準醫療研究

Taiwan Proteomics Society

TPS

台灣蛋白體學會

主持人

廖寶琦 特聘教授

國立成功大學

工業衛生學科暨環境醫學研究所

Moderator

Distinguished Prof. Pao-Chi Liao

Department of Environmental and Occupational Health,
National Cheng Kung University

陳璿宇 教授
Prof. Hsuan-Yu Chen



研究員/中央研究院 統計科學研究所
Research Fellow / Institute of Statistical Science,
Academia Sinica

Education

2003/09-2007/01; Graduate Institute of Epidemiology, National Taiwan University; Ph.D.

Experience

- 2023 - Present, Research Fellow, Institute of Statistical Science, Academia Sinica
- 2023 - Present, Professor (Joint appointment), National Taiwan University, Taipei Medical University, Tzu Chi University, National Chung Hsing University
- 2007 - 2023, Post Doctoral, Assistant, Associate Research Fellow, Institute of Statistical Science, Academia Sinica

Honors and Awards

- The 17th Wang Ming-Ning Memorial Foundation (王民寧紀念基金會) award for outstanding Ph.D. dissertation (優秀博士論文獎), 2007
- Distinguished Contribution Award in Technology Transfer, National Science Council, Taiwan, 2008
- 2007, 2008, 2012, 2013, 2015, 2020, Significant Research Achievements of Academia Sinica

Research Interests

Data Science, Precision Medicine, Multi-omics and multimodal deep learning

Selected Publications

1. 2026, “Navigating luminal heterogeneity: etiology-based proteogenomic subtyping for targeted treatment strategies in breast cancer”, *Molecular Cancer*, https://doi.org/10.1186/s12943-026-02701-x **Corresponding author**
2. 2026, “Integrative proteogenomics maps multifactorial aetiology, progression and therapeutic vulnerabilities in gastric cancer”, *Gut*, 75: 886-904 **Corresponding author**
3. 2025, “Integrative Analysis of Lung Adenocarcinoma Across Diverse Ethnicities and Exposures”, *Cancer Cell*, 45: 1731-1757 **Corresponding author**
4. 2025, “Composite microRNA-genetic risk score model links to migraine and implicates its pathogenesis”, *Brain*, 148: 2178-2188 **Corresponding author**
5. 2020, “Proteogenomics of Non-smoking Lung Cancer in East Asia Delineates Molecular Signatures of Pathogenesis and Progression”, *Cell*, 182, 226-244. **Corresponding author**
6. 2015, “R331W Missense Mutation of Oncogene YAP1 Is a Germline Risk Allele for Lung Adenocarcinoma With Medical Actionability.”, *Journal of Clinical Oncology*, 33, 2303-2310. **First author**
7. 2007, “A five-gene signature and clinical outcome in non-small-cell lung cancer.”, *New England Journal of Medicine*, 356, 11-20. **First author**

From Multi-Omics to Multimodal Deep Learning: Advancing AI-Driven Precision Medicine

Hsuan-Yu Chen[1,2,3*]
[1] *Institute of Statistical Science, Academia Sinica*
[2] *Graduate Institute of Clinical Medicine, National Chung Hsing University*
[3] *Center for Cancer Research, Kaohsiung Medical University*

Artificial intelligence (AI) is transforming biomedical research and precision medicine. However, the foundation of any successful AI application is high-quality and comprehensive data. Without appropriate biomedical data, there is no meaningful AI. Multi-omics technologies, including genomics, transcriptomics, proteomics, and other molecular profiling platforms, provide multidimensional information to characterize disease heterogeneity, biological mechanisms, and individual differences. The integration of these data with clinical and environmental information therefore provides an essential foundation for developing AI-driven precision medicine.

In this talk, I will first introduce our multi-omics studies and demonstrate how large-scale molecular data can be translated into clinically relevant knowledge. I will then present our deep learning strategy that transforms high-dimensional omics data into structured image representations. By applying convolutional neural networks to genome-, transcriptome, proteome-wide images, we aim to capture complex global patterns and interactions beyond conventional analyses based on individual biomarkers or preselected features. These approaches provide a new framework for multimodal representation and prediction from biomedical big data.

Finally, I will discuss our ongoing development of autonomous agentic AI for drug discovery. Multiple specialized AI agents can collaboratively perform target identification, molecular and protein-structure modeling, candidate design, computational evaluation, and prioritization. Importantly, computational predictions are connected to wet-lab experiments, and experimental results are returned to the AI system to guide subsequent design. This iterative AI design-experimental validation-learning-redesign cycle establishes a closed-loop framework toward autonomous biomedical discovery and next-generation precision medicine.

Prof. Hyun Joo An



Graduate School of Analytical Science and Technology,
Chungnam National University
Director / Asia-Pacific Glycomics Reference Site,
Chungnam National University
Homepage) www.agrs.kr
E-mail) hjan@cnu.ac.kr

Professional Experience

- 2024 ~ Present, Vice president for International Affairs, Chungnam National University (CNU), Korea
- 2020. 09 ~ 2023. 09, Dean, Graduate School of Analytical Science and Technology (GRAST), CNU
- 2011. 03 ~ Present, Associate Professor / Professor, GRAST, CNU
- 2011. 06 ~ Present, Director, Asia-Pacific Glycomics Reference Site

Current Positions in Academic Societies

- Vice President, Korean Human Proteome Organization (KHUPO)
- Vice President, Korean Society of Glycoscience
- National Representative of Korea, Asia Glycoscience and Glycotechnology Consortium (AGGC)
- Member, Academic Affairs Committee, Korean Chemical Society (AGCC)
- Diversity Representative (Eastern Region), Human Proteome Organization (HUPO) Council (2025–2027)

Selected Recent Publications

1. Hwang HJ, *et al.* Therapy-induced senescent cancer cells contribute to cancer progression by promoting ribophorin 1-dependent PD-L1 upregulation. ***Nature Communications*** 16, 353 (2025).
2. Oh MJ, *et al.* MS-Based Glycome Characterization of Biotherapeutics With N- and O-Glycosylation. ***Mass Spectrometry Reviews***, online ahead of print (2025).
3. Lee J, *et al.* Deciphering mouse brain spatial diversity via glyco-lipidomic mapping. ***Nature Communications*** 15, 8689 (2024).
4. Pham DT, *et al.* Novel Siglec-1/β-CD/rGO Impedimetric Biosensor for Detecting Neu5Ac Expression in Red Blood Cells. ***Biosensors and Bioelectronics***, 117235 (2024).
5. Lee J, *et al.* Advancements in protein glycosylation biomarkers for ovarian cancer through mass spectrometry-based approaches. ***Expert Review of Molecular Diagnostics*** 24(4), 249–258 (2024).
6. Oh MJ, *et al.* Equivalence assessment of biotherapeutics with N- and O-glycosylation sites by sequential intact glycoform mass spectrometry (IGMS). ***Journal of Pharmaceutical and Biomedical Analysis*** 234, 115558 (2023).
7. Maixner F, *et al.* Microscopic evidence of malaria infection in visceral tissue from Medici family, Italy. ***Emerging Infectious Diseases*** 29(6), 1280 (2023).
8. Oh MJ, *et al.* In-depth investigation of altered glycosylation in human haptoglobin associated with cancer by mass spectrometry. ***Mass Spectrometry Reviews*** 42(2), 496–518 (2023).
9. Yoon JH, *et al.* Brain lipidomics: From functional landscape to clinical significance. ***Science Advances*** 8(37), eadc9317 (2022).
10. Lee J, *et al.* Spatial and temporal diversity of glycome expression in mammalian brain. ***Proceedings of the National Academy of Sciences*** 117(46), 28743–28753 (2020).

What Gene Editing Cannot Predict: Glycan Remodeling in Xenogeneic Tissues and Blood

Hyun Joo An[1,2]*

1 Asia-Pacific Glycomics Reference Site (AGRS)

2 Graduate School of Analytical Science and Technology (GRAST), Chungnam National University, Daejeon 34134, Republic of Korea

Xenogeneic organs and red blood cells offer a route around the shortage of human donors, yet both are limited by antibody-mediated rejection against glycan epitopes absent in humans: terminal alpha-galactose (α-Gal), N-glycolylneuraminic acid (Neu5Gc), and the Sd^a antigen. Pigs lacking GGTA1, CMAH, and B4GALNT2 were engineered to remove these determinants. Whether three deletions eliminate the intended epitopes without rewiring the remaining glycome cannot be inferred from genotype alone, because glycans are not template-encoded; their structures emerge from competition among glycosyltransferases for shared substrates.

We applied isomer-sensitive porous graphitized carbon nano-liquid chromatography-tandem mass spectrometry (nanoLC-MS/MS) glycomics with site-specific glycoproteomics to heart, kidney, and pancreas tissues and to red blood cells from wild-type and triple-knockout pigs, using human red blood cells as the clinical reference. Immunogenic glycan burden in wild-type tissue proved strongly organ-dependent, with α-Gal and Neu5Gc showing distinct tissue preferences, arguing against a single editing strategy serving every transplantable organ. In red blood cells the two species differed in kind rather than degree: human cells were dominated by blood group antigen-associated structures, whereas wild-type porcine cells carried greater structural diversity, with multiple non-human determinants co-occurring within individual glycans. At glycosylation sites on band 3, the principal erythrocyte membrane glycoprotein, knockout cells retained the same occupied sites and comparable glycan scaffolds, with remodeling confined to terminal capping.

This asymmetry delineates what is predictable and what is not. Substitutions arising at the nucleotide sugar donor pool follow from biosynthetic logic; the fate of substrates freed by the remaining deletions does not, and only measurement resolves it. Rather than treating glycans as one more omics layer to concatenate, we propose encoding biosynthetic network constraints as structural priors, an approach suited to data that are few in number but heavily constrained in form, and close with the open problems we cannot solve alone.

Dr. Je-Hyun Baek



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E-mail: jhbaek@mf.seegene.com
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Department: R&D Center for Clinical Mass Spectrometry
Affiliation: Seegene Medical Foundation, Office: 320 Cheonho-daero, Seongdong-gu, Seoul, Korea
LinkedIn: https://www.linkedin.com/in/je-hyun-baek-190a2642/

Education

2001.09-2007.08	Ph.D.	Korea University (Life Science & Biotechnology)
1998.03-2000.02	Master	Ajou University (Biochemistry)
1991.03-1998.02	Bachelor	Ajou University (Chemistry)

Professional Experience

2018.06-present	Research Director	Seegene Medical Foundation, R&D Center for Clinical Mass Spectrometry
2013.04-2018.01	Head of R&D Division	Diatech Korea, R&D Division
2012.03-2013.03	Research Professor	Seoul National University, Medical Research Center
2010.02-2012.01	Postdoc	Department of Neurobiology, Physiology and Behavior, UC Davis (USA)

Academic/Professional Society

2024.01~present	Committee Member	Korea Human Proteome Organization (KHUPO)
2024.01~present	Chief of Translational Medicine Study Group	Korean Society for Mass Spectrometry (KSMS)
2014.09~present	Vice-President	Korea Association of Research for Pharmaceutical Analysis (Korea Drug Research Association)

Publications (Recent 2 years)

1. Integrated multiomics reveals starvation-driven keratin degradation and persistence in Fervidobacterium islandicum AW-1. Jae-Yoon Sung, Ji-Yeon Kim, Hyeon-Su Jin, Je-Hyun Baek, Nicole Enjeh Kim, Hyun Ho Song, Seong-Hun Bong, Yong-Jik Lee, Byoung-Chan Kim, Do Yup Lee, Dong-Woo Lee. *iScience*. 2026 Apr 20;29(5):115755.
2. GAFAD: An LC-MS/MS-Based Model for Early Hepatocellular Carcinoma Detection Beyond GALAD's Limitation. Hyojin Kim, Wonseok Oh, Juri Park, Saeyoung Lee, Won Suk Yang, Soon Sun Kim, Jae Youn Cheong, Je-Hyun Baek*. *Clin Mol Hepatol*. 2026 32(3):1225-1239.
3. Development and Validation of a Lectin-independent Liquid Chromatography-Tandem Mass Spectrometry Method for Serum Glycosylated Alpha-fetoprotein Analysis and Comparison with a Liquid-phase Binding Assay. Hyojin Kim, Juri Park, Hanseul Suh, Saeyoung Lee, Yoonha Park, Won Suk Yang, Dohsik Minn, Soon Sun Kim, Jae Youn Cheong, Je-Hyun Baek*. *Ann Lab Med* 2026 Jan 1;46(1):62-71.

4. Direct Identification of IMP and VIM Metallo-β-Lactamases in Gram-Negative Bacteria Using A-MALDI. Dong Huey Cheon, Seohyun Hwang, Yoon Kyung Choi, Saeyoung Lee, Won Suk Yang, Bo Kyung Kim, Min Jin Kim, Sun Hwa Lee, Je-Hyun Baek*. *Rapid Commun Mass Spectrom*. 2025 Oct 15;39(19):e10095.

Beyond Genomic Liquid Biopsy: Tumor-Proximal BALF Extracellular Vesicle Multi-Proteomics for Precision Lung Cancer Care

Wonseok Oh [1], Jae Young Hur [2,3], Yoonha Park [1], Hee Joung Kim [2,4], In Ae Kim [2,4], Wan Seop Kim [2,3], Won Suk Yang [1], Kye Young Lee [2,4*], Je-Hyun Baek [1*]

[1] *R&D Center for Clinical Mass Spectrometry, Seegene Medical Foundation, Seoul, Republic of Korea.*
[2] *Precision Medicine Lung Cancer Center, Konkuk University Medical Center, Seoul, Republic of Korea.*
[3] *Department of Pathology, Konkuk University School of Medicine, Seoul, Republic of Korea.*
[4] *Department of Pulmonary Medicine, Konkuk University School of Medicine, Seoul, Republic of Korea.*

Precision oncology increasingly depends on molecular information to guide treatment, yet adequate tumor tissue is often difficult to obtain in patients with small, inaccessible, or early-stage lung lesions. Plasma-based liquid biopsy offers a less invasive alternative, but limited tumor-derived material can restrict its sensitivity, particularly in early disease. Building on our previous work with bronchoalveolar lavage fluid (BALF) extracellular vesicles (EVs) for genomic liquid biopsy, we sought to move beyond mutation detection toward a tumor-proximal multi-proteomic approach that captures the biological state of lung cancer. Paired lesional BALF EV samples were analyzed using mass-spectrometry-based proteomics, allowing lesion-associated molecular changes to be explored. The resulting molecular profiles revealed cancer-associated alterations linked to disease stage and genomic background, while additional glycoproteomic and phosphoproteomic information provided complementary insight into tumor biology beyond DNA alterations alone. To translate these high-dimensional data into clinically tractable signatures, we combined customized DIA processing, biologically guided feature selection, and machine-learning approaches. This strategy enabled the identification of compact candidate marker panels with potential utility for lung cancer detection and stage stratification. Our findings suggest that BALF EVs may serve not only as carriers of tumor-derived genetic material but also as multidimensional reporters of the local tumor microenvironment. This work represents a translational step from genomic liquid biopsy toward tumor-proximal multi-proteomics, with the goal of improving early detection, molecular stratification, and precision treatment decisions in patients for whom conventional tissue-based diagnosis remains challenging.

Session 2：AI for Bioinformatics

TBSB

台灣生物資訊與系統生物學會

主持人

張資昊 教授

臺北醫學大學

醫學資訊研究所

Moderator

Prof. Tzu-Hao Chang

Graduate Institute of Biomedical Informatics,
Taipei Medical University



何信瑩 副院長
Prof. Shinn-Ying Ho

國立陽明交通大學 工程生物科學學院
特聘教授/國立陽明交通大學 生物資訊及系統生物研究所
Vice Dean / College of Engineering Bioscience,
National Yang Ming Chiao Tung University
Distinguished Professor / Institute of Bioinformatics & Systems Biology,
National Yang Ming Chiao Tung University

Education

國立交通大學 資訊工程博士（1992）

Experience

- 國立交通大學生物科技學院/生物資訊所 教授兼副院長/所長
- 台灣生物資訊與系統生物學會 理事長 (2022/01-2026/01)
- 國立交通大學衍生新創研究服務公司（數冠科技）創辦人

Honors and Awards

- 國立陽明交通大學學術卓越貢獻表揚(2024, 2025)。
- 榮獲第 18-22 連續五屆國家新創獎(2021-2025)。
- 全球 Top 2%終身頂尖科學家 (AI & Image Processing, Bioinformatics)。

Research Interests

人工智慧、演化學習、生物資訊、智慧醫療、生醫數據分析

Selected Publications

1. T-Y Chen, S-Y Ho*, TargetPrior: A miRNA-Signature Embedded Evolutionary Learning Framework for Prioritizing Drug Targets in Acute Myeloid Leukemia, *Bioinformatics*, 2026 (accepted).
2. TY Wu, YL Ho, SY Sathipati, YJ Ho, TY Chen, TY Lee, SY Ho*, Network molecular nutrition framework for exploring the oncogenic mechanisms of nutrient deficiency in cervical cancer, *Clinical Nutrition Open Science*, vol. 60, page 100638, 2026.
3. I Lee, H-C Chu, W-H Hui, W-K Seto, S-Y Ho*, Y-H Huang*, Evolutionary learning-derived multimodal models integrating comprehensive imaging features to predict early recurrence of hepatocellular carcinoma after curative resection, *Journal of Hepatology*, Vol. 82, pp. S478-S479, 2025.
4. IC Lee, YP Tsai, YC Lin, TC Chen, CH Yen, NC Chiu, HE Hwang, CA Liu, S-Y Ho*, Y-H Huang*, A hierarchical fusion strategy of deep learning networks for detection and segmentation of hepatocellular carcinoma from computed tomography images, *Cancer Imaging* 24 (1), 43, 2024
5. S. Yerukala Sathipati, M.-J. Tsai, S. K Shukla, S-Y Ho*, Artificial intelligence-driven pan-cancer analysis reveals miRNA signatures for cancer stage prediction, *Human Genetics and Genomics Advances*, 4 (3), 2023.

Evolutionary Learning in Bioinformatics: From Signature Discovery to Intelligent Systems

Shinn-Ying Ho* (何信瑩)
Institute of Bioinformatics and Systems Biology, National Yang Ming Chiao Tung University, Taiwan

High-dimensional biological data present major challenges for machine learning, particularly in identifying informative molecular signatures, modeling complex regulatory interactions, and developing robust predictive models. Over the past two decades, our research has advanced evolutionary learning (EL) by integrating efficient evolutionary optimization with machine learning to address these challenges. This talk introduces the methodological foundations of EL, including orthogonal-array-based genetic algorithms and particle swarm optimization. We highlight representative applications of EL in discovering compact and informative molecular signatures, particularly miRNA- and lncRNA-based signatures for cancer classification, prognosis, and treatment response prediction. We further discuss signature stability through the SISig framework and its recent extension using GPU-accelerated parallel evolutionary learning to efficiently discover robust consensus molecular signatures. Building upon these developments, we present TargetPrior, which integrates miRNA signatures, gene regulatory networks, and biological knowledge to prioritize candidate therapeutic targets. Applications in liver cancer imaging and recurrence prediction further illustrate the translational potential of AI in medical informatics. Finally, we explore the application of EL to large language model compression, highlighting its potential for developing scalable intelligent systems in bioinformatics and precision medicine.



陳鯨太 助理教授
Assistant Prof. Ching-Tai Chen

臺北市立大學 資訊科學系
Department of Computer Science, University of Taipei

Education

國立交通大學 生物資訊及系統生物研究所

Experience

- 助理教授, 臺北市立大學 資訊科學系
- 助理教授, 亞洲大學 生物資訊與醫學工程學系
- 博士後研究員, 中央研究院 資訊科學所
- 博士後研究員, 日本 產業総合技術研究所

Honors and Awards

- 臺北市立大學 114 年度教師研究成果獎勵
- 指導學生榮獲「2026 智創未來 - 生成式 AI 創意競賽」邏輯思維設計組金牌獎
- 指導學生參加「2025 生成式人工智慧創新應用國際競賽」獲得佳作
- 指導學生參加「2025 全國 AI 專題創意競賽」榮獲跨領域組第三名
- 指導學生參加「2025 全國 AI 專題創意競賽」榮獲跨領域組佳作
- 指導學生參加中央研究院統計所舉辦「2025 資料科學漫步」榮獲優選獎
- 指導學生參加「中技社 AI 創意競賽」榮獲佳作
- 指導學生參與「智創未來 : 2025 生成式 AI 創意設計競賽」榮獲企業創新獎、與邏輯思維設計組第三名
- 指導學生獲得國科會舉辦「2024 GenAI Stars 生成式 AI 百工百業應用選拔」創意創客組優選獎
- 指導學生獲得「智創未來 : 2024 生成式 AI 創意設計競賽」邏輯思維設計組金牌獎&企業創意獎
- 指導學生獲得「智創未來 : 2024 生成式 AI 創意設計競賽」問題解決組金牌獎&企業創意獎

Research Interests

生物資訊、人工智慧、自然語言處理、計算蛋白質體學

Selected Publications

1. Yi-Yun Chou, Jing-Chen Yang, Tun-Chu Tsai, Ching-Tai Chen*. 2025. "IQUP identifies quantitatively unreliable spectra with machine learning for isobaric labeling-based proteomics." *Scientific Reports* 15:31900.
2. Saptashwa Datta, Jen-Chieh Yu, Yi-Hsiang Lin, Yun-Chen Cheng and Ching-Tai Chen*. 2025. "AAGP integrates physicochemical and compositional features for machine learning-based prediction of anti-aging peptides." *Scientific Reports* 15: 29036.
3. Yu, Jen-Chieh, Kuan Ni, and Ching-Tai Chen*. 2024. "ENCAP: Computational Prediction of Tumor T Cell Antigens with Ensemble Classifiers and Diverse Sequence Features." *PLOS ONE* 19 (7): e0307176.

4. Lee, Yun-Chen, Jen-Chieh Yu, Kuan Ni, Yu-Chuan Lin, and Ching-Tai Chen*. 2024. "Improved Prediction of Anti-Angiogenic Peptides Based on Machine Learning Models and Comprehensive Features from Peptide Sequences." *Scientific Reports* 14 (1): 14387.
5. You-Qian Lee*, Ching-Tai Chen*, Chien-Chang Chen, Chung-Hong Lee, Peitsz Chen, Chi-Shin Wu, Hong-Jie Dai. 2024. "Unlocking the Secrets Behind Advanced Artificial Intelligence Language Models in De-Identifying Chinese-English Mixed Clinical Text," *Journal of Medical Internet Research* 26 (1): e48443.
6. Tzu-Yun Kuo, Jen-Hung Wang, Yung-Wen Huang, Ting-Yi Sung*, Ching-Tai Chen*. 2023. "Improving quantitation accuracy in isobaric-labeling mass spectrometry experiments with spectral library searching and feature-based peptide-spectrum match filter." *Scientific Reports* 13 14119 (2023).
7. Jen-Hung Wang, Wai-Kok Choong, Ching-Tai Chen*, Ting-Yi Sung*. 2022. "Calibr improves spectral library search for spectrum-centric analysis of data independent acquisition proteomics." *Scientific Reports* 12 (1), 1-13.
8. Ching-Tai Chen*, Jen-Hung Wang, Cheng-Wei Cheng, Wei-Che Hsu, Chu-Ling Ko, Wai-Kok Choong, Ting-Yi Sung*. 2021. "Multi-Q 2 software facilitates isobaric labeling quantitation analysis with improved accuracy and coverage." *Scientific Reports* 11 (1), 1-12.

Beyond Balanced Benchmarks: Imbalance-Aware Prediction of Blood-Brain Barrier Penetrating Peptides

Ching-Tai Chen[1*], Tadeo Oran Pontjoharyo[2]
[1] *Department of Computer Science, University of Taipei, Taipei, Taiwan*
[2] *Department of Bioinformatics and Medical Engineering, Asia University, Taichung, Taiwan*

The blood-brain barrier (BBB) blocks most therapeutic compounds from reaching the central nervous system (CNS), making BBB-penetrating peptides (BBPs) attractive shuttle molecules for treating conditions such as Alzheimer's disease and glioblastoma. Computational screening can greatly accelerate BBP discovery, but existing predictors suffer from two key limitations: artificially balanced evaluation datasets that overstate real-world performance, and training sets containing fewer than 500 experimentally validated positive sequences.

We present a machine learning prediction framework that mimics realistic BBP screening conditions by constructing datasets with positive-to-negative (PN) ratios ranging from 1:1 to 1:15. Rather than relying on a fixed feature set, the framework tailors its feature representation to the degree of class imbalance: an adaptive selection pipeline derives a customized subset from thousands of sequence descriptors for each PN ratio. Feature analysis revealed that global composition and physicochemical properties dominate in balanced settings, whereas local dipeptide and spatial co-occurrence patterns become essential as the number of negatives grows.

Beyond model design, data scarcity remains a fundamental ceiling on predictive performance. We will also briefly discuss our ongoing exploration of protein language model (PLM)-based data augmentation as a feasible route to expanding limited BBP training data, with preliminary observations suggesting that generated sequences preserve the key biochemical characteristics of experimentally validated BBPs.

The novelty of this work lies in imbalance-aware feature design and optimization. The proposed method outperforms existing predictors on four independent test datasets with varying imbalanced conditions, improving Matthew's correlation coefficient by 2.7%–5.0% and accuracy by 1.5%–4.1%. By delivering reliable performance under realistic screening conditions, we offer a generalizable tool for large-scale BBP discovery, and together with PLM-driven augmentation, points toward a more scalable path for accelerating CNS drug delivery research.

黃耀廷 系主任
Prof. Yao-Ting Huang

教授/國立中正大學 資訊工程學系
Department Chair / Institute of Computer Science and Information Engineering, National Chung Cheng University



Education

Ph.D., Computer Science and Information Engineering, National Taiwan University (2003–2006)

Experience

- Director, Taiwan Genomics and Genetics Society, 2017–2026
- Director, Taiwan Society of Precision Health and Toxicogenomics, 2022–2026
- Head, Division of Information Education, Office of Information Technology, National Chung Cheng University, 2017-2025.
- Associate Editor, Journal of Information Science and Engineering, 2020-

Honors and Awards

Outstanding Research Award, National Chung Cheng University, 2023.

Research Interests

Bioinformatics; algorithms; clinical metagenomic next-generation sequencing; long-read sequencing; infectious disease; cancer genomics.

Selected Publications

1. Huang Y-T*, Liu P-Y, Shih P-W. Homopolish: a method for the removal of systematic errors in Nanopore sequencing by homologous polishing. *Genome Biology*, 2021.
2. Lin J-H, Chen L-C, Yu S-C, Huang Y-T*. LongPhase: an ultra-fast chromosome-scale phasing algorithm for small and large variants. *Bioinformatics*, 2022.
3. Chiou C-S, Chen B-H, Wang Y-W, Kuo N-T, Chang C-H, Huang Y-T*. Correcting modification-mediated errors in nanopore sequencing by nucleotide demodification and reference-based correction. *Communications Biology*, 2023.
4. Huang Y-T, Liu P-Y. Emergence of carbapenem resistance in persistent Shewanella algae bacteremia: the role of PdsS G547W mutation in adaptive subpopulation dynamics. *Annals of Clinical Microbiology and Antimicrobials*, 2024.
5. Huang P-H§, Huang Y-T§, et al. Genomic Heterogeneity and Lineage-Specific Beta-Lactamases in Recurrent Achromobacter Bloodstream Infection Patients. *Emerging Microbes & Infections*, 2025.

Long-Read Sequencing for Precision Medicine: From Tumor Variant Phasing to Infectious Disease Diagnosis

Yao-Ting Huang[1*]
[1] *Department of Computer Science and Information Engineering, National Chung Cheng University, Taiwan*

Long-read sequencing enables integrated analysis of genetic variation and DNA methylation, supporting precision medicine across human genomics and oncology. This presentation examines haplotype reconstruction in tumor genomes and applies related strategies to resolve complex, low-frequency molecular signals in infectious diseases.

For tumor genome analysis, we develop LongPhase-S to reconstruct somatic haplotypes from paired tumor–normal sequencing, enabling tumor DNA fraction estimation and variant recalibration. We further develop LongPhase-TO for tumor-only sequencing by incorporating loss of heterozygosity, thereby addressing allelic imbalance in heterogeneous tumors. These methods support sensitive detection and phasing of low-frequency variants in complex tumor samples, providing a framework for resolving tumor heterogeneity and distinguishing somatic variants from sequencing errors in samples containing admixed normal cells.

Beyond tumor genomics, distinguishing disease-associated signals from complex biological backgrounds is also a central challenge in infectious disease diagnosis. We address this challenge through multimodal approaches that integrate pathogen detection with host molecular signals and clinical context. Such multimodal strategies may improve discrimination between causative pathogens, colonization, and incidental findings. Together, these studies demonstrate how somatic variant detection, haplotype phasing, and complementary molecular evidence can resolve heterogeneous signals and improve interpretation across tumor genomics and infectious disease diagnostics.

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2. 入口氣流風速測試(Inflow)
3. 氣流煙霧型態測試
4. 進氣濾網洩漏測試
5. 排氣濾網洩漏測試
6. 現場設置測試
7. 照明照度測試
8. 噪音層級測試
9. UV燈強度測試
10. 振動測試

Session 3：精準醫學檢測與多體學判讀

台灣精準醫學學會
Taiwan Precision Medicine Society

TPMS

台灣精準醫學學會

主持人

王智弘 總院長

臺北市立聯合醫院

理事長/臺灣精準醫學學會

Moderator

Dr. Chih-Hung Wang

Superintendent / Taipei City Hospital
President / Taiwan Precision Medicine Society



陳燕麟 主任
Dr. Yen-Lin Chen

三軍總醫院 精準醫學暨基因體中心
主治醫師/三軍總醫院 病理部
助理教授/輔仁大學 醫學系
Director / Center for Precision Medicine and Genomics,
Tri-Service General Hospital
Visiting Staff / Department of Pathology,
Tri-Service General Hospital
Assistant Prof. / School of Medicine, Fu Jen Catholic University

Education
- 輔仁大學醫學系學士
- 輔仁大學化學所博士

Experience
- 耕莘醫院病理科主治醫師
- 日本國家神經精神中心研究員
- 牛津大學 Botnar 研究所研究員
- 三軍總醫院病理部主治醫師
- 中華民國肌萎縮症病友協會常務理事

Honors and Awards
- 新北市優秀勞工獎
- 新北市第四屆醫療公益獎
- 乳癌防治基金會第三屆優秀論文獎
- 基因檢測公益計畫發起者
- 輔仁大學傑出校友
- 法鼓山關懷生命智慧獎
- 罕病基金會首屆罕見疾病貢獻獎

Research Interests
基因檢測、肌肉病理

Selected Publications
1. Chang CH, Chou CY, Raben TG, et.al. Polygenic height prediction for the Han Chinese in Taiwan. *NPJ Genom Med.* 2025 Feb 5;10(1):7. doi: 10.1038/s41525-025-00468-6.
2. Chen YL, Lin WN, Pan PY, et.al. Generation of a novel Dysferlin microdeletion knock-in mouse model mimicking muscular dystrophy-like pathology. *Sci Rep.* 2026 Apr 1;16(1):15322. doi: 10.1038/s41598-026-46635-2.
3. Lee PY, Shih CY, Cheng CY, et.al. The Prognostic Value of CIP2A and Its Association with CD31, E-Cadherin, and pAMPK in Lung Cancer. *Int J Mol Sci.* 2025 Aug 28;26(17):8362. doi: 10.3390/ijms26178362.
4. Wang WK, Lin HY, Lin CH, et.al. GRK6 palmitoylation dictates triple-negative breast cancer metastasis via recruiting the β-Arrestin 2/MAPKs/NF-κB signaling axis. *Breast Cancer Res.* 2024 Dec 31;26(1):193. doi: 10.1186/s13058-024-01953-z.

5. Tsai MK, Chen YL, Chen HC, et.al. IgG4-related tubulointerstitial nephritis. *Ann Rheum Dis.* 2024 Mar 12;83(4):537-538. doi: 10.1136/ard-2023-224899.

Hidden in Plain Sight: Uncovering Diagnostic Clues in Whole-Exome Sequencing

Yen-Lin CHEN [1,2 *]
[1] *Department of Pathology, Tri-Service General Hospital*
[2] *Taiwan Muscular Dystrophy Association*

Whole-exome sequencing (WES) has become an important tool for diagnosing rare and inherited diseases. However, despite continuous advances in sequencing technologies and bioinformatics, translating genetic findings into a definitive clinical diagnosis remains challenging. In real-world practice, critical diagnostic clues are not always immediately apparent and may only emerge through repeated analysis, integration of clinical information, and stepwise diagnostic reasoning. In this presentation, we will share our clinical experiences with WES-based diagnosis at Tri-Service General Hospital. Through selected real-world cases, we will illustrate the diagnostic journey from initial observations and uncertainties to further investigation, reassessment, and the gradual discovery of meaningful clues. Rather than focusing solely on final diagnostic outcomes, we will highlight the challenges, unexpected findings, and practical lessons encountered throughout the process. We will also explore how overlooked information within existing genomic data may provide additional diagnostic insights. By sharing these experiences, we aim to encourage broader perspectives on genomic interpretation, emphasize the importance of critical thinking and continuous reassessment, and demonstrate the potential value of WES beyond conventional variant analysis.

張廷彰 教授級顧問醫師
Dr. Ting-Chang Chang



林口長庚紀念醫院 婦產部
Professor-Level Consulting Physician /
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CURRENT APPOINTMENTS
- 林口長庚紀念醫院婦產部教授級顧問醫師（2020 年起）
- 長庚大學醫學院教授
- 長庚紀念醫院學術教授

EDUCATION & TRAINING
- 中國醫藥學院醫學系
- 美國哈佛大學 公共衛生碩士
- 美國布萊格姆婦女醫院及新英格蘭滋養細胞疾病中心 婦癌研究員

PROFESSIONAL EXPERIENCE
- 長庚紀念醫院北院區婦產部主任（2013–2019）
- 林口長庚紀念醫院婦癌科主任（2000–2007）
- 臺灣精準醫學學會創會理事長（2015–2019）
- 臺灣癌症登記學會創會理事長（2006–2012）

EXPERTISE
- 婦癌診斷、治療與婦科手術
- 人體臨床試驗設計、執行及資料分析
- 實證醫學、生物統計與癌症登記

SPEAKER INTRODUCTION
張廷彰教授為婦癌及臨床研究領域專家，現任林口長庚紀念醫院婦產部教授級顧問醫師，曾任婦產部主任、婦癌科主任，並為臺灣精準醫學學會創會理事長。

AI 在臨床醫療數據分析的應用



陳昱光 主任
Dr. Yu-Guang Chen

三軍總醫院 再生醫療中心
Director / Center for Cell Therapy and Regenerative Medicine,
Tri-Service General Hospital

CURRENT APPOINTMENTS

- 三軍總醫院再生醫療中心主任
- 三軍總醫院內科部血液腫瘤科主治醫師
- 國防醫學大學醫學系助理教授
- 中華民國血液暨骨髓移植學會副秘書長

EDUCATION & TRAINING

- 英國倫敦大學學院（UCL）癌症中心 分子腫瘤學博士
- 國防醫學院醫學系

PROFESSIONAL EXPERIENCE

- 內科、血液病、腫瘤內科及血液暨骨髓移植專科醫師資格
- 三軍總醫院內科部住院醫師及總醫師

EXPERTISE

- 白血病、淋巴瘤與多發性骨髓瘤之診斷及治療
- 骨髓移植、細胞治療及血液疾病照護
- 實體癌症標靶治療與精準腫瘤醫療

SPEAKER INTRODUCTION

陳昱光醫師現任三軍總醫院再生醫療中心主任、血液腫瘤科主治醫師及國防醫學大學助理教授，專長為血液腫瘤、骨髓移植、細胞治療與精準腫瘤學。

Precision Medicine in Hematology Practice

Yu-Guang Chen, MD
Regenerative Medicine Center, Division of Hematology and Oncology, Tri-Service General Hospital, Taipei, Taiwan

Hematology has become one of the most data-dense fields of precision medicine. Germline sequencing, somatic mutation profiling, flow cytometry, transcriptomic signatures and measurable residual disease (MRD) assays now inform nearly every step of patient care. The real clinical challenge has shifted from generating these data to interpreting them correctly at the bedside. In this talk, we use a series of clinical cases to explore the application of precision medicine across hematologic diseases. These cases span both benign and malignant hematology, from inherited red cell disorders to acute leukemia and cellular immunotherapy. Through these cases, we show how multi-omic results support accurate diagnosis, refine risk stratification, guide disease monitoring and inform treatment selection. In benign hematology, germline testing can clarify long-standing diagnostic uncertainty and spare patients unnecessary investigations. In hematologic malignancies, integrated genomic and immunophenotypic data define disease subtypes and match patients to targeted agents or immunotherapy. Serial MRD assessment further allows clinicians to anticipate relapse and to time transplantation or pre-emptive intervention. Each testing platform has distinct strengths, and no single assay captures the full clinical picture. Results therefore need to be interpreted together with clinical findings and morphology. We also discuss common pitfalls in clinical interpretation. These include variants of uncertain significance, clonal hematopoiesis, discordance between testing platforms, assay sensitivity limits and reimbursement constraints in Taiwan. Looking ahead, single-cell technologies and artificial intelligence may further improve the speed and accuracy of interpretation. Precision hematology works best when laboratory, clinical and bioinformatic expertise meet in a shared interpretive framework.

Keywords precision medicine, next-generation sequencing, multi-omics, measurable residual disease, hematologic diseases

Session 4：Evolutionary genomics

TSECB

台灣演化與計算生物學會

主持人

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附設醫院醫學研究部

Moderator

Dr. Chia-Lang Hsu

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National Taiwan University
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呂美曄 博士
Dr. Mei-Yeh Jade Lu

研究技師/中央研究院 生物多樣性研究中心
Senior Research Specialist / Biodiversity Research Center, Academia Sinica



Education

- **Ph.D.**, Dept. of Plant Biology, University of Illinois at Urbana-Champaign, USA
- **M.S.**, Dept. of Botany and Plant Sciences, University of California, Riverside, USA
- **B.S.**, Dept. of Horticulture, National Taiwan University, Taiwan

Experience

- **Senior Research Specialist**, Biodiversity Research Center, Academia Sinica (2021–Present)
- **Head & Manager**, High-Throughput Genomics Core Facility, Academia Sinica (2008–Present)
- **Associate Research Specialist**, BRC, Academia Sinica (2015–2021)
- **Visiting Scientist Scholar**, Joint Genome Institute (JGI), DOE, USA (2017)
- **Assistant Research Specialist**, BRC, Academia Sinica (2011-2015)
- **Postdoctoral Fellow**, University of Iowa (1998–2004) & Academia Sinica (2005–2010)

Honors and Awards

- **ASCF Core Facility Grant,** High-Throughput Genomics Core Facility, Academia Sinica (2019-2021, 2022–2026, 2027-2031)
- **Academia Sinica Infectious Disease Research (AS-IDR) Grant** (Co-PI, 2021–2025)
- **MOST Genomic Analyses of Drug-Resistant Mycobacterium tuberculosis Grant**
- **MOST Genomic Analysis of Worldwide Blast Fungal Isolates Grant**
- **NSTC/MOST Gender and Technology Project**
- **10x Genomics Certified Service Provider** (Single-Cell & Visium CytAssist) (2024~)
- **10x Genomics ATAC-seq Grant Program Award** (2019)
- **Excellent Performance Award for High Throughput Sequencer**, Academia Sinica (2015~2017)
- **Roche 454 GS FLX+, Global Grant Award for free run** (2013)
- **Postdoctoral Fellowship, American Heart Association, USA** (1999–2001)

Research Interests

- High-Throughput Sequencing Technologies (2nd & 3rd -Gen major platforms)
- Single-Cell & Spatial Transcriptomics/Omics (10x Visium HD, Xenium, Long-Read Iso-Seq)
- Plant Functional Genomics & C4 Transcriptomics
- 3D Genomes & Microbial Genomics

Selected Publications

1. Kong WL, Wu CC, Chen Y H, Li JY, Ding K, Lee Y C,.., Chang YM, Li WH*, & Lu **MYJ***. (2026). Profiling maize embryonic leaf development and discovering new genes using high-resolution spatial long-read isoform sequencing. ***Nayzwd Paar yx*** (*in press*). DOI: 10.1038/s41477-026-02364-y.

2. Wu CC, Larsson L, Hsieh KT, Yu CP, Chen YH, Ding KH, Yang HC, Lundeberg J, Ho CK*, Wu SH*, **Lu MYJ*** & Li WH* (2026). Serial Spatial Transcriptomes Reveal Regulatory Transitions in Maize Leaf Development. ***Pcary Bjsydbjrsosiãj.***, DOI: 10.1111/pbi.70515.
3. Liu WY, Yu CP, Chang CK, Chen HJ, Li MY, Chen YH, Shiu SH, Ku MSB*, Tu SL*, **Lu MYJ***, & Li WH* (2022). Regulators of early maize leaf development inferred from transcriptomes of laser capture microdissection (LCM)-isolated embryonic leaf cells. ***PNAS***, 119(35): e2208795119.
4. **Lu, MYJ***, Fan WL, Wang WF, Chen T, Tang YC, Chu FH, ... & Li WH* (2014). Genomic and transcriptomic analyses of the medicinal fungus Antrodia cinnamomea for its metabolite biosynthesis and sexual development. ***PNAS***, 111(44): E4743-E4752.
5. Gómez Luciano LB, Tsai IJ, .., **Lu MJ***, Nakayashiki H*, Li WH*. Blast Fungal Genomes Show Frequent Chromosomal Changes, Gene Gains and Losses, and Effector Gene Turnover. ***MsoBjsoEàso*** 2019;36(6):1148-1161.
6. Chang YM, Lin HH, Liu WY, Yu CP, Chen HJ, Wartini PP, ... **Lu MYJ**, ... & Li WH* (2019). Comparative transcriptomics method to infer gene coexpression networks and its applications to maize and rice leaf transcriptomes. ***PNAS***, 116(8): 3091-3099.

When Spatial Meets Long Reads: Unveiling Novel Genes and Alternative Splicing in Maize Embryonic Leaves

Wye-Lup Kong[1], Chi-Chih Wu[1,2], Yi-Hua Chen[1], Jeng-Yi Li[1], Kai-Hsuan Ding[1], Yi-Chen Lee[1,3], Zhi Thong Soh[1], Chun-Shiu Wu[1], Yao-Ming Chang[4], Wen-Hsiung Li[1,5]*, and Mei-Yeh Jade Lu[1]*

[1] Biodiversity Research Center, Academia Sinica, Taipei 115201, Taiwan
[2] Institute of Plant and Microbial Biology, Academia Sinica, Taipei, 115201, Taiwan
[3] Department of Agronomy, National Taiwan University, Taipei, 115201, Taiwan
[4] Institute of Biomedical Science, Academia Sinica, Taipei 115201, Taiwan
[5] Department of Ecology and Evolution, University of Chicago, Chicago, IL 60637, United States

Dissecting transcriptomic diversity at both spatial and full-length isoform levels is essential for understanding cell-type specification, regulatory evolution, and developmental novelties in plants. While maize (Zea mays) leaf architecture and C4 photosynthesis have been extensively studied, conventional short-read spatial profiling often misses critical alternative splicing events, structural variations, and novel isoforms.

In this talk, we present what happens when spatial transcriptomics meets long-read isoform sequencing. By integrating PacBio HiFi/Kinnex Iso-Seq with high-resolution spatial profiling (Spatial Iso-Seq / MAS Iso-Seq), we captured full-length transcript isoforms directly across micro-architectural zones of developing maize embryonic leaves.

Our spatial isoform mapping uncovers thousands of previously unannotated splice variants, tissue-specific isoform preferences, and key regulatory transition networks driving early leaf vascular and mesophyll differentiation. This synergistic multi-omics strategy provides an unprecedented multi-dimensional view of plant functional genomics, opening new opportunities for evolutionary genomics and modern crop biotechnology. (Nat. Plants 2026, https://doi.org/10.1038/s41477-026-02364-y)

Reference: Kong, W. L., Wu, C. C., Chen, Y. H., Li, J. Y., Tin, K. X., Lee, Y. C., Soh, Z. T., Wu, C. S., Chang, Y. M., Li, W. H., & Lu, M. Y. J. Profiling maize embryonic leaf development and discovering new genes using high-resolution spatial long-read isoform sequencing. ***Nayzwd Pcary yx*** 2026, **12**, 1860–1878.

陳志冠 助理教授
Assistant Prof. Chih-Kuan Chen



國立東華大學 生化暨分子醫學科學系
Department of Biochemistry and Molecular Medicine,
National Dong Hwa University

Education
- 博士 臺灣大學生態學與演化生物學研究所
- 碩士 台灣大學植物科學所
- 學士 嘉義大學生物資源系

Experience
- 助理教授 國立東華大學生化暨分子醫學科學系
- 博士後研究 Department of Pathology, University of Southern California, CA
- 博士後研究 Biodiversity Research Center, Academia Sinica, Taiwan
- 研究助理 Academia Sinica NGS core facility

Honors and Awards
- 2026 Outstanding research poster award, 9th Annual Skin Symposium, USA.
- 2018-2019 Postdoctoral Research Abroad Program (PRAP), (MOST 107-2917-I-564-024), Taiwan.
- 2017-2018 Postdoctoral Scholars (Regular). Academia Sinica, Taiwan.
- 2017 Invited speaker. 89th Annual Meeting of the Genetics Society of Japan, Japan.
- 2017 Wellcome Trust Travel Awards. 18th International Congress of Developmental Biology, Singapore.
- 2017 Best Papers Prize of 88th Annual Meeting of Genetics Society of Japan, Japan.
- 2016 Exchange student travel award, Genetic Society of Japan (GSJ), Japan.
- 2016 The best Ph.D. student poster award, 2016 NTU & NTNU Joint Symposium on Ecology and Evolutionary Biology, Taiwan.

Research Interests
Single-cell omics analyses, Evolutionary biology, Avian skin biology

Selected Publications
1. **Lin, G. W., C. K. Chen (co-first)**, T. X. Jiang, Y. C. Liang, P. C. Tang, P. Wu, R. B. Widelitz, C. F. Chen and C. M. Chuong (2025). "Molecular insights into region-specific sexual dichromatism: Comparative transcriptome analysis of red cheek pigmentation in zebra finches." PLoS Genet **21**(5): e1011693. (SCI IF: 4.5; GENETICS & HEREDITY; Ranking: 36/171 =21.05%)
2. **Chen, C. K.**, Y. M. Chang, T. X. Jiang, Z. Yue, T. Y. Liu, J. Lu, Z. Yu, J. J. Lin, T. D. Vu, T. Y. Huang, H. I. Harn, C. S. Ng, P. Wu, C. M. Chuong and W. H. Li (2024). "Conserved regulatory switches for the transition from natal down to juvenile feather in birds." Nat Commun **15**(1): 4174. (SCI IF: 14.7; Discipline: MULTIDISCIPLINARY SCIENCES; Ranking: 8/134=5.9%)
3. **Chen, C. K.**, W. T. Juan, Y. C. Liang, P. Wu and C. M. Chuong (2021). "Making region-specific integumentary organs in birds: evolution and modifications." Curr Opin Genet Dev **69**: 103-111. (SCI IF: 4.665; Discipline: GENETICS & HEREDITY; Ranking: 48/175=27.4%)

4. **Chen, C. K.**, H. F. Chuang, S. M. Wu and W. H. Li (2019). "Feather Evolution from Precocial to Altricial Birds." Zool Stud **58**: e24. (SCI IF: 1.042; Discipline: ZOOLOGY; Ranking: 128/170=75.2%)
5. **Chen, C. K.**, C. P. Yu, S. C. Li, S. M. Wu, M. J. Lu, Y. H. Chen, D. R. Chen, C. S. Ng, C. T. Ting and W. H. Li (2017). "Identification and evolutionary analysis of long non-coding RNAs in zebra finch." BMC Genomics **18**(1): 117. (SCI IF: 4.142; Discipline: GENETICS; Ranking: 58/174=33%)
6. **Chen, C. K.**, C. S. Ng, S. M. Wu, J. J. Chen, P. L. Cheng, P. Wu, M. J. Lu, D. R. Chen, C. M. Chuong, H. C. Cheng, C. T. Ting and W. H. Li (2016). "Regulatory Differences in Natal Down Development between Altricial Zebra Finch and Precocial Chicken." Mol Biol Evol. (SCI IF: 12.1; Discipline: EVOLUTIONARY BIOLOGY; Ranking: 2/50=4%)

Diversification of Feather Transitions and Their Underlying Molecular Control

Chih-Kuan Chen[1,2,3], Gee-Way Lin[2,3,4], Wen-Hsiung Li[3,5*], Cheng-Ming Chuong[2,3*]
[1] *Department of Biochemistry and Molecular Medicine, National Dong Hwa University, Taiwan.*
[2] *Department of Pathology, Keck School of Medicine, University of Southern California, CA, USA.*
[3] *The iEGG and Animal Biotechnology Center, National Chung Hsing University, Taiwan.*
[4] *Department of Biochemistry and Molecular Cell Biology, Taipei Medical University, Taiwan.*
[5] *Biodiversity Research Center, Academia Sinica, Taiwan.*

Feathers are key avian innovations that support thermoregulation, flight, communication, and sexual signaling. Understanding how feather morphology and coloration arise during development is therefore central to understanding avian adaptation and evolution. Here, we investigate two major developmental transitions in birds. First, we examined the primary transition from natal down feathers, specialized for heat conservation, to juvenile feathers, which enable individuals to live independently. Time-ordered gene co-expression network analysis, epigenetic profiling, and functional perturbation identified four major molecular regulations. Extracellular matrix reorganization promotes peripheral pulp formation, which is required for feather branching. α-SMA (*ACTA2*) compartmentalizes dermal papilla stem cells involved in feather renewal cycles. LEF1 functions as a key Wnt-signaling hub that drives rachis formation, converting radial downy feathers into bilaterally symmetrical juvenile feathers. In addition, the recruitment of scale keratins strengthens the feather sheath, with SOX14 acting as an important regulator. These mechanisms are conserved between chickens and zebra finches, suggesting an ancient developmental program that may have originated in feathered dinosaurs. Second, we investigated the secondary transition that generates region-specific coloration and sexual dichromatism in zebra finches. The proportions of eumelanin and pheomelanin determine cheek coloration. Comparison of the sexually dimorphic cheek region and the monomorphic scalp region identified transcription factors associated with localized pigmentation. Several Z-chromosome genes were linked to male red cheek coloration, supporting a cell-autonomous genetic control of the sexual dimorphism. Comparative motif analyses further revealed that conserved craniofacial domains can give rise to convergently evolved color diversification across passerines. Together, these findings reveal how conserved developmental programs establish feather architecture, while regulatory innovations generate the diversity of avian pigmentation and sexual dimorphism.

References:

1. Chen, C. K.; Chang, Y. M.; Jiang, T. X.; Yue, Z.; Liu, T. Y.; Lu, J.; Yu, Z.; Lin, J. J.; Vu, T. D.; Huang, T. Y.; et al. Conserved regulatory switches for the transition from natal down to juvenile feather in birds. *Nat Commun* **2024**, *15* (1), 4174. DOI: 10.1038/s41467-024-48303-3 From NLM Medline.
2. Lin, G. W.; Chen, C. K.; Jiang, T. X.; Liang, Y. C.; Tang, P. C.; Wu, P.; Widelitz, R. B.; Chen, C. F.; Chuong, C. M. Molecular insights into region-specific sexual dichromatism: Comparative transcriptome analysis of red cheek pigmentation in zebra finches. *PLoS Genet* **2025**, *21* (5), e1011693. DOI: 10.1371/journal.pgen.1011693 From NLM Medline.



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國立臺灣大學 生態學與演化生物學研究所
Institute of Ecology and Evolutionary Biology,
National Taiwan University

Education

- 美國杜克大學博士
- 國立臺灣師範大學學士

Experience

- 奧地利孟德爾分子植物生物學研究所博士後
- 國立臺灣大學博士後

Honors and Awards

- 2025 國科會傑出研究獎
- 2025 正瀚生技公司青年學者獎
- 2024 中央研究院年輕學者研究成果獎
- 2024 臺灣植物學會林秋榮教授植物科學創新研究獎
- 2024 臺灣大學學術勵進青年講座
- 2023 國科會吳大猷先生紀念獎
- 2022 傑出人才發展基金會年輕學者創新獎
- 2021 楊祥發院士傑出農業科學年輕學者獎
- 2020 EMBO Global Investigator Award
- 2018 臺灣植物學會新秀獎

Research Interests

基因體學、演化遺傳學

Selected Publications

1. C-M Yeh, C-C Chien, Y-T Cheng, P-M Yeh, H-W Chen, Y-W Huang, H-W Chen, Y-K Tu, C-R Lee*. 2026 June. The genome and transcriptome of Musa itinerans unveil novel candidate genes for fusaric acid resistance. *The Plant Journal* 126(6):e70991
2. C-C Chien, T Seiko, C Muto, H Ariga, Y-C Wang, C-H Chang, H Sakai, K Naito*, C-R Lee*. 2025 May. A single domestication origin of adzuki bean in Japan and the evolution of domestication genes. *Science* 388, eads2871
3. H-W Chen, C-C Chien, C-R Lee*. 2024 Aug. Distinct types of selection and genetic architecture shape molecular variation during the domestication of vegetable crops. *Plant Physiology* 195(4):2533-2541
4. Y-P Lin, H-W Chen, P-M Yeh, SS Anand, J Lin, J Li, T Noble, R Nair, R Schafleitner, M Samsonova, E Bishop-von-Wettberg, S Nuzhdin, CT Ting, RJ Lawn, C-R Lee*. 2023 Oct. Demographic history and distinct selection signatures of two domestication genes in mungbean. *Plant Physiology* 193(2):1197-1212
5. P-W Ong, Y-P Lin, H-W Chen, C-Y Lo, M Burlyaeva, T Noble, R Nair, R Schafleitner, M Vishnyakova, E Bishop-von-Wettberg, M Samsonova, S Nuzhdin, C-T Ting, C-R Lee*. 2023 June. Environment as a limiting factor of the historical global spread of mungbean. *eLife* accepted doi:10.7554/eLife.85725

6. J Li, C-R Lee*. 2023 July. The role of gene presence-absence variations on genetic incompatibility in Asian rice. *New Phytologist* 239(2):778-791
7. LN Carley, JP Mojica, B Wang, C-Y Chen, Y-P Lin, KVSK Prasad, E Chan, C-W Hsu, R Keith, CL Nunez, CF Olson-Manning, CA Rushworth, MR Wagner, J Wang, P-M Yeh, M Reichelt, K Ghattas, J Gershenzon, C-R Lee*, T Mitchell-Olds*. 2021 Aug. Ecological factors influence balancing selection on leaf chemical profiles of a wildflower. *Nature Ecology & Evolution* 5:1135-1144
8. Y-P Lin, T Mitchell-Olds, C-R Lee*. 2021 Apr. The ecological, genetic, and genomic architecture of local adaptation and population differentiation in Boechera stricta. *Proceedings of the Royal Society* B 288:20202472

The origin and spread of adzuki bean across Asia

Kai Chun Lee [1], Chih-Wei Lin [1], Chih-Cheng Chien [2], Cheng-Ruei Lee [1*]
[1] *Institute of Ecology and Evolutionary Biology, National Taiwan University, Taipei, Taiwan*
[2] *Research Center of Genetic Resources, National Agriculture and Food Research Organization, Tsukuba, Ibaraki, Japan*

Understanding the origin and expansion of crops helps us trace the sources of crop genetic variation to design better breeding programs. While population genetics studies and archaeology are commonly used methods to investigate crop evolutionary processes, their results do not always agree. In the Asian crop adzuki bean (*Vigna angularis*), while archaeological evidence points to the earliest traces of cultivation in Japan, previous genetic studies suggest the highest genetic variation in central China. Using wild and cultigen accessions across Asia, we showed that domestication happened in Japan, followed by later expansion into China. A large-scale wild-cultigen hybridization happened in China later, generating the higher genetic variation that previously confounded scientists. We mapped, validated, and dated several key mutations affecting domestication traits. Interestingly, these mutations seem to have occurred and increased in frequency long before the domestication events, suggesting early selection by the Jomon population, who were previously thought to be hunter-gatherer-fishers. After domestication, adzuki has recently spread to the high-latitude regions beyond the ancestral range, and we mapped several mutations affecting its successful flowering in these regions. While the spread to high-latitude regions appears independent in China and Japan, these spreads utilize independent combinations of the same set of old mutations that originated in Japan, demonstrating a complex process of parallel evolution.

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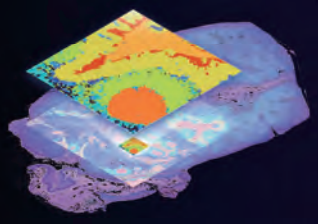
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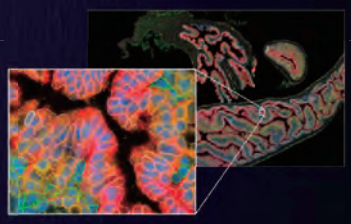


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- ✓ 結合一體化自動化流程與雲端分析

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Session 5：癌症微環境之多體學剖析：從細胞訊號到精準治療

TGMBS

台灣基因醫學暨生物標記學會

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莊智弘 教授

高雄醫學大學

醫學檢驗生物技術學系

Moderator

Prof. Chih-Hung Chuang

Department of Medical Laboratory Science and Biotechnology,
Kaohsiung Medical University



王慧菁 所長
Prof. Lily Hui-Ching Wang

國立清華大學 分子與細胞生物研究所
國立清華大學 醫學科學系
Director / Institute of Molecular and Cellular Biology,
National Tsing Hua University
Prof. / Department of Medical Science,
National Tsing Hua University

Education

- 1998-2004 國立成功大學，基礎醫學研究所/博士
- 1993-1997 國立成功大學，生物系/理學士

Experience

- 2026/08~國立清華大學 分子與細胞生物研究所 所長
- 2025/05~國立清華大學 藥品與醫材法規科學碩士在職學位學程 主任
- 2022/08~國立清華大學 學士後醫學系 教授
- 2021/08~國立清華大學 分子與細胞生物研究所 教授
- 2022~2025 國立清華大學 教學發展中心 學習與研究發展組 組長
- 2015~2021 國立清華大學分子與細胞生物研究所 副教授
- 2010~2015 國立清華大學分子與細胞生物研究所 助理教授
- 2009~2010 瑞士巴賽爾大學生物中心 計畫主持人
- 2006~2009 德國馬克斯普朗克生化研究所 博士後研究
- 2004~2006 國家衛生研究院臨床研究組 博士後研究

Honors and Awards

- 2026 未來科技獎 Future Tech Award
- 2026 財團法人肝病防治學術基金會 研究獎助
- 2026 國立清華大學生命科學暨醫學院教師傑出研究獎
- 2025 國家新創獎 National Innovation Award
- 2025 乳癌防治基金會優秀論文獎
- 2025 全國大專院校精準健康產業創新創業競賽-智慧健康組金獎
- 2025 耳目醫新精準健康 Demo Day 全國生醫創新創業競賽 金獎
- 2024 全國大專院校精準健康產業創新創業競賽-智慧健康組金獎
- 2024 未來科技獎 Future Tech Award
- 2023 國立成功大學醫學院 校友傑出成就獎
- 2023 財團法人肝病防治學術基金會 研究獎助
- 2022 Sigma Xi 科學研究學會榮譽會員
- 2020 國立清華大學 傑出教學獎
- 2019 財團法人肝病防治學術基金會研究獎助
- 2016 吳健雄學術基金會 台灣女科學家新秀獎
- 2008 Max-Planck Gesellschaft ein Forbildungsstipendium
- 2007 國科會菁英計畫專案留學獎學金

- 2006　財團法人成杏基金會優秀論文獎
- 2005　財團法人宋瑞樓教授學術基金會優秀論文獎

Research Interests

細胞生物學，腫瘤生物學，活細胞顯微攝影，新藥開發

Selected Publications

1. Pan MY, Lee TH, Peng TN, Wu EY, Chiu YT, Chiu CC, Nguyen HV, Pao YS, Chao MH, Lee RK, Wu HC, Wang LH*. Programmable Nanobody-Targeting Chimeras Enable Intracellular Viral Protein Degradation. ***Advanced Science*** (Weinh). 2026 Jul 17:e76687. doi: 10.1002/advs.76687.
2. Gurme ST, Su YT, Koparde B, Wang LH*, Lee GB*. An integrated microfluidic system for automated extraction of in vitro transcribed mRNAs via probe-coated magnetic beads. ***Lab Chip***. 2026 Jan 20;26(2):471-480. doi: 10.1039/d5lc00938c.
3. Wang HY, Shen SY, Wang LH*, Lee GB*. An integrated microfluidic system with shear force control for an automatically modified mRNA display technique for screening high-specificity peptide probes. ***Lab Chip***. 2026 Jan 6;26(1):143-153. doi: 10.1039/d5lc00914f.
4. Chiu YT, Huang YS, Pan MY, Chang CY, Chiu CC, Nguyen HV, Lee TH, Sun YJ, Li WH, Wang LH*. SARS-CoV-2 Evolution in Humans Enables Its Transmission to Nonhuman Primates. ***Mol Biol Evol***. 2025 Oct 29;42(11):msaf281. doi: 10.1093/molbev/msaf281.
5. Pao YS, Liao KJ, Shiau YC, Chao MH, Li MC, Lin LM, Chang HH, Yeh HW, Chen YJ, Chiu YT, Pan MY, Chang YH, Shen SY, Lin SY, Cheng HC, Lin YC, Sun YJ*, Kuo CC*, Hsieh HP*, **Wang LH***. KIF2C promotes paclitaxel resistance by depolymerizing polyglutamylated microtubules. ***Developmental Cell*. 2025** Aug 4;60(15):2097-2113.e8. doi: 10.1016/j.devcel.2025.03.004.
6. Yang JY, Wu YH, Pan MYC, Chiou YT, Lee RKL, Li TN, **Wang LH***. Chemical-induced degradation of PreS2 mutant surface antigen via the induction of microautophagy. ***Antiviral Research*** 2022 Nov; 207:10547. doi: 10.1016/j.antiviral.2022.105417
7. Wu YH, Yang Y, Chen CH, Hsiao CJ, Li TN, Liao KJ, Watashi K, Chen BS, **Wang LH***. Aerobic glycolysis supports hepatitis B virus biosynthesis through interaction between viral surface antigen and pyruvate kinase isoform M2. ***PLOS Pathogens***. 2021 Mar 15;17(3):e1008866. doi: 10.1371/journal.ppat.1008866. PMID: 33720996; PMCID: PMC8009439
8. Bhattacharjee MJ, Lin JJ, Chang CY, Chiou YT, Li TN, Tai CW, Shiu TF, Chen CA, Chou CY, Chakraborty P, Tseng YY, **Wang LH***, Li WH*. Identifying primate ACE2 variants that confer resistance to SARS-CoV-2. ***Molecular Biology and Evolution***. 2021, 25;38(7):2715-2731. doi: 10.1093/molbev/msab060.

Targeting KIF2C to Overcome Paclitaxel Resistance: How a Microtubule Depolymerase Reads the Tubulin Code

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Paclitaxel and other microtubule-targeting agents (MTAs) remain central to triple-negative breast cancer (TNBC) treatment, yet their efficacy is limited by acquired chemoresistance and cross-resistance. Through integrative transcriptomic analyses, we identified kinesin family member 2C (KIF2C/MCAK) as consistently overexpressed across TNBC subtypes and associated with poor prognosis. Paclitaxel-resistant TNBC cells exhibited elevated KIF2C expression and increased tubulin tyrosination and polyglutamylation. Mechanistically, polyglutamylation strengthened electrostatic interactions between tubulin C-terminal tails and KIF2C, enabling preferential depolymerization of paclitaxel-stabilized microtubules. To overcome this resistance mechanism, we developed 7S9, a selective, cell-permeable KIF2C inhibitor that stabilizes KIF2C–tubulin binding and blocks microtubule depolymerization. At sublethal concentrations, 7S9 restored paclitaxel-induced microtubule bundling and mitotic cell death, enhancing sensitivity to taxanes by over 90-fold and to ixabepilone, eribulin, and vinorelbine by at least 20-fold. In syngeneic 4T1 models, 7S9 combined with paclitaxel achieved synergistic tumor growth inhibition in chemoresistant tumors. Analysis of tissue microarrays from 482 patients further linked cytoplasmic KIF2C expression to advanced tumor stage and lymph node metastasis. Together, these findings reveal a KIF2C–tubulin modification axis driving MTA resistance and establish KIF2C inhibition as a promising strategy to overcome therapeutic cross-resistance in TNBC and potentially other female tumors.

References: Pao YS, Liao KJ, Shiau YC, Chao MH, Li MC, Lin LM, Chang HH, Yeh HW, Chen YJ, Chiu YT, Pan MY, Chang YH, Shen SY, Lin SY, Cheng HC, Lin YC, Sun YJ, Kuo CC, Hsieh HP, **Wang LH**. KIF2C promotes paclitaxel resistance by depolymerizing polyglutamylated microtubules. Developmental Cell. 2025 Aug 4;60(15):2097-2113.e8. doi: 10.1016/j.devcel.2025.03.004.

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Education

- 2005/06~ 2010/03 PhD Pharmaceutical Sciences The Ohio State University
- 1998/09~ 2002/06 BS. Pharmacy National Taiwan University

Experience

- 2018/02 ~ current: Associate Professor, NCKU
- 2015/02 ~ 2018/01: Assistant Professor, NCKU
- 2013/02 ~ 2015/01: Research Assistant Professor, NCKU
- 2012/10 ~ 2013/01: Postdoctoral Research Scientist, NCKU
- 2010/05 ~ 2012/09: Postdoctoral Fellow, German Cancer Research Center

Honors and Awards

- 2022 Awards for Teaching and Mentoring: Outstanding Award Grand Review and Competition for PhD Student and Postdoctoral Research
- 2020 Original Research Award, Hsu-Yuan Educational Foundation
- 2017 Young Investigator Award, Hsu-Yuan Educational Foundation
- 2011 Alexander von Humboldt Postdoctoral Research Fellowship

Research Interests

Cancer biology, the tumor microenvironment, epigenetics and RNA m^5C modifications

Selected Publications

1. Chang CJ, **Huang PH**, et al., Macrophage activation determines muscle wasting in pancreatic cancer. ***Oncogene***, 2025
2. Chen SY, Fang YH, et al., Transcriptome- wide RNA 5-methylcytosine profiles of human iPSCs and iPSC-derived cardiomyocytes. ***Scientific Data***, 2024
3. Ding LY, Chang CJ, et al., Stromal Rigidity Stress Accelerates Pancreatic Intraepithelial Neoplasia Progression and Chromosomal Instability via Nuclear Protein Tyrosine Kinase 2 Localization. ***Am J Pathol.***, 2024
4. Chen SY and **Huang PH**. Enrichment of mRNA and bisulfite-mRNA library preparation for next-generation sequencing. ***J Vis Exp.***, 2023
5. Chen SY, Chen KL, Ding LY, Yu CH, et al., RNA bisulfite sequencing reveals NSUN2- mediated suppression of epithelial differentiation in pancreatic cancer. ***Oncogene***, 2022
6. Ding LY et al., Epigenetic silencing of AATK in acinar to ductal metaplasia in murine model of pancreatic cancer. ***Clinical Epigenetics***, 2020

From tumor dedifferentiation to systemic wasting: Roles of NSUN2-mediated m^{5}C modification and macrophage-derived CHI3L1 in pancreatic cancer

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The 5-methylcytosine (m^{5}C) methyltransferase NOP2/Sun RNA Methyltransferase 2 (NSUN2) is upregulated in pancreatic cancer and is associated with tumor progression. Transcriptome-wide RNA bisulfite sequencing shows that NSUN2-mediated m^{5}C modification of mRNA suppresses epithelial differentiation and drives tumor growth. This intrinsic epitranscriptomic reprogramming not only sustains the aggressive nature of pancreatic cancer cells but also influences the surrounding desmoplastic TME, setting the stage for systemic disease complications. While the intrinsic epitranscriptomic reprogramming sustains primary tumor aggressiveness, overcoming PDAC lethality also requires addressing its extrinsic, systemic complications. As aggressive tumors interact with the host immune system, we used single-cell RNA sequencing in tumor-bearing models to analyze the tumor-stroma-derived systemic signaling that trigger skeletal muscle wasting. Specific macrophage populations in the TME are revealed as primary mediators of this cachexia. These tumor-associated macrophages secrete Chitinase 3-Like 1 (CHI3L1), which actively suppresses myotube formation by upregulating a deleterious signaling cascade, including HDAC3 and NF-κB, in skeletal muscle. Targeted blockade of the CHI3L1-HDAC3 axis mitigates muscle wasting and suppresses metastatic progression.

In summary, we elucidate two critical but distinct axes of pancreatic cancer pathobiology: NSUN2-mediated m^{5}C RNA modifications that drive an undifferentiated, aggressive tumor phenotype, and a pro-inflammatory macrophage CHI3L1–HDAC3 cascade that triggers severe skeletal muscle wasting. Combining therapeutic strategies that target both the intrinsic epitranscriptomic landscape and extrinsic macrophage-driven cachexia offers a comprehensive approach to improving outcomes in PDAC.

李佳陽 教授
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Education

國立清華大學生醫工程與環境科學系博士

Experience

- 2022/2~迄今 教授 高雄醫學大學醫學研究所
- 2024/8~迄今 副研發長 高雄醫學大學研究發展處
- 2025/12~迄今 執行長 高雄醫學大學疾病多體學研究中心

Honors and Awards

- 第 21 屆國家新創獎
- 2024FITI 創新創業激勵計畫創業潛力獎
- 114 財團法人肝病防治學術基金會研究獎助金

Research Interests

基因體學、代謝性疾病、腫瘤生物學、分子免疫學

Selected Publications

1. Lai YH, Liu WL, Lee TY, Kuo CW, Liu YR, Huang CY, Chen YH, Chen IL, Wu SZ, Wang SC, Lee PY, Liu CC, Lo J, Chang YC, Kuo HF, Hsieh CC, **Li CY***, Liu PL*. Magnolol regulates miR-200c-3p to inhibit epithelial–mesenchymal transition and retinoblastoma progression by modulating the ZEB1/E-cadherin axis in vitro and in vivo. *Phytomedicine*. 2023 Feb:110:154597.
2. Lo J, Wu HE, Liu CC, Chang KC, Lee PY, Liu PL, Huang SP, Wu PC, Lin TC, Lai YH, Chang YC, Chen YR, Lee SI, Huang YK, Wang SC*, **Li CY***. Nordalbergin exerts anti-neuroinflammatory effects by attenuating MAPK signaling pathway, NLRP3 inflammasome activation and ROS production in LPS-activated BV2 microglia. *International Journal of Molecular Sciences*. 2023 April 14;24(8):7300.
3. Wu HE, Su CC, Wang SC, Liu PL, Cheng WC, Yeh HC, Chuu CP, Chen JK, Bao BY, Lee CH, Ke CC, Chen YR, Yu YH, Huang SP*, **Li CY***. Anti-cancer effects of morusin in prostate cancer via inhibition of Akt/mTOR signaling pathway. *American Journal of Chinese Medicine*. 2023 Apr 29:1-21.
4. Chien HT#, **Li CY**#, Su WH, Chang KC, Chen CS, Liu YT, Chen CY, Dai CY, Wang SC. Multi-omics profiling of chemotactic characteristics of brain microglia and astrocytoma. *Life Sciences*. 2023 Jul 5;330:121855.
5. Yazal T, Lee PY, Chen PR, Chen IC, Liu PL, Chen YR, Tzu-Chieh Lin TC, Chen YT, Huang SP, Yeh HC, Liu CC, Lo J, Wu HE, S Wang SC*, **Li CY***. Kurarinone exerts anti-inflammatory effect via reducing ROS production, suppressing NLRP3 inflammasome, and protecting against LPS-induced sepsis. *Biomedicine & Pharmacotherapy*. 2023 Oct; 167:115619.
6. Ke CC, Jassi C, Chuang CH, Ko CY, Huang SP, Kuo SH, **Li CY***, Hsieh YJ*. Exosomes Derived from Irradiated-Prostate Cancer Cells Promote Cancer Progression. *Journal of Medical and Biological Engineering*. 2024 Aug 44;607–617.

7. Yazal T, **Li CY***. Serum Iron Overload Triggers the SMAD Pathway and Induces Hepcidin Expression in Hepatocytes through SMURF1. *Journal of Clinical and Translational Hepatology*. 2024 Sep 28;12(9):761-762.
8. Chen PR#, **Li CY**#, Yazal T, Chen IC, Liu PL, Chen YT, Liu CC, Lo J, Lin TC, Chang CT, Wu HE, Chen YR, Cheng WC, Chiu CC, Chen CS, Wang SC*. Protective effects of nordalbergin against LPS-induced endotoxemia through inhibiting MAPK/NF-κB signaling pathway, NLRP3 inflammasome activation, and ROS production. *Inflammation Research*. 2024 Oct;73(10):1657-1670.
9. Chen TE, Lo J, Huang SP, Chang KC, Liu PL, Wu HE, Chen YR, Chang YC, Liu CC, Lee PY, Lai YH, Wu PC, Wang SC*, **Li CY***. Glaucine inhibits hypoxia-induced angiogenesis and attenuates LPS-induced inflammation in human retinal pigment epithelial ARPE-19 cells. *European Journal of Pharmacology*. 2024 Oct 15;981:176883.
10. Chen PR, Su TS, Chang KC, Chiu CC, Huang SP, Yeh HC, Wang SC*, **Li CY***. Anti-cancer effects of Psoralea corylifolia: Insights into cancer hallmarks. *Phytochemistry Reviews*. 2025 June 13. (Accepted)

Unraveling the Molecular Mechanisms of Tirzepatide in MASLD: A Multi-Omics Perspective

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[2] *Center for Disease Multi-Omics Research, Kaohsiung Medical University*

Metabolic dysfunction-associated steatotic liver disease (MASLD) is a progressive metabolic disorder characterized by hepatic steatosis, metabolic dysregulation, chronic inflammation, and fibrotic remodeling. Despite recent advances in therapeutic development, the molecular mechanisms underlying disease progression and treatment response remain incompletely understood. Tirzepatide, a dual glucose-dependent insulinotropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1) receptor agonist, has demonstrated promising therapeutic effects on metabolic dysfunction and metabolic dysfunction-associated steatohepatitis (MASH). However, the molecular pathways mediating its hepatic benefits and the biological determinants of therapeutic responsiveness require further investigation.

In this study, we aim to investigate the molecular signatures associated with tirzepatide treatment and identify key biological pathways involved in hepatic metabolic regulation, inflammatory responses, mitochondrial function, and fibrotic remodeling. Cross-species integration of human and experimental datasets may further help distinguish conserved disease-associated molecular signatures from treatment-responsive pathways. A comprehensive understanding of tirzepatide-associated molecular alterations may facilitate the identification of potential therapeutic targets and biomarkers for monitoring disease progression and treatment response.

Session 6: Exosomes in Human Diseases

Taiwan Society for Extracellular Vesicles

TSEV

台灣胞外體學會

主持人

沈湯龍 教授

國立臺灣大學

植物病理與微生物學系

Moderator

Prof. Tang-Long Shen

Department of Plant Pathology and Microbiology,
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國立成功大學 基礎醫學研究所
Institute of Basic Medical Sciences,
National Cheng Kung University

Education

- Chung Shan Medical University, Taiwan (B.S.)
- National Cheng Kung University, Taiwan (M.S.)
- University of Colorado Denver, Anschutz Medical Campus, USA (Ph.D.)

Experience

- 2012-2014 Postdoctoral Fellow
  Department of Pharmacology, University of Colorado Denver, Anschutz Medical Campus, USA
- 2014-2017 Postdoctoral Fellow
  Department of Basic Medical Science, College of Medicine, National Cheng Kung University, Taiwan
- 2017-2022 Assistant Investigator
  Institute of Molecular Medicine, National Cheng Kung University, Taiwan
- 2022-present Assistant professor
  The Institute of Basic Medical Science National Cheng Kung University, Taiwan

Honors and Awards

- 2025-Future Tech Award (未來科技獎). Multi-Omics-Guided Precision Oncology for Pancreatic Cancer (Team Member).
- 2024-Investigator Award of Hsu-Yuan Education Foundation (財團法人徐燕教育基金會 優秀年輕學者獎).
- 2022-Outstanding Alumni of the Department of Biomedicine, Chung Shan Medical University, Taiwan
- 2021-Outstanding Paper Award of the 31st Wang Ming-Ning Memorial Foundation Award (第 31 屆王民寧獎國內醫藥相關研究所暨學術研究機構博士後研究員優秀論文獎)
- 2013-AACR-Aflac, Incorporated Scholar-in-Training Awards, 2013 AACR Advances in Breast Cancer Research

Research Interests

1. Mechanistic studies of the tumor microenvironment in pancreatic cancer and urothelial carcinoma
2. Functional roles and regulatory mechanisms of extracellular vesicles in cancer progression and intercellular communication

Selected Publications

1. Hau-Chern Jan, Ying-Chieh Su, Steven Kuan-Hua Huang, Chu-An Wang, Ju-Ming Wang, Chien-Feng Li. HDAC4 overexpression predicts aggressive phenotypes and poor outcomes in urothelial carcinoma. *Cancer Treat Res Commun.* 2026 Feb 22:47:101148.
2. Chu-An Wang†, Ya-Chin Hou†, Yi-Kai Hong, Yu-Jing Tai, Chieh Shen, Pei-Chi Hou, Jhao-Lin Fu, Cheng-Lin Wu, Siao Muk Cheng, Daw-Yang Hwang, Yung-Yeh Su, Yan-Shen Shan*, Shaw-Jenq Tsai*. Intercellular TIMP-1-CD63 signaling directs the evolution of immune escape and metastasis in KRAS mutated pancreatic cancer cells. *Mol Cancer* 2025 Jan 18;24(1):25.
3. Yu-Ying Chao, Ruei-Ci Lin, Ping Jui Su, Chu-An Wang, Ting-Yuan Tu, Ya-Chin Hou, Yi-Tzui Tsai, I-Chen Peng, Shaw-Jenq Tsai, Yan-Shen Shan, Chia-Yih Wang. Melanophilin-induced primary cilia promote pancreatic cancer metastasis. *Cell Death & Disease.* 2025 Jan 16;16(1):22.
4. Lian-Chung Yu*, Chu-An Wang*, Che-Yuan Hu, Kun-Che Lin, Chien-Hui Ou, Hau-Chern Jan. Preoperative systemic inflammation response index enhances the prognostic value of tumor multifocality in upper tract urothelial carcinoma. *Oncol Lett.* 2024 Jul 15;28(3):436.
5. Chu-An Wang, Chien-Feng Li, Rho-Chi Huang, Yo-Hua Li, Jing-Ping Liou, Shaw-Jenq Tsai. Suppression of Extracellular Vesicle VEGF-C-mediated Lymphangiogenesis and Pancreatic Cancer Early Dissemination By a Selective HDAC1/2 Inhibitor. *Mol Cancer Ther.* 2021 Sep;20(9):1550-1560.
6. Chu-An Wang and Shaw-Jenq Tsai. Regulation of lymphangiogenesis by extracellular vesicles in cancer metastasis. *Exp Biol Med (Maywood).* 2021 Oct;246(19):2048-2056.
7. Wan-Ning Li, Kuei-Yang Hsiao, Chu-An Wang, Ning Chang, Pei-Ling Hsu, Chung-Hsien Sun, Shang-Rung Wu, Meng-Hsing Wu, Shaw-Jenq Tsai. Extracellular vesicle-associated VEGF-C promotes lymphangiogenesis and immune cells infiltration in endometriosis. *PNAS* October 13, 2020 117 (41) 25859-25868.
8. Chu-An Wang, Yi-Hern Chang, Pei-Chi Hou, Yu-Jing Tai, Wan-Ning Li, Pei-Ling Hsu, Shang-Rung Wu, Wen-Tai Chiu, Chien-Feng Li, Yan-Shen Shan, Shaw-Jenq Tsai. DUSP2 regulates extracellular vesicle-VEGF-C secretion and pancreatic cancer early dissemination. *J Extracell Vesicles.* 2020 Apr 4;9(1):1746529.

Macrophage-Tumor Crosstalk Remodels Extracellular Vesicles to Drive Pancreatic Cancer Progression

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[2] *Department of Physiology, College of Medicine, National Cheng Kung University, Tainan, Taiwan*

Pancreatic ductal adenocarcinoma (PDAC) develops within a complex microenvironment in which macrophage–tumor interactions promote disease progression. We previously found that sustained ERK activation, driven in part by loss of the phosphatase DUSP2, enhances extracellular vesicle (EV)-mediated release of VEGF-C, thereby promoting lymphangiogenesis and lymphovascular dissemination. We further identified macrophages as a key source of signals that reinforce ERK activity in PDAC cells through the tetraspanin CD63. This interaction increases CD63 expression and EV production, whereas CD63 knockdown suppresses macrophage-induced EV release and alters EV cargo composition. In vivo, PDAC-derived EVs target fibroblasts, and CD63 depletion reduces tumor desmoplasia. Together, these findings identify macrophage–tumor crosstalk as a feed-forward mechanism that sustains ERK signaling and reshapes tumor-derived EV output. This interaction promotes PDAC progression through at least two complementary routes: enhancing VEGF-C-associated EV signaling to support lymphangiogenesis and metastatic dissemination, and increasing CD63-dependent EV production and cargo remodeling to regulate fibroblast activity and desmoplasia.

References:

1. Chu-An Wang†, Ya-Chin Hou†, Yi-Kai Hong, Yu-Jing Tai, Chieh Shen, Pei-Chi Hou, Jhao-Lin Fu, Cheng-Lin Wu, Siao Muk Cheng, Daw-Yang Hwang, Yung-Yeh Su, Yan-Shen Shan*, Shaw-Jenq Tsai*. Intercellular TIMP-1-CD63 signaling directs the evolution of immune escape and metastasis in KRAS mutated pancreatic cancer cells. *Mol Cancer* 2025 Jan 18;24(1):25.

謝明佑 教授
Prof. Ming-You Shie



中國醫藥大學 生物醫學工程學系
Department of Biomedical Engineering,
China Medical University

Education

國立成功大學 生物醫學工程系 博士

Experience

- 中國醫藥大學附設醫院 異種移植轉譯研發中心 研發副主任
- 中國醫藥大學附設醫院 外泌體及多維生醫轉譯開發中心 研究員
- 中國醫藥大學 生物醫學工程系 副教授

Honors and Awards

- 2026 生醫材料暨藥物釋放學會-傑出年輕學者獎
- 2025 未來科技獎
- 2024 年國家新創獎
- 2023 中國民國牙醫師公會全國聯合會-學術研究特優
- World's Top 2% Scientists, 2020-2025

Research Interests

再生醫學
生物製造
工程化外泌體開發

Selected Publications

1. Kuo TY, Chen CY, Lin YH, Cho DY, Lin TL, **Shie MY***. Lithium-doped calcium silicate scaffolds induce miR-130a-rich M2 macrophage EVs accelerating diabetic bone healing through angiogenesis-osteogenic coupling and mitochondrial repair. *Chemical Engineering Journal* 2026;543:178698.
2. **Shie MY***, Chiu CD, Chen Y, Lin YH, Yu MH, Chiu YP, Chiu PF, Chen CY, Chen YW, Chen MC, Cho DY*. Engineering integrin αvβ8-targeted extracellular vesicles to deliver BDNF mRNA for motor recovery in spinal cord injury. *Journal of Nanobiotechnology* 2026;24:305.
3. **Shie MY***, Chen MC, Chen Y, Huang SW, Lin YH, Yu MH, Pan CM, Chen CY, Chen YW, Hsieh CL, Chiu SC, Kao ST, Lin JG, Cho DY*. Engineered extracellular vesicles-mediated curcumin delivery in brain microenvironment modulating lysosomes, mitochondria, and microglia reprogram for Parkinson's disease therapy. *Journal of Nanobiotechnology* 2026;24:46.
4. **Shie MY***, Huang SW, Chen Y, Chen MC, Pan CM, Chen CY, Lin YH, Yu MH, Kan KW, Chiu SC, Ho HC, Chen YW, Cho DY. Engineering HLA-G-targeted extracellular vesicles nanoplatform for enhanced cancer therapy through precise cancer drug delivery. *Nature Communications* 2025;16:11308.
5. Yu MH, Lin YH, Liu EW, Hsu-Jiang TY, Cho DY, Lee JJ, **Shie MY***. Mitochondrial biogenesis modulation by silicon-stimulated mesenchymal stem cells-derived extracellular vesicles drives angio- and lymphangiogenesis in chronic wound healing. *Materials Today Bio* 2025;35:102629.

Therapeutic Applications of Engineered Extracellular Vesicles in Brain Disorders

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[2] *Xenotransplantation Translational Research Center, China Medical University Hospital, Taichung 404327, Taiwan*

Parkinson's disease (PD), major depressive disorder (MDD), and Alzheimer's disease (AD) share several interconnected pathological features, including neuroinflammation, impaired neurotrophic support, oxidative stress, and, particularly in neurodegenerative disorders, pathological protein accumulation. This study established a modular engineered extracellular vesicle (EV) platform incorporating interchangeable surface-targeting ligands and therapeutic cargos. The EVs were loaded with messenger RNA encoding brain-derived neurotrophic factor (BDNF) and evaluated for therapeutic applications across these three brain disorders. For PD, dopamine transporter-targeted EVs enhanced cellular uptake by SH-SY5Y cells, reduced both Ser129 phosphorylation and total accumulation of α-synuclein, and restored tyrosine hydroxylase and DJ-1 expression, as well as neuronal morphology. For MDD, integrin-targeted EVs ameliorated stress-induced depressive-like behaviors, neurochemical imbalance, neuroinflammation, and hippocampal pathological changes. Evaluation of AD-related pathological features further demonstrated that integrin-targeted EVs reduced amyloid-β and phosphorylated Tau accumulation, attenuated oxidative stress and ferroptosis, and promoted synaptic preservation and neurovascular repair. Collectively, these findings indicate that engineered EVs can be tailored through disease-relevant targeting ligands and therapeutic cargo delivery to coordinately regulate neurotrophic signaling, proteostasis, and neuroimmune responses. This modular platform therefore holds promise as a precision therapeutic strategy for multiple brain disorders.

廖秀蓉 博士
Dr. Hsiu-Jung Liao



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National Yang Ming Chiao Tung University

Education
國立台灣大學免疫學所博士

Experience
- 東京大學，免疫學所，博士後研究員
- 中央研究院，基因體中心，博士後研究員

Honors and Awards
- 2026，未來科技獎
- 2025，國家新創獎
- 2025, AI 大健康醫療應用創新大賽 最佳潛力獎
- 2024，未來科技獎
- 2015，獲得國科會千里馬計畫補助

Research Interests
免疫學，蝕骨細胞，自體免疫疾病，骨關節疾病，細胞治療

Selected Publications
1. **Hsiu-Jung Liao***, Tzu-Chi Liao, Yu-Feng Wu. (2026, Feb) Integrating Regenerative Therapies and Artificial Intelligence for Precision Management of Diabetic Foot Ulcers. *Advances in Wound Care* **(IF: 5.6)**
2. Tzu-Yi Huang*, Niann-Tzyy Dai*, **Hsiu-Jung Liao***, Ly Hien Doan, Tai-Shan Cheng, Wen-Yu Hsieh, I-Hsuan Lin, Yu-Tang Huang, Chia-Ching Liaw, Hsiao-Sheng Liu, Wei-Ming Cheng, Chun-Li Su, Ching-Wei Yang, Jin-Mei Lai, Mei Nan Tuan, Hui-Kang Liu, Chi-Ying F. Huang. (2026, Jan) Dibifree, a dietary phytomix, improves glycemic control and adiposity via modulation of the gut–pancreas–adipose–immune axis in type 2 diabetes. *Food Research International*. Volume 223, Part 1, January 2026, 117820 **(IF: 8)**
3. **Hsiu-Jung Liao**, Lin-Ping Cheng, Yu-Ching Hsieh, Chien-Sheng Wu, Hung-Chih Yang, I-Tsu Chyuan, Ping-Ning Hsu. (2025, Jul) Interleukin 27 (IL-27) productions in the liver microenvironment confer an immunosuppressive role and enhance hepatitis B viral persistence. *Journal of Microbiology, Immunology and Infection*. Jul 16:S1684-1182(25)00136-7 **(IF: 3.7)**
4. I-Tsu Chyuan, **Hsiu-Jung Liao**, Tse-Hua Tan, Huai-Chia Chuang, Yu-Chuan Chu, Meng-Hsun Pan, Chien-Sheng Wu, Ching-Liang Chu, Bor-Ching Sheu and Ping-Ning Hsu. (2024, Mar) Association of TRAIL receptor with phosphatase SHP-1 enables repressing T cell receptor signaling and T cell activation through inactivating Lck. *Journal of Biomedical Science*. 31:33 (**IF: 10.9**)
5. Sin-Yu Chen, Yi-Ling Chen, Po-Chen Li, Tai-Shan Cheng, Yeh-Shiu Chu, Yi-Shan Shen, Hsin-Tung Chen, Wei-Ni Tsai, Chien-Ling Huang, Martin Sieber, Yuan-Chieh Yeh, Hsiao-Sheng Liu, Chi-Ling Chiang, Chih-Hung Chang, Andrew S. Lee, Ly James Lee*, **Hsiu-Jung Liao***, Hon-Kan Yip* and Chi-

Ying F. Huang*. (2024, Mar) Engineered extracellular vesicles carrying let-7a-5p for alleviating inflammation in acute lung injury. *Journal of Biomedical Science*. 31:30 (**IF: 10.9**)
6. **Hsiu-Jung Liao**, Hwei-Fang Tsai, Chien-Sheng Wu, I-Tsu Chyuan, Ping-Ning Hsu (2019, Jan). TRAIL inhibits RANK signaling and suppresses osteoclast activation via inhibiting lipid raft assembly and TRAF6 recruitment. *Cell Death and Disease*, 10(2):77. **(IF: 9.696)**
7. I-Tsu Chyuan, Hwei-Fang Tsai, **Hsiu-Jung Liao**, Chien-Sheng Wu, Ping-Ning Hsu (2018, Sep). An apoptosis-independent role of TRAIL in suppressing joint inflammation and inhibiting T-cell activation in inflammatory arthritis. *Cellular Molecular Immunology*, 15(9):846-857. **(IF: 22.1)**
8. **Hsiu-Jung Liao**, I-Tsu Chyuan, Chien-Sheng Wu, Shu-Wha Lin, Kun-Hug Chen, Hwei-Fang Tsai, Ping-Ning Hsu. (2015, Jan). Increased neutrophil infiltration, IL-1 production and a SAPHO syndrome-like phenotype in PSTPIP2-deficient mice. *Rheumatology (Oxford)*, 54(7), 1317-1326. **(IF: 7.580)**

AI-Engineered Extracellular Vesicles: A New Frontier for Disease-Modifying Osteoarthritis Therapy Extracellular Vesicles

Hsiu-Jung Liao
Far Eastern Memorial Hospital, National Yang Ming Chiao Tung University

Osteoarthritis (OA) is a prevalent and multifactorial degenerative joint disease characterized by progressive cartilage destruction, chronic inflammation, immune dysregulation, functional impairment, and persistent pain. Current treatments primarily alleviate symptoms but do not effectively modify the underlying disease process, underscoring the urgent need for innovative disease-modifying therapies. Mesenchymal stem cell–derived extracellular vesicles (MSC-EVs) represent a promising cell-free therapeutic platform; however, their clinical translation remains limited by biological heterogeneity, variable molecular cargo, and insufficient control over therapeutic potency. In this study, we developed an artificial intelligence (AI)–guided peptide-priming strategy using CatiPep®, an AI-engineered bioactive molecule designed to reprogram MSC functional states and enhance the therapeutic properties of their secreted EVs. CatiPep® was identified through an integrated discovery platform combining bioinformatics, systems biology, and predictive modeling. MSCs were primed with CatiPep® under GMP-compatible culture conditions, followed by scalable EV isolation and comprehensive characterization of physicochemical properties, miRNA profiles, and protein cargo. Functional activity was evaluated using in vitro assays of chondrogenesis and inflammatory regulation, together with preclinical OA models assessing cartilage regeneration, pain-related behavior, and joint function. CatiPep® priming substantially altered MSC-EV molecular cargo, enriching anti-inflammatory and chondrogenesis-associated signals while suppressing inflammatory and cartilage-catabolic pathways. Compared with unprimed EVs, CatiPep®-primed EVs enhanced SOX9 and type II collagen expression, reduced inflammatory mediator production, and promoted anabolic chondrocyte activity. In OA animal models, these engineered EVs improved cartilage integrity, locomotor performance, joint function, and pain-related outcomes, while demonstrating a favorable safety profile and consistent production quality. Collectively, these findings establish CatiPep® as an AI-enabled molecular switch capable of transforming MSC-EVs into multifunctional, disease-modifying biologics. By integrating AI-guided molecular discovery, precision MSC priming, and extracellular vesicle engineering, this platform provides a scalable, standardized, and clinically translatable strategy for OA and potentially other aging-associated degenerative diseases.

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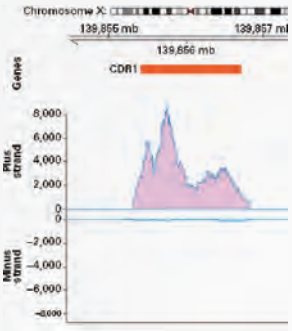
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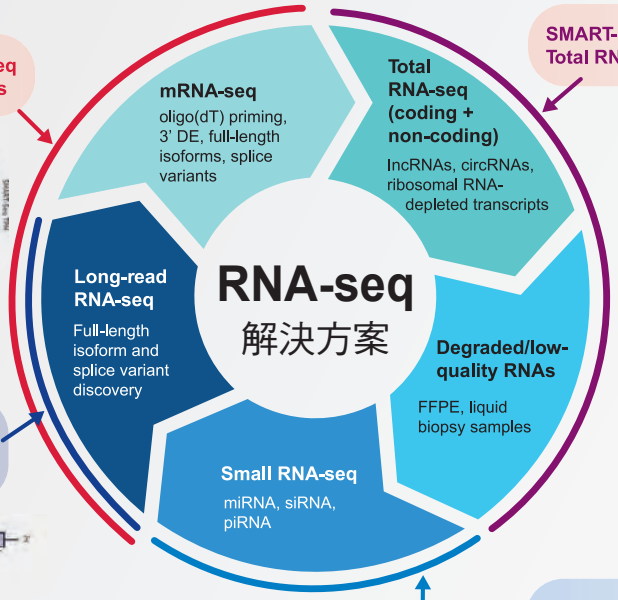
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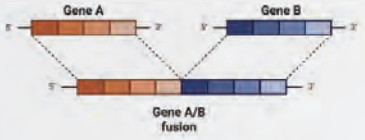


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Session 7：Systems Biology

TSECB

台灣演化與計算生物學會

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國立中興大學 生命科學系
Assistant Prof. / Department of Life Sciences,
National Chung Hsing University



Education

SEPTEMBER 2010 - JUNE 2017 PhD, Genome and Systems Biology, National Taiwan University, Taiwan

Experience

- **August 2024 – present**
  Assistant Professor and YuShan Young Fellow, National Chung Hsing University, Taichung, Taiwan
- **January 2024 – July 2024**
  Senior Research Scientist, Department of Platform Technology, MiNA Therapeutics, London, UK
- **July 2022 – December 2023**
  Research Scientist, Department of Platform Technology, MiNA Therapeutics, London, UK
- **August 2017 – June 2022**
  Postdoc, Laboratory of Telomeric Chromatin in Health and Disease – Joachim Lingner group, EPFL, Switzerland

Honors and Awards

- **YuShan Young Fellow program**, MOE, Taiwan, 2025-2028
- **EMBO Postdoctoral Fellowship**, EMBO, Germany, 2019-2021
- **Outstanding Publication Award**, National Taiwan University, Taiwan, 2016
- **Travel Fellowship**, Institute of Molecular Biology, Academia Sinica, Taiwan, 2016
- **Travel Fellowship**, Ministry of Education, Taiwan, 2016
- **1st Place Award, Research Poster Competition**, Ciliate Molecular Biology Conference, Italy, 2015

Research Interests

Telomere replication, small-activating RNA, ageing and ageing-related diseases, multi-omics, molecular medicine

Selected Publications

1. CYG Lin*, AC Näger*, T Lunardi, A Vančevska, G Lossaint, J Lingner. The Human Telomeric Proteome During Telomere Replication. *Nucleic Acids Res.* 2021 Nov 8:gkab1015. doi: 10.1093/nar/gkab1015. Epub ahead of print.
2. CYG Lin*, JL Chao, HK Tsai, D Chalker, MC Yao. Setting boundaries for genome-wide heterochromatic DNA deletions through flanking inverted repeats in *Tetrahymena thermophila*, *Nucleic Acids Research*, 2019; 47(10), 5181–5192, doi:10.1093/nar/gkz209

Nucleolar Proteins Coordinate Telomere Maintenance in Telomerase-Positive and ALT Cancer Cells

Lily Yong-Sin Huang[1], Li-Mei Chung[1], Yu-Tung Chou[1], Chih-Yi Gabriela Lin[1,2*]
[1]*Department of Life Sciences, College of Life Sciences, National Chung Hsing University, Taiwan*
[2]*Doctoral Program in Microbial Genomics, National Chung Hsing University and Academia Sinica, Taiwan*

Telomeres protect chromosome ends from being recognized as DNA damage through the shelterin complex, which consists of TRF1, TRF2, RAP1, TIN2, TPP1, and POT1. Efficient telomere replication requires coordinated interactions between shelterin and the DNA replication machinery; however, the regulatory factors that bridge these processes remain poorly understood. Using proteomic approaches, we previously identified several candidate regulators of telomere replication. Here, we investigated the roles of two nucleolar proteins, CCDC86 and REXO4, in telomere maintenance using telomerase-positive HeLa cells and ALT (Alternative Lengthening of Telomeres)-positive, telomerase-negative U2OS cells. REXO4 depletion increased telomeric DNA content in HeLa cells but reduced telomeric DNA content in U2OS cells, suggesting distinct roles depending on the telomere maintenance mechanism. Moreover, REXO4 knockdown elevated TERRA (telomeric repeat-containing RNA) expression, consistent with its recently reported role in regulating R-loops. In contrast, CCDC86 depletion had only modest effects on telomeric DNA content and TERRA expression. Transcriptomic analysis revealed two distinct regulatory pathways: CCDC86 depletion specifically reduced TRF1 and its binding partner TIN2 expression while largely preserving TRF2 levels, whereas REXO4 depletion preferentially downregulated TRF2 with minimal effects on TRF1. Supporting a functional connection between CCDC86 and the TRF1 pathway, siCCDC86 and siTRF1 induced highly overlapping transcriptional responses in U2OS cells, sharing 120 significantly enriched pathways, including suppression of immune and metabolic programs and activation of DNA replication-associated gene sets. Together, our findings reveal two distinct nucleolar regulatory axes that converge on shelterin components and differentially modulate telomere maintenance in telomerase-positive and ALT cancer cells.

林暐翔 助研究員
Dr. Wei-Hsiang Lin



中央研究院 分子生物研究所
Assistant Research Fellow / Institute of Molecular Biology, Academia Sinica

Education

- **New York University** 2008 Sep – 2015 May PhD in Biology
  Advisor: Dr. Edo Kussell
  Thesis title: Stochastic switching as an adaptive strategy in fluctuating environments: From single-cell dynamics to population evolution in microorganisms
- **National Taiwan University** 2002 Sep – 2007 Jun Double major in: B.S. in Life Science, B.S. in Physics

Experience

- **Academia Sinica**
  2023 Sep – present Assistant Research Fellow (Principal Investigator)
- **Yale University/Stanford University, Christine Jacobs-Wagner's lab**
  2016 Jun – 2023 May Postdoctoral associate
- **New York University, Edo Kussell's lab**
  2015 Jun – 2016 Mar Postdoctoral associate
- **Academia Sinica, Cheng-Ting Chien's lab**
  2007 Jun – 2008 Aug Research Assistant

Research Interests

Microfluidic culture, microbiology, single-cell microscopy.
Mathematical biology, biological physics.

Selected Publications

1. Genome concentration limits cell growth and modulates proteome composition in *Escherichia coli*. (2024) Jarno Mäkelä, Alexandros Papagiannakis, Wei-Hsiang Lin, Michael Charles Lanz, Skye Glenn, Matthew Swaffer, Georgi K Marinov, Jan M Skotheim, Christine Jacobs-Wagner. ***Eqed*** 13, RP97465
2. Connecting single-cell ATP dynamics to overflow metabolism, cell growth, and the cell cycle in *Escherichia coli*. (2022) Wei-Hsiang Lin and Christine Jacobs-Wagner. ***Czwwdr yBjsosi ã*** 32(18):3911-3924
3. Origin of exponential growth in nonlinear reaction networks. (2020) Wei-Hsiang Lin*, Edo Kussell, Lai-Sang Young and Christine Jacobs-Wagner* ***Pwsbddcjr i xseyj d Nayjsr aoAbacdq ã se Sbydr bdx*** 117 (45), 27795-27804
4. Complex interplay of physiology and selection in the emergence of antibiotic resistance. (2016) Wei-Hsiang Lin and Edo Kussell. ***Czwwdr yBjsosi ã*** 26(11):1486-1493

Studying ATP and growth dynamics at the single-cell level in *E. coli*

Wei-Hsiang Lin[1,*]
[1] *Institute of Molecular Biology, Academia Sinica, Taipei, Taiwan*

ATP (adenosine triphosphate) is the common energy carrier for all living systems and is essential for cell metabolism and RNA synthesis. Using microfluidic cell culture with an ATP fluorescence biosensor, we measured the long-term ATP dynamics for *E. coli* cells for multiple days. We found that intracellular ATP level is coupled with bacterial cell cycle, and the ATP oscillation is attenuated in the absence of overflow metabolism. ATP dynamics and instantaneous cell growth exhibit cross-correlation with time-delay. In addition to constant environment, we measured the growth rate and ATP level response under nutrient switch. Overall, single-cell microscopy reveals detail regulation and dynamical pattern of cell growth and metabolism.

References:

1. Lin, W.-H., Jacobs-Wagner, C. Connecting single-cell ATP dynamics to overflow metabolism, cell growth, and the cell cycle in *Escherichia coli*. *Current Biology* 32 (18), 3911-3924. e4

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Dr. Jie-Ning Li



國立成功大學 細胞生物與解剖學研究所
Assistant Prof. / Department of Cell Biology and Anatomy,
National Cheng Kung University

Education

- **Ph.D.** in Institute of Basic Medical Sciences, College of Medicine, National Cheng Kung University (NCKU), Tainan, Taiwan 2014.09-2021.01
- **M.S.** in Department of Medical Laboratory Science and Biotechnology, College of Medicine, National Cheng Kung University (NCKU), Tainan, Taiwan 2013.09-2014.07
- **B.S.** in Department of Medical Laboratory Science and Biotechnology, College of Medicine, National Cheng Kung University (NCKU), Tainan, Taiwan 2009.09-2013.06

Experience

- **Assistant professor** in Department of Cell Biology and Anatomy, College of Medicine, National Cheng Kung University 2026.08-present
- **Postdoctoral Research Fellow** in Department of Medical Laboratory Science and Biotechnology, College of Medicine, National Cheng Kung University 2021.02-2026.07
- **Research Consultant** in Breast Medical Center, National Cheng Kung University Hospital 2023.01-2025.07

Research Interests

Breast Cancer, Precision Oncology, RNA Damage Response, RNA Surveillance System, Noncoding RNA

Selected Publications

1. **Jie-Ning Li***, Ching-Hung Lin, Ming-Yang Wang, Pai-Sheng Chen. (2026, Jun Accepted) Tumor Hypoxia: Molecular Pathways and Therapeutic Interventions. ***Drug Discovery Today*** **(5-year IF=8.9; Ranking 4.8%, PHARMACOLOGY & PHARMACY) *First author**
2. **Jie-Ning Li***, Ming-Yang Wang, Chiao Lo, Laising Yen, Chien-Hung Yu, Yu-Jhen Lyu, Pai-Sheng Chen. (2026, Feb). A non-canonical role for HIF-1α: redirecting DGCR8 to the RNA exosome for snoRNA degradation and translational modulation. ***Nucleic Acids Res.*** 2026 Feb; 2026;54(3):gkag070. doi:10.1093/nar/gkag070 **(5-year IF=17.6; Ranking 4.3%, BIOCHEMISTRY & MOLECULAR BIOLOGY) *First author**
3. **Jie-Ning Li***, Ming-Yang Wang, Jhen-Wei Ruan, Yu-Jhen Lyu, Yi-Hsiu Weng, Pownraj Brindangnanam, Mohane Selvaraj Coumar, Pai-Sheng Chen. (2024, Oct). A transcription-independent role for HIF-1α in modulating microprocessor assembly. ***Nucleic Acids Res.*** 2024 Oct; 2024;52(19):11806-11821. doi: 10.1093/nar/gkae792 **(5-year IF=17.6; Ranking 4.3%, BIOCHEMISTRY & MOLECULAR BIOLOGY) *First author**
4. **Jie-Ning Li***, Zhu-Jun Loh, Hui-Wen Chen, I-Ying Lee, Jui-Hung Tsai, Pai-Sheng Chen. (2024, Oct). SnoRNA U50A Mediates Everolimus Resistance in Breast Cancer Through mTOR Downregulation. ***Transl Oncol.*** 2024 Oct:48:102062. doi: 10.1016/j.tranon.2024.102062. Epub 2024 Aug 1 **(5-year IF=4.6; Ranking 27.9%, ONCOLOGY) *First author**

Beyond Transcription: Novel Mechanisms of HIF-1α in Small RNA Regulation, the Exoswitch

Jie-Ning Li (Jessy Li) Assistant Professor
Department of Cell Biology and Anatomy, College of Medicine, National Cheng Kung University, Tainan, 70101, Taiwan

Hypoxia-inducible factor 1-alpha (HIF-1α) is known as a transcriptional regulator of cellular adaptation to hypoxia and other microenvironmental stresses. However, how HIF-1α coordinates nuclear RNA-processing machineries independently of transcription remains incompletely understood. Here, by integrating two complementary studies, we define a transcription-independent mechanism through which HIF-1α reprograms DGCR8-centered RNA metabolism. HIF-1α preferentially associates with monomeric DGCR8, prevents DGCR8 dimerization, and disrupts the assembly of the Drosha/DGCR8 complex (Microprocessor), thereby suppressing primary microRNA processing. Rather than simply rendering DGCR8 inactive, HIF-1α redirects DGCR8 toward an RRP6-containing RNA exosome, establishing a dynamic Microprocessor-to-exosome switch, or "Exoswitch". This molecular transition is accompanied by the redistribution of DGCR8 and RRP6 from the nucleolus to the nucleoplasm and a shift in DGCR8 substrate preference from primary microRNAs to small nucleolar RNAs. Consequently, the newly configured DGCR8-RNA exosome complex promotes snoRNA degradation without significantly altering snoRNA transcription or precursor processing. The resulting depletion of snoRNAs reduces ribosomal RNA pseudouridylation and 2'-O-methylation, compromises rRNA processing, and decreases global translation efficiency. These responses are induced by hypoxia and growth-factor signaling and are evolutionarily conserved across human cells, Caenorhabditis elegans, and Drosophila melanogaster, with corresponding DGCR8 complex-switching patterns observed in human breast cancer tissues. Collectively, our findings position HIF-1 α as a systems-level coordinator of post-transcriptional and translational adaptation. By reallocating DGCR8 between distinct ribonucleoprotein complexes, HIF-1 α simultaneously suppresses microRNA biogenesis, enhances snoRNA turnover, and adjusts ribosome output to changing environmental conditions. This work expands the functional scope of HIF-1α beyond canonical transcription and identifies DGCR8 as a context-dependent molecular switch linking microenvironmental signals to RNA processing, RNA surveillance, ribosome biogenesis, and protein synthesis.

Session 8：代謝體學新紀元：從精密質譜分析到精準醫學實踐

TGMBS

台灣基因醫學暨生物標記學會

主持人

黃友利 副校長

高雄醫學大學

Moderator

Prof. Yeou-Lih Huang

Senior Vice President
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徐丞志 執行長
Dr. Cheng-Chih Hsu

創辦人/雷文虎克生物技術
副教授/國立臺灣大學 化學系暨研究所
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National Taiwan University
Founder / Leeuwenhoek Biotechnology

Education

- 2014 Ph.D. in Chemistry and Biochemistry, University of California, San Diego, USA
- 2007 Master's in Chemistry, National Taiwan University
- 2005 B.Sc. in Chemistry, National Taiwan University

Experience

- 2021 Founder of Leeuwenhoek Biotechnology Co., Ltd.
- 2020 Associate Professor, Department of Chemistry, National Taiwan University
- 2015-2020 Assistant Professor, Department of Chemistry, National Taiwan University
- 2014-2015 Postdoctoral Researcher, Department of Chemistry, Stanford University

Honors and Awards

- 2018 National Taiwan University Teaching Excellence Award
- 2018 Ministry of Science and Technology FutureTech Breakthrough Award
- 2019 Recipient of the Ministry of Science and Technology Young Scholar Program (Columbus Program)
- 2020 National Taiwan University Teaching Excellence Award
- 2020 Taiwan Society for Mass Spectrometry (TSMS) Young Scholar Research Award
- 2020 Chinese Chemical Society (CSLT) Outstanding Young Chemist Award Medal
- 2021 National Health Research Institutes Research Grant Program (CDG) Recipient
- 2023 Ministry of Science and Technology FutureTech Award
- 2024 Ta-You Wu Memorial Award

Research Interests

Medical Mass Spectrometry Imaging
Human Microbiome Metabolomics
Ambient Ionization Mass Spectrometry and its Clinical Diagnosis Applications

Selected Publications

1. Laura Min Xuan Chai, Ching Kao, Ming-Yang Wang, **Cheng-Chih Hsu*** "Untargeted Swab Touch Spray-Mass Spectrometry Analysis with Machine Learning for On-Site Breast Surgical Margin Assessment." (2025), ***Anal. Chem. 97,*** *4, 1960–1965*
2. Qiang Lyu, Rou-An Chen, Hsiao-Li Chuang, Hsin-Bai Zou, Lihong Liu, Li-Kang Sung, Po-Yu Liu, Hsin-Yi Wu, Hsin-Yuan Chang, Wan-Ju Cheng, Wei-Kai Wu, Ming-Shiang Wu, **Cheng-Chih Hsu*** "Bifidobacterium alleviate metabolic disorders via converting methionine to 5'-methylthioadenosine." (2024), ***Gut Microbes***, 16(1).
3. Wei-Chieh Wang, Chu-Hsuan Huang, Hsin-Hsiang Chung, Pei-Lung Chen, Fung-Rong Hu, Chang-

Hao Yang, Chung-May Yang, Chao-Wen Lin, **Cheng-Chih Hsu***, Ta-Ching Chen*" Metabolomics facilitates differential diagnosis in common inherited retinal degenerations by exploring their profiles of serum metabolites" ***Nat. Commun.,*** 15, 3562

4. Kai-Wen Cheng, Pin-Rui Su, Kate Jo-Ann Feller, Miao-Ping Chien*, **Cheng-Chih Hsu*** "Investigating the Metabolic Heterogeneity of Cancer Cells Using Functional Single-Cell Selection and nLC Combined with Multinozzle Emitter Mass Spectrometry" (2024), ***Anal. Chem.*** 96, 2, 624−629 (featured on front cover)
5. Ying-Chen Huang, Hsin-Hsiang Chung, Ewelina P Dutkiewicz, Chih-Lin Chen, Hua-Yi Hsieh, Bo-Rong Chen, Ming-Yang Wang*, **Cheng-Chih Hsu*** "Predicting Breast Cancer by Paper Spray Ion Mobility Spectrometry Mass Spectrometry and Machine Learning." (2020) ***Anal. Chem***. 92, 1653 1657. (featured on front cover)

Generalizability of machine-learning–identified metabolites for breast cancer diagnosis across MS platforms and medical centers

Laura Min Xuan Chai[1], Un Cheng Loi[1], Tzu-Yin Wan[1], Hsiang-Wei Huang[2], Li-Wei Tsai[2,3,4], Wei-Ning Chiu[5], Chiao Lo[5], Ming-Yang Wang[2], Cheng-Chih Hsu[1,6]

[1]Department of Chemistry, National Taiwan University, Taiwan, 10617, Taiwan
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[4]Department of Surgery, National Taiwan University College of Medicine, Taipei, Taiwan
[5]Department of Surgery, National Taiwan University Hospital, Taipei, Taiwan
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Data accumulation presents a major challenge for mass-spectrometry (MS)-based clinical applications, particularly for fresh tissue specimens. Leveraging existing databases across different ionization methods, instruments, and clinical sites would be highly advantageous, enabling institutions to adopt instrumentation that matches their analytical needs, infrastructure constraints, or existing resources, without requiring extensive model retraining. Hence, we evaluated the cross-platform generalizability of a published SVM-RFE model trained on 684 breast core-needle biopsies (CNB) analyzed via miniaturized mass spectrometry coupled with paper spray ionization (MiniMAP) for breast cancer diagnosis. Model performance was assessed via data generated from three MS setups and cohorts from two clinical sites.

Breast CNBs were collected at National Taiwan University Cancer Centre (NTUCC) and National Taiwan University Hospital (NTUH) under IRB approval (201705092RIND) with informed consent. Specimens were extracted by sonication and the supernatants were analyzed in positive mode, *m/z* 500–1000, using MS1 scan settings on three platforms: MiniMAP on single quadrupole Advion Interchim Scientific puriFlash® (A), electrospray ionization (ESI) with triple quadrupole Waters Xevo TQD (W), and nano-ESI kit with miniaturized ion-trap Purspec Cell (P). Spectra were processed to match the original model workflow (scan averaging, m/z alignment, TIC normalization, and 1-Da binning). The pre-trained MiniMaP SVM-RFE classifier was applied without retraining, in which it had a 92.6% specificity and 77.5% sensitivity. A slight drop in sensitivity was observed on platform W and P. To improve prediction performance, chemometric calibration-transfer models were trained on paired measurements (A↔W, A↔P) and applied before classification.

References: Huang, H.-C.; Chung, H.-H.; Yu, J.-Y.; Chen, B.-R.; Wang, M.-Y.; Hsu, C.-C. Development and Multicenter Validation of On-Site Breast Cancer Diagnosis Using Paper Spray Ionization Miniature Mass Spectrometry. *Commun. Med*. **2025**, 5, 259.

詹華蓁 助理教授
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合聘助理教授/高雄醫學大學 醫學檢驗生物技術學系
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Kaohsiung Medical University

Education

- 2002 - 2007 Master Graduate Institute of Medicine, Kaohsiung Medical University
- 2011 - 2015 Ph.D. Graduate Institute of Medicine, Kaohsiung Medical University

Experience

- 2007.07 - 2007.10 China Medical University Hospital (CMUH), Taichung, Taiwan
  Medical Scientist (醫學檢驗部醫檢師)
- 2008.01 - 2008.07 Kaohsiung Medical University Hospital (KMUH), Kaohsiung
  Research Assistant (專任研究助理)
- 2008.07 - 2009.07 Athero & Vascular Medicine, Baylor College of Medicine, Texas, USA
  Research Assistant II (研究助理 II)
- 2009.07 - 2010.08 Vascular and Medicinal Research, Texas Heart Institute, Texas, USA
  Research Assistant II (研究助理 II)
- 2010.10 - 2013.07 China Medical University Hospital (CMUH), Taichung, Taiwan
  Chief of technician (技術組長)
- 2013.08 - 2020.07 Center for Lipid Biosciences, KMUH, Taiwan
  Assistant Investigator (醫學助理研究員)
- 2020.08 - 2021.09 Center for Lipid Biosciences, KMUH, Taiwan
  Associate Investigator (醫學副研究員)
- 2021.08 - 2022.07 Fooyin University, Department of Medical Laboratory Science and Biotechnology, Kaohsiung, Taiwan. 輔英科技大學醫學檢驗生物技術學系
  Assistant Professor (專案助理教授)
- 2021.08 - Present I-Shou University, Department of Medical Laboratory Science, Kaohsiung, Taiwan. 義守大學醫學檢驗生物技術學系
  Assistant Professor (專任助理教授)
- 2024.07 - Present Mackay Memorial Hospital, Division of Thoracic Surgery, Taipei
  Consultant
- 2025.02 - Present Kaohsiung Medical University, College of Health Sciences, Medical Laboratory Science and Biotechnology, Taiwan
  Assistant Professor (合聘助理教授)

Honors and Awards

- The Second Prize in the 2015 Metabolomics symposium in human disease and diagnosis Poster Competition Award

- 高雄醫學大學跨校研究學習成效競賽獲得佳作
- Finalist of Young Investigator Competition of the University of Cologne-Kaohsiung Medical University Joint Symposium on Cancer, Aging, Neurodegeneration, 2017

Selected Publications

1. Ting-Chieh Lai, Shao-Chi Hung, **Hua-Chen Chan**, Chih-Sheng Chu, Gopal Kedihithlu Marathe, Guenter Schwarz, Liang-Yin Ke. Lysophosphatidylcholine as a mechanistic and therapeutic nexus in atherosclerotic cardiovascular disease. *Biomed Pharmacother.* 2026 Apr:197:119069. [IF= 7.5; R/C=24/195= 12.3 % in MEDICINE, RESEARCH & EXPERIMENTAL]
2. Chang-Yi Yen, Chang-Chi Lin, Ping-Chang Lin, Ching-Yi Tsai, Li-Teh Liu, Chen-Hsuan Lin, Chun-Hong Chen, Chia-Chun Tseng, **Hua-Chen Chan**, Cheng-Chin Wu, Tsan-Teng Ou, Jih-Jin Tsai, Jeng-Hsien Yen. Association Between Chikungunya Infection and Rheumatoid Arthritis in Taiwan-A Cross-Sectional Study. *Int J Rheum Dis.* 2026 Aug;29(8):e70784. [IF= 2.0; R/C=36/58= 62.07 % in RHEUMATOLOGY]
3. You Shan Ngui, Yi-Wen Huang, Shao-Chi Hung, Shih-Han Huang, Chieh-Ju Wen, Yi-An Chang, Hsing-Chen Wu, Hsiu-Chuan Chan, Mei-Lin Chan, Liang-Yin Ke, **Hua-Chen Chan**. Development of an isoform-discriminating RT-qPCR assay for transcript-level profiling of human ASAH2 in colorectal cancer cell models. *Mol Biol Rep.* 2026 Aug 18;53(1):1425. [IF= 3.2; R/C=185/328= 56.92 % in BIOCHEMISTRY & MOLECULAR BIOLOGY]
4. **Hua-Chen Chan**, Hsiu-Chuan Chan, Liu-Fang Wang, Mei-Lin Chan, Wen-Chien Huang, Daniel Bender, Yu-Min Ko, Ming-Lung Yu, Guan-Ming Ke, Mei-Chuan Chou*, Ching-Kuan Liu*, Liang-Yin Ke*. The lysophosphatidylcholine-HIF-1α axis enhances apolipoprotein E sialylation and promotes electronegative LDL accumulation. *iScience*. 2026. June 19; 29(6):116167. [IF= 4.1; R/C=22/136= 16.2 % in Multidisciplinary Science]
5. Yi-Wen Huang*, **Hua-Chen Chan***, Jing-Yi Khoo, Mei-Lin Chan, Daniel Bender, Vinoth Kumar Ponnusamy, Abdel Ali Belaidi, Liang-Yin Ke (2025, Dec). Assessing the critical role of ceramide in the pathogenesis of Alzheimer's disease and its clinical significance. *Neurochem Int.* 2026 Jan;192:106104. NSTC 113-2314-B-214-002. [IF= 4.0; R/C=88/314= 28 % in Neurosciences]
6. Shao-Chi Hung, Chih-Chieh Chen, **Hua-Chen Chan**, Mei-Lin Chan, You Shan Ngui, Jia-Bin Mei, Yi-Wen Huang, Vyala Hanumanthareddy Chaithra, Gopal Kedihithlu Marathe, Liang-Yin Ke (2025, Oct). Structural and molecular dynamics insights into the competitive inhibition of the platelet-activating factor receptor by acyl-PAF. *J Biol Chem.* 2025; 301(12):110831. NSTC 113-2314-B-214-002.
7. Shao-Chi Hung, Te-Fu Chan, Hsiu-Chuan Chan, Chia-Ying Wu, Mei-Lin Chan, Jie-Yang Jhuang, Ji-Qin Tan, Jia-Bin Mei, Shi-Hui Law, Vinoth Kumar Ponnusamy, **Hua-Chen Chan**, Liang-Yin Ke. Lysophosphatidylcholine Impairs the Mitochondria Homeostasis Leading to Trophoblast Dysfunction in Gestational Diabetes Mellitus. *Antioxidants (Basel)*. 2024 Aug19;13(8):1007. [IF= 6.0; R/C=8/72= 11.11 % in CHEMISTRY, MEDICINAL]

Beyond LDL Cholesterol: Lysophosphatidylcholine as Cargo and Driver of Pathogenic Electronegative Lipoproteins

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Jointly Appointed Faculty, Kaohsiung Medical University, Kaohsiung, Taiwan

Low-density lipoprotein cholesterol (LDL-C) is a well-established therapeutic target for cardiovascular disease; however, LDL-C concentration alone does not fully reflect the pathogenic properties of circulating lipoproteins or the residual vascular risk that persists despite lipid-lowering therapy. Lysophosphatidylcholine (LPC) is a bioactive phospholipid enriched in atherosclerotic lesions and modified lipoproteins, yet its molecular heterogeneity, lipoprotein context, and contribution to pathogenic lipoprotein remodeling remain incompletely understood.

In our recent study, targeted lipid profiling showed that the major circulating LPC species, LPC(16:0), LPC(18:0), and LPC(18:1), remained elevated in patients with hyperlipidemia despite effective statin-associated reductions in LDL-C. Functional studies further demonstrated marked acyl-chain–dependent endothelial responses: saturated LPC(18:0) and LPC(16:0) induced greater cytotoxicity, cytochrome c release, and mitochondrial disruption than unsaturated LPC(18:1). Molecular modeling also suggested distinct LPC species–dependent interaction patterns with the lipid-sensing receptor GPR132, supporting the concept that LPC should not be regarded as a biologically uniform lipid class.

The pathological relevance of LPC may additionally depend on its lipoprotein carrier. Highly electronegative LDL, designated L5, exhibits potent vascular and thrombotic activities. In patients with ST-elevation myocardial infarction, L5 promotes platelet activation and aggregation, whereas in systemic lupus erythematosus, LPC-enriched L5 is associated with endothelial–monocyte interactions, vascular inflammation, and early vascular aging. These observations suggest that L5 functions as a pathogenic delivery platform for bioactive lipid cargo.

Recent findings further reveal that LPC is not only transported by electronegative lipoproteins but may also promote their formation. In hepatocytes, LPC activates the HIF-1α pathway and enhances the expression of glycosylation-related enzymes, including GALNT2 and ST3GAL1, thereby increasing apolipoprotein E glycosylation and sialylation. This modification increases lipoprotein electronegativity and promotes the accumulation of electronegative VLDL and LDL.

Collectively, these findings support a model in which LPC acts both as a molecular-species–dependent vascular mediator and as an upstream driver of pathogenic lipoprotein remodeling. Species-resolved LPC profiling and characterization of LPC-associated lipoprotein modifications may therefore provide additional insight into cardiovascular risk beyond conventional LDL-C measurement.



朱俐潔 副教授
Associate Prof. Lichieh Julie Chu

長庚大學 生物醫學研究所
Graduate Institute of Biomedical Sciences,
Chang Gung University

Education

國立陽明交通大學醫學生物技術研究所 博士

Experience

- 2026 – now 副教授 • 長庚大學生物醫學研究所
- 2021 – now 理事 • 台灣蛋白體學會
- 2021 – 2026 助理教授 • 長庚大學生物醫學研究所
- 2020 – 2022 理事 • 台灣生物資訊與系統生物學會
- 2018 – 2021 秘書長 • 台灣蛋白體學會
- 2014 – 2021 助理研究員 • 長庚大學分子醫學研究中心
- 2011 – 2014 博士後研究員 • 長庚大學分子醫學研究中心
- 2008 – 2008 訪問學者 • 美國華盛頓大學

Honors and Awards

- 第 19 屆台塑企業應用技術研討會研發論文優勝
- The Phi Tau Phi Scholastic Honor Society

Research Interests

癌症生物學、生物標誌、(多)體學與系統生物學、胞外體研究

Selected Publications

1. **Chu LJ**, Tsai CJ, Chien CC, Hsiao YC, Yu JS, Tsai JY, Yeh TS. Plasma POSTN Derived From Bile Proteome Is a Promising Biomarker for Cholangiocarcinoma With Efficacy Comparable and Complementary to CA19.9. ***United European Gastroenterol J.*** 2025 Jun 13(7):1155
2. **Chu LJ**, Hsu YT, Aoh Y, Cheng CY, Wang HH, Wu MT, Li HH, Lai XY, Hua KF, Chen CM, Ju TC. Neuroprotective Effects of Glucosamine in Huntington's Disease Through NLRP3 Inflammasome Inhibition. ***J Neuroimmune Pharmacol.*** 2025 20(1):104.
3. Liu CC*, **Chu LJ***, Yeh YM, Lin HC, Chen LC, Huang CY, Chiu SF, Cheng FW, Lin WN, Huang KY. Immunomodulatory roles of autophagic flux and IFIT in human ectocervical cells upon Trichomonas vaginalis infection. ***Int Immunopharmacol.*** 2025 16(155):114643.
4. **Chu LJ**, Chang YT, Chien CY, Chung HC, Wu SF, Chen CJ, Liu YC, Liao WC, Chen CH, Chiang WF, Chang KP, Wang JS, Yu JS. Clinical validation of a saliva-based matrix metalloproteinase-1 rapid strip test for detection of oral cavity cancer. ***Biomed J.*** 2023 47(1):100594.
5. Chang YT, **Chu LJ**, Liu YC, Chen CJ, Wu SF, Chen CH, Chang IY, Wang JS, Wu TY, Dash S, Chiang WF, Chiu SF, Gou SB, Chien CY, Chang KP, Yu JS. Verification of saliva matrix metalloproteinase-1 as a strong diagnostic marker of oral cavity cancer. ***Cancers*** 2020 12(8):2273

Integrated Multi-Omics Analysis of Molecular Pathways Driving Malignant Transformation in Oral Potentially Malignant Disorders

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[1] *Graduate Institute of Biomedical Sciences, Chang Gung University, Taoyuan, Taiwan.*

[2] *Institute of Molecular Medicine, National Tsing Hua University, Hsinchu, Taiwan.*

[3] *Molecular Medicine Research Center, Chang Gung University, Taoyuan, Taiwan.*

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Oral cancer is prevalent among Taiwanese men who chew betel quid, typically arising from the malignant transformation of oral precancerous lesions. However, the mechanisms driving the progression of these lesions remain incompletely understood, and a scarcity of research materials has hindered early prevention and intervention efforts. To investigate the alterations occurring in epithelial cells during malignant transformation, this study utilized the DOK cell line, a model of mild oral precancerous lesions, subjecting it to long-term, continuous exposure to arecoline to establish cell lines with varying degrees of malignancy. A multi-omics integration approach encompassing transcriptomics, proteomics, metabolomics, and miRNA analysis employed to examine the molecular and pathway-level changes throughout the progression. As the duration of exposure increased, the expression of c-Myc and NRF2, along with cell proliferation, migration, and tumorigenic potential, all intensified, indicating an escalation in malignancy. The integrated multi-omics data revealed several key alterations associated with this malignant progression, including enhanced antioxidant capacity, shifts in polyamine and taurine metabolic pathways, and distinct changes in tryptophan metabolism during the precancerous stage. These findings were validated in the cell model and corroborated by observations in clinical oral cancer samples. The results suggest that the progression of oral precancerous lesions is closely linked to enhanced antioxidant capacity and metabolic reprogramming, and specific molecules identified in this process may serve as potential biomarkers for the prevention, monitoring, and management of oral cancer.

Session 9: Plant and Microbial EVs

Taiwan Society for Extracellular Vesicles

TSEV

台灣胞外體學會

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Moderator

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Department of Food Safety/Hygiene and Risk Management,
National Cheng Kung University
Affiliated Associate Prof. / Department of Public Health,
National Cheng Kung University
Director / Taiwan Association for Lactic Acid Bacteria

Education
國立臺灣大學 生化科技學系 博士

Experience
- 國立成功大學 食品安全衛生暨風險管理研究所 助理教授
- 工業技術研究院生醫與醫材研究所 研究員
- Purdue University 醫藥科學系 博士後研究員
- 國立臺灣大學生化科技學系 博士後研究員

Honors and Awards
- 吳大猷先生紀念獎（2026）
- 國立成功大學產學合作成果特優教師（2025, 2026）
- 第 63 屆台灣農業化學會-學術獎（2025）
- 台灣保健食品學會-傑出學術研究獎（2025）
- 財團法人成杏醫學文教基金會-優秀論文獎（2024, 2025）
- 第 21 屆國家新創獎-學研新創獎（2024）
- 臺灣綜合大學年輕學者創新研發成果-傑出獎（2024）
- 未來科技獎（2024）
- 北美國立成功大學校友基金會跨領域研究獎勵優良獎（2024）
- 李國鼎科技與人文講座-金質獎（2023）
- 國立成功大學醫學院年度新人最佳論文獎（2023）

Research Interests
食品微生物；次世代益生菌；胞外體；腸道菌相

Selected Publications
1. BH Lee, RY Ferng, HY Chien, CH Huang, **WH Hsu***. Harnessing *Rhizopus oligosporus* mycelium as a sustainable protein source: Functional properties and optimization of a meat-analogue “chicken popcorn”. Food Hydrocolloids, 174, 112325, 2026.
2. **WH Hsu**, YZ Chen, YT Chiang, YT Chang, YW Wang, KT Hsu, YY Hsu, PT Wu, BH Lee*. Polystyrene nanoplastics disrupt the intestinal microenvironment by altering bacteria-host interactions through extracellular vesicle-delivered microRNAs. Nature Communications, 16, 5026, 2025.
3. BH Lee, YC Lin, YJ Zheng, TL Shen, TY Cheng, CC Huang, **WH Hsu***. Nanoplastics indirectly compromise lettuce growth in hydroponic systems via microbial extracellular vesicles derived from *Curvibacter fontanus*. Journal of Hazardous Materials, 480, 136457, 2024.

4. BH Lee, YZ Chen, TL Shen, TM Pan, **WH Hsu***. Proteomic characterization of extracellular vesicles derived from lactic acid bacteria. Food Chemistry, 427, 136685, 2023.

Extracellular Vesicle-Mediated Host-Microbe Crosstalk

Wei-Hsuan Hsu
Associate Professor
Department of Food Safety/Hygiene and Risk Management
National Cheng Kung University, Tainan, Taiwan

The composition and dynamics of the gut microbiota are intimately associated with host growth, digestion, and immune regulation, exerting a profound influence on human health. In recent years, global research efforts have increasingly focused on the microbiome's role in disease pathogenesis and therapy, establishing it as a central theme in modern biomedicine. Evaluating structural and functional alterations in the gut microbiota offers great potential for precision and personalized medicine. The gut ecosystem represents a highly interactive network between the host and microorganisms, where microbial communities adapt dynamically to environmental changes to maintain host homeostasis. Understanding the reciprocal regulation among microbial populations and the crosstalk between microbes and their host remains a key to decoding the microbiota–health relationship. Extracellular vesicles (EVs), nanosized bi-layered vesicles released by both prokaryotic and eukaryotic cells, have emerged as critical mediators of intercellular and interspecies communication. These vesicles carry diverse bioactive molecules that can enter target cells and modulate gene expression, thereby serving as information carriers across biological kingdoms. Microbial EVs, in particular, facilitate communication between microbes and between microbes and the host, influencing gut microbiota composition and host physiology. Elucidating how microbial EVs modulate microbial networks and host health represents a novel and promising direction for advancing microbiome-based therapeutics and precision health applications.

潘敏雄 副院長
Prof. Min-Hsiung Pan



國立臺灣大學 生物資源暨農學院
特聘教授/國立臺灣大學 食品科技研究所
Associate Dean / College of Bioresources and Agriculture,
National Taiwan University
Distinguished Professor / Institute of Food Science and Technology,
National Taiwan University

Education
- Ph.D. in Biochemistry, Institute of Biochemistry, College of Medicine, NTU (1996–2000)
- M.S. in Food Chemistry, Department of Food Science, National Taiwan Ocean University (NTOU) (1993–1995)
- B.S. in Food Chemistry, Department of Food Science, NTOU (1991–1993)

Experience
- Associate Dean, College of Bioresources and Agriculture, NTU (2026–present)
- Director, Bachelor Program of Biotechnology and Food Nutrition, NTU (2026–present)
- Distinguished Professor, Institute of Food Science and Technology, NTU (2015–present)
- Director, Institute of Food Science and Technology, NTU (2018–2024)

Honors and Awards
- Fellow, The Institute of Food Science & Technology (IFST)
- Eminence Chair Professorships of Fubon (2026/8/1~2029/7/31)
- Research.com Best Scientist — Biology & Biochemistry (National Ranking 14) and Chemistry (26), 2026.
- World's Top 2% Scientists — Stanford University, 2020-2026 (Annual and Career Impact).
- Fellow, International Academy of Food Science and Technology (IAFoST), 2024.
- MOST Outstanding Research Award (2018) and Highly Cited Researcher, Clarivate Analytics (2018).

Research Interests
- Isolation, purification, and structural characterization of dietary bioactive compounds and plant-derived extracellular vesicles.
- Analytical methods for food analysis, bioactive compound identification, and quality evaluation.
- Bioavailability, epigenetic regulation, and molecular mechanisms of phytochemicals in cellular and animal models.
- Functional foods and dietary supplements for cancer chemoprevention, obesity, liver disease, gut microbiota, and metabolic health.

Selected Publications

1. Kuo, Y. R.; Lin, T. L.; Ho, P. Y.; Chen, W. J.; Chu, H.F.; Tseng, M. C.; Wei, Y. S.; **Pan, M. H.*** *Lactiplantibacillus plantarum*-Fermented Broccoli Attenuates Metabolic Dysfunction and Improves Skeletal Muscle Quality in High-Fat/High-Fructose Diet-Fed Mice. ***Journal of Agricultural and Food Chemistry***, 2026; 74 (39): 31112–31126.
2. Koh, Y. C.; Ruan, J. Y.; Wan, C. H.; **Pan, M. H.*** *Morus alba* leaf tea consumption reduces hyperlipidaemia, adipocyte hypertrophy and inflammation in HFD-induced obese mice. ***Journal of Traditional and Complementary Medicine***, 2026.
3. Lin, Y. C.; Hsu, P. K.; **Pan, M. H.*** Regulatory Effects of Pumpkin Seed Extract on Glucose Metabolism and Insulin Signaling in Diabetic Models. ***ACS Omega***, 2026. 11(22), 32263-32272.

Regulatory Effects and Potential Applications of Plant-Derived Nanovesicles in Intestinal Inflammation and Oxidative Stress

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[1] *Institute of Food Sciences and Technology, National Taiwan University, 10617, Taipei, Taiwan.*
[2] *Department of Medical Research, China Medical University Hospital, China Medical University, 40402, Taichung City, Taiwan.*
[3] *Department of Food Nutrition and Health Biotechnology, Asia University, 41354, Taichung City, Taiwan.*

Plant-derived nanovesicles (PDNVs) are nanoscale vesicles containing various bioactive components, including lipids, proteins, and nucleic acids. Recent studies have suggested that PDNVs possess diverse biological activities and may contribute to the maintenance of physiological homeostasis, highlighting their potential applications in disease prevention and health promotion. In our research, we have focused on nanovesicles isolated from turmeric (*Curcuma longa*) and lotus leaves (*Nelumbo nucifera*) and investigated their effects on intestinal inflammation and oxidative stress. The results showed that both turmeric- and lotus leaf-derived nanovesicles attenuated intestinal inflammation and histopathological damage in dextran sulfate sodium (DSS)-induced mouse models of colitis. These effects were associated with modulation of tight junction protein expression, including occludin, claudin-2, and claudin-4, suggesting improved intestinal epithelial barrier integrity. Additionally, both types of nanovesicles attenuated TLR4/MyD88-associated inflammatory signaling and altered gut microbiota composition. Moreover, turmeric-derived nanovesicles (TDNVs) regulated Nrf2 and HO-1 protein expression in colonic tissues, suggesting a potential role in antioxidant defense. In a *Caenorhabditis elegans* model, TDNVs enhanced resistance to oxidative and thermal stress, reduced reactive oxygen species (ROS) levels and lipofuscin accumulation. These effects may be partly attributed to increased nuclear translocation of DAF-16 and elevated expression of its downstream stress-response genes, *hsp-16.2* and *sod-3*. Our studies suggest that turmeric- and lotus leaf-derived nanovesicles exert diverse biological activities through multiple biological pathways involved in intestinal barrier regulation, inflammatory responses, antioxidant defense, and stress resistance. These findings also provide valuable insights into the potential applications of PDNVs in functional food development and disease prevention.

沈湯龍 教授
Prof. Tang-Long Shen



國立臺灣大學 植物病理與微生物學系
Department of Plant Pathology and Microbiology,
National Taiwan University

Education

- 1997-2002 PhD. in Cancer Cell Biology in the Department of Molecular Medicine, Cornell University, USA
- 1987-1991 BS in Plant Pathology in the Department of Plant Pathology and Entomology, National Taiwan University, Taiwan

Experience

- Professor, Department of Plant Pathology and Microbiology, National Taiwan University
- Associate Dean, College of Bioresources and Agriculture, National Taiwan University
- Director, Global Innovation Joint-Degree Program on Agro-biomedicine (GIP-TRIAD), College of Medicine, National Taiwan University
- Professor, Graduate Program of Genomics and Systems Biology, National Taiwan University
- Professor, Institute of Biotechnology, National Taiwan University
- Adjunct Professor in Graduate Institute of Biomedical Sciences, China Medical University
- Adjunct Professor, School of Integrative and Global Majors (SIGMA), University of Tsukuba, Tsukuba, Japan

Honors and Awards

- Blind Prevention Foundation Scholar Travel Award, 2010
- The 6th Breast Cancer Outstanding Research Award, Breast cancer prevention foundation, 2016
- 第 16 屆國家新創獎 Development of a small molecule enhancement for erythropoiesis, 2019
- 第 18 屆國家新創獎獲獎項目為「利用腸腦軸線概念開發改善睡眠之植萃原料 Bugu-S™」, 2021
- Austrian Schweitzer International Foundation [World Medical Award], 2023

Research Interests

Microbial metabolites, cancer biology, signal transduction, exosome, viroid

Selected Publications

H-index:46 Citation >15,500 (As of 2026/8/01)

1. Shang-Min Chou, Wan-Jung Lai, Tzu-Wen Hong, Jui-Ya Lai, Sheng-Hong Tsai, Yen-Hsun Chen, Sz-Hsien Yu, Cheng-Hsiang Kao, Richard Chu, Shih-Torng Ding, Tsai-Kun Li and **Tang-Long Shen*** (2014) Synergistic property of Cordycepin in cultivated *Cordyceps militaris*-mediated apoptosis in human leukemia cells ***Phytomedicine*** 21 (12): 1516–1524
2. Yu-Ling Tai, Pei-Yu Chu, I-Rui Lai, Hui-Yuan Tseng, Jun-Yang Liou, Jun-Lin Guan, **Tang-Long Shen*** (2015) An EGFR/Src-dependent b4 Integrin/FAK complex contributes to the malignancy of breast cancer. ***Scientific Reports* 5**, 16408; doi: 10.1038/srep16408
3. Yen-Wen Wang, Tzu-Wen Hong, Yu-Ling Tai, ..., Tsai-Kun Li, Shean-Shong Tzean and **Tang-Long Shen*** (2015) Evaluation of an epitypified *Ophiocordyceps formosana* (*Cordyceps s.l.*) for its

pharmacological potential. ***Evidence-based complementary and alternative medicine*** Volume 2015, Article ID 189891

4. Bruno Costa-Silva, Nicole M. Aiello, Swarnima Singh, Haiying Zhang, Basant Kumar Thakur, Annette Becker, Gonçalo Rodrigues, **Tang-Long Shen**, K...Hector Peinado, Ben Stanger, Jacqueline Bromberg, and David Lyden (2015) Pancreatic cancer exosomes initiate pre-metastatic niche formation in the liver**. *Nature Cell Biol. 17(6):816-26.***
5. Ayuko Hoshino*, Bruno Costa-Silva*, **Tang-Long Shen***, Goncalo Rodrigues, Ayako Hashimoto, Milica Tesic Mark, Henrik Molina, S.... Jacqueline Bromberg, David Lyden (2015) Tumor exosome integrins determine organotropic metastasis. ***Nature 527(7578):329-35*** (***co-first authors**)
6. Yu-Ling Tai, Ko-Chien Chen, Jer-Tsong Hsieh, **Tang-Long Shen*** (2018) Exosomes in cancer development and clinical applications. ***Cancer Science*** 109(8):2364-2374. doi: 10.1111/cas.13697.
7. Tai YL, Chu PY, Lee BH, Chen KC, Yang CY, Kuo WH, **Shen TL*** (2019) Basics and applications of tumor-derived extracellular vesicles. ***J Biomed Sci.*** 26(1):35. doi: 10.1186/s12929-019-0533-x
8. Siou-Ru Shen, Wei-Jen Chen, Hui-Fang Chu, Shiuan-Huei Wu, Yu-Ru Wang, **Tang-Long Shen*** (2021) Amelioration of 5-fluorouracil-induced intestinal mucositis by *Streptococcus thermophilus* ST4 in a mouse model. ***PLOS One*** 16(7):e0253540. doi: 10.1371/journal.pone.0253540
9. Li-Chun Chang, Yi-Chiung Hsu, Han-Mo Chiu, Koji Ueda, Ming-Shiang Wu, Chiun-How Kao, and **Tang-Long Shen**.(2021) Exploration of the Proteomic Landscape of Small Extracellular Vesicles in Serum as Biomarkers for Early Detection of Colorectal Neoplasia. ***Front Oncol.*** 2021; 11: 732743. doi: 10.3389/fonc.2021.732743
10. Chang, Li-Chun, Han-Mo Chiu, Ming-Shiang Wu, and **Tang-Long Shen*** (2022) The Role of Small Extracellular Vesicles in the Progression of Colorectal Cancer and Its Clinical Applications. ***International Journal of Molecular Sciences*** 23 (3): 1379. https://doi.org/10.3390/ijms23031379
11. Wei-Jen Chen, Yu-Shan Wei, Hsin-I. Lo, Hui-Fang Chu, Ming-Chung Tseng, Bao-Hong Lee, **Tang-Long Shen*** (2024) Liquid-state fermented *Ganoderma lucidum* GANO99 regulates gut microbiota and concomitantly modulates the behavioral deficits and neurohistopathological hallmarks of Alzheimer's disease in a preclinical transgenic mouse model of Alzheimer's disease. ***Journal of Food Biochemistry*** Article ID （5）6676977
12. Chuang WY, Lin YC, Shrestha B, Jennifer J, Stadler M, Tzean SS, Wu S, Ko CC, Hsieh SY, Wu ML, Wang SC, **Shen TL*,** Ariyawansa HA* (2024). Phylogenetic diversity and morphological characterization of cordycipitaceous species in Taiwan. ***Studies in Mycology*** 109: 1–56. doi: 10.3114/sim.2024.109.01
13. Yasuhiro Shinkai, Yunjie Ding, Toru Matsui, George Devitt, Masahiro Akiyama, **Tang-Long Shen**, Motohiro Nishida, Tomoaki Ida, Takaaki Akaike, Sumeet Mahajan, Jon M Fukuto, Yasuteru Shigeta, Yoshito Kumagai (2025) Growth inhibitory factor/metallothionein-3 is a sulfane sulfur-binding protein ***eLife*** 12:RP92120
14. Chang, L.-C., **Shen, T.-L.**, and Ochiya, T. (2026) Novel Liquid Biopsy in Gastrointestinal Cancers. ***Journal of Gastroenterology and Hepatology***, 41: 546–555. https://doi.org/10.1111/jgh.70219.

Beyond the Cell: Extracellular Vesicles in Viroid-Associated Systemic Pathogenesis

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Viroids are small, circular, single-stranded noncoding RNAs that cause economically important diseases in plants. *Citrus exocortis viroid* (CEVd) infects citrus and a broad range of solanaceous hosts, including tomato. However, the mechanisms underlying its systemic spread and symptom development remain incompletely understood. Here, we investigated the potential involvement of tomato-derived extracellular vesicles (tEVs) in CEVd-associated systemic pathogenesis. tEVs were isolated from upper stems and young leaves of CEVd-infected and mock-inoculated tomato plants at six weeks post-inoculation (wpi). Particle concentration, protein and RNA content, and transmission electron microscopy (TEM) analyses revealed no substantial differences in EV abundance or morphology between infected and mock-inoculated plants. Nevertheless, CEVd genomic RNA was detected specifically in tEV preparations from infected plants, suggesting a potential association between CEVd and extracellular vesicles. Small RNA sequencing further identified viroid-derived small RNAs (vd-sRNAs) enriched in the variable region of the CEVd genome, with predicted host targets predominantly associated with immune responses and hormone signaling. Comparative proteomic analysis revealed increased abundance of several proteins involved in defense and catabolic processes, including catalase, subtilisin-like proteases, and pathogenesis-related (PR) proteins, in tEVs from infected leaves. To assess the biological activity of CEVd-associated tEVs, we inoculated tomato plants with tEV preparations derived from CEVd-infected plants. At four wpi, treated plants exhibited significantly reduced biomass and leaf area, accompanied by downregulation of hormone-related genes in systemic leaves compared with mock controls. Collectively, these findings reveal that CEVd infection alters the molecular cargo of tomato-derived extracellular vesicles and that EV preparations from infected plants can induce systemic disease-associated phenotypes. Our study provides new insights into the potential involvement of extracellular vesicles in viroid-associated systemic pathogenesis and host defense regulation.

Key word: *Citrus exocortis viroid*、viroid pathogenesis、extracellular vesicle、viroid-derived small RNAs



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AI 智慧自動化

自動辨識托盤類型、自動對焦定位與 AI 分析，一鍵導出標準化定量報告。

f/0.70 大光圈與深度冷凍

配備極致 f/0.70 超大光圈鏡頭與深度冷卻科研級 CCD 相機，有效降低背景雜訊，輕鬆捕捉 fg 級極微弱訊號，獲奧運藥檢採用。

9 通道多重螢光

支援 400nm ~ 900nm 可見光與近紅外光，搭配強大 N.M.F. 光譜拆分演算法 (Spectral Unmixing)，消除螢光干擾。

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Session 10：質譜多體學大數據生醫分析

Taiwan Proteomics Society

TPS

台灣蛋白體學會

主持人

王家義 副研發長

國立成功大學

研究發展處

Moderator

Prof. Chia-Yih Wang

Associate Vice President / Office of Research and Development,
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于松桓 助理教授
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Education

- 2013 - 2018 Institute of Molecular Infection Biology, University of Würzburg, Germany
- 2006 - 2008 Institute of Bioinformatics and System Biology, National Chiao Tung University
- 2002 - 2006 Department of Life Science, National Tsing Hua University

Experience

- 2024 - present Consultant, Genomic and Proteomic Core Laboratory, Kaohsiung Chang Gung Memorial Hospital
- 2021 - present Assistant Professor Institute of Precision Medicine, College of Medicine, National Sun Yat-sen University
- 2021 - present School of Medicine, College of Medicine, National Sun Yat-sen University, Taiwan
- 2018 - 2021 PostDoc, Max Planck Institute of Biochemistry, Germany
- 2011 - 2013 Research assistant, Institute of Bioinformatics and System Biology, National Chiao Tung University

Honors and Awards

- 2025 Prominent Faculty – Academic Research Award, National Sun Yat-sen University
- 2022 Excellent tutor Awards in the College of Medicine, National Sun Yat-sen University
- 2021-2023 New Faculty Award, National Sun Yat-sen University

Research Interests

Bioinformatics; Genomics; Proteomics; Machine Learning

Selected Publications

1. PH Wu, SC Chen, CJ Chien, J Lin, HY Lee, YT Lin, TC Weng, PC Hsu, MT Wu, SH Yu, “Epigenetic Signatures of Phthalate Exposure and Potential Risks: A DNA Methylation Analysis using Infinium MethylationEPIC BeadChip” *Environmental Epigenetics*, 2025, dvaf020 (selected as the cover image)
2. SH Yu, SC Chen, PS Wu, PI Kuo, TA Chen, HY Lee, MH Lin, “Quantification Quality Control Emerges as A Crucial Factor to Enhance Single Cell Proteomics Data Analysis” *Molecular & Cellular Proteomics*, 2024, 100768
3. MH Lin, PS Wu, TH Wong, IY Lin, J Lin, J Cox, SH Yu, “Benchmarking differential expression, imputation and quantification methods for proteomics data”, *Briefings in Bioinformatics*, 23(3), 2022
4. SH Yu, P Kyriakidou and J Cox, “Isobaric matching between runs and novel PSM-level normalization in MaxQuant strongly improve reporter ion-based quantification” *Journal of proteome research* 19 (10), 3945-3954, 2020

ANALomics: An Integrated Computational Toolkit for Multi-Omics Analysis Across Genomics, Transcriptomics, and Multi-Layered Proteomics

Johnathan Lin[1], Chih-Wen Lu[1], Sebastian Paul Perner[2], Yanlong Ji[2], Chih-Yu Chang[1], Kuan-Ting Pan[3], Sung-Huan Yu[1, 4*]

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The rapid advancement of high-throughput sequencing and mass spectrometry technologies has revolutionized biological research. However, fully dissecting complex biological mechanisms and disease pathologies requires a holistic view that spans multiple molecular layers—from genomic variants identified via DNA sequencing and transcriptomic profiles from RNA-seq, to post-translational modification (PTM) dynamics captured by phosphoproteomics and glycoproteomics. Despite the abundance of multi-omics data, researchers often face significant technical hurdles due to fragmented analytical pipelines, incompatible data formats, and a lack of standardized multi-layer integration platforms.

To address these challenges, we present ANALomics, a comprehensive bioinformatics platform engineered for end-to-end multi-omics analysis and cross-layer integration. ANALomics seamlessly unites DNA sequencing (variant calling), RNA-seq, global proteomics, phosphoproteomics, and glycoproteomics within a unified computational ecosystem. The toolkit encompasses a full suite of benchmarked and optimized analytical workflows, including quality control, normalization, missing value imputation, co-expression network construction, differential expression, dimensionality reduction, and multi-layer functional enrichment analysis. Crucially, ANALomics features a suite of interactive visualization tools, enabling users to effortlessly cross-examine, align, and compare multi-omic layers in real time. ANALomics provides an accessible, robust, and visual solution for researchers seeking to transform complex multi-omics datasets into actionable biological insights and translational discoveries.



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國立陽明交通大學 生物資訊及系統生物研究所 博士

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- 工業技術研究院 生醫與醫材研究所 博士後研究員
- 高雄醫學大學附設醫院 助理研究員
- 國立中山大學 醫學科技研究所 助理教授
- 國立中山大學 醫學科技研究所 副教授

Honors and Awards

- 114 學年度 醫學院優良導師
- 114 學年度 學術研究績優教師
- 111 學年度 全校優良導師

Research Interests

生物資訊、分子模擬、計算生物學、基因體分析、生醫 AI 模型開發

Selected Publications

1. Liu YY, Huang LW, Su MF, Chen BR, **Chen CC*** (2026) Global integrative analysis of whole-genome-based predictions of antimicrobial resistance trends in five major nosocomial pathogens. *Microbiology Spectrum* (accepted).
2. Hung SC, **Chen CC**, Chan HC, Chan ML, Ngui YS, Mei JB, Huang YW, Vyala Hanumanthareddy Chaithra, Gopal Kedihithlu Marathe, Ke LY* (2025) Structural and molecular dynamics insights into the competitive inhibition of the platelet-activating factor receptor by acyl-PAF. *Journal of Biological Chemistry* 301, 110831.
3. Liu YY, Hsu CY, Yang YC, Huang CH, **Chen CC*** (2023) ProbioMinServer: an integrated platform for assessing the safety and functional properties of potential probiotic strains. *Bioinformatics Advances* 3, vbad153.
4. **Chen CC***, Liu YY, Yang YC, Hsu CY (2022) 5NosoAE: a web server for nosocomial bacterial antibiogram investigation and epidemiology survey. *Nucleic Acids Research* 50, W21-W28.

Machine Learning and Deep Learning of MALDI-TOF Mass Spectrometry Data for Predicting Vancomycin Resistance in Enterococci

Chia-Yu Lin[1], Chien-Hsun Huang[2], Jiun-Nong Lin[3,4], Chih-Chieh Chen[1*]

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Vancomycin-resistant Enterococcus (VRE) is an important multidrug-resistant pathogen, making rapid identification of vancomycin resistance essential for clinical management and infection control. Proteomic mass spectrometry enables rapid and high-throughput bacterial profiling, but its complex and high-dimensional data require advanced computational methods.

In this study, we developed an integrated machine learning and deep learning framework for predicting vancomycin resistance in Enterococcus using proteomic mass spectrometry data. A total of 938 mass spectrometry spectra were processed through feature alignment, normalization, and dimensional standardization. Multiple models, including ensemble learning algorithms, kernel-based classifiers, convolutional neural networks, and recurrent neural networks, were developed and evaluated using training, validation, and independent test sets.

Model performance ranged from 0.81 to 0.89 for accuracy, 0.88 to 0.97 for AUC, and 0.87 to 0.92 for F1-score. Ensemble learning achieved the highest performance, with an AUC of 0.97 and an F1-score of 0.92. These results demonstrate the potential of integrating proteomic mass spectrometry with machine learning and deep learning for rapid prediction of vancomycin resistance in clinical microbiology.



李宗夷 理事長
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Experience

- 2025/11 至今: 台灣生物資訊與系統生物學會 / 理事長
- 2025/10 至今: 國立陽明交通大學生物資訊中心 / 主任
- 2025/10 至今: 國立陽明交通大學工程生物學院 / 特聘教授
- 2025/08 至今: 國立陽明交通大學生物資訊及系統生物研究所 / 所長
- 2024/01-2025/10: 台灣生物資訊及系統生物學會 / 秘書長
- 2023/02-2025/07: 國立陽明交通大學生物資訊及系統研究所 / 教授
- 2022/04-2022/12: 晨伽生物科技股份有限公司 / 共同創辦人
- 2018/08-2022/12: 香港中文大學醫學院 / 副教授
- 2015/08-2018/07: 元智大學資訊工程學系 / 教授
- 2012/08-2015/07: 元智大學資訊工程學系 / 副教授
- 2009/08-2012/07: 元智大學資訊工程學系 / 助理教授

Honors and Awards

- 2026 – 第十九屆台灣管理學會論文大賞: 特優論文獎
- 2025 – **傑出人才基金會-第十三屆年輕學者創新獎**
- 2025 – 第十八屆台灣管理學會論文大賞: AI 應用，特優論文獎
- 2025 – **World's Top 2% Scientists in 2024 (全球前 2%科學家)**
- 2024 – **World's Top 2% Scientists in 2023 (全球前 2%科學家)**
- 2023 – **教育部玉山青年學者**
- 2023 – **World's Top 2% Scientists in 2022 (全球前 2%科學家)**
- 2023 – **華仁青年講座教授**
- 2022 – **World's Top 2% Scientists in 2021 (全球前 2%科學家)**
- 2021 – **香港中文大學 2021 年度師風師德獎**
- 2019 – 香港中文大學**校長青年學者**
- 2016 – 元智大學研究傑出獎
- 2015 – 第十三屆有庠傑出教授獎

Research Interests

生物大數據、人工智慧、生物資訊、智慧醫療平台

Selected Publications

1. Peilin Xie, Xingchen Liu, Jiahui Guan, Zhihao Zhao, Jianzhi Wu, Tzong-Yi Lee, Junwen Wang, Lantian Yao, Bingyu Cui, Ying-Chih Chiang, 2026 Aug, "MOSAIC enables selective transfer and fairness-aware generalization in cross-species DNA methylation prediction", ***Nucleic Acids Research*** 54 (15), gkag822.
2. Guan-Ting Chen, Lei-Chen Liang, Yun Tang, Yu-Chen Chen, Chi-Nga Chow, Michael Anekson Widjaya, Wei-Chih Huang, Tzong-Yi Lee, 2026 July, "miRSiC: a regulatory-aware machine learning framework for microRNA expression inference across bulk and single-cell transcriptomes", ***Briefings in Bioinformatics***, 27(4), bbag457.
3. Xiao-Xuan Cai, Jing-Shan Liao, Jia-Jun Ma, Yu-Xuan Pang, Yi-Gang Chen, Yang-Chi-Dung Lin, Yi-Dan Chen, Xin Cao, Yi-Cheng Zhang, Tao-Sheng Xu, Tzong-Yi Lee, Hsi-Yuan Huang, Hsien-Da Huang, 2026 Feb, "SiCmiR Atlas: Single-Cell miRNA Landscape Reveals Hub-miRNA and Network Signatures in Human Cancers", ***Advanced Science***, e14446.
4. Yen-Peng Chiu, Lantian Yao, Yun Tang, Chia-Ru Chung, Yuxuan Pang, Ying-Chih Chiang, Tzong-Yi Lee, 2026 Jan "ANIA: an inception-attention network for predicting minimum inhibitory concentration of antimicrobial peptides", ***Briefings in Bioinformatics***, 27(1), bbag023.
5. Cheng-Pei Lin, Yann-Jen Ho, Yen-Peng Chiu, Yun Tang, You Sheng Paik, Guan-Ting Chen, Wei-Chih Huang, Tzong-Yi Lee, 2026 Jan, "MoAGNN: a multi-omics hierarchical graph neural network for subtype classification and prognosis prediction in lung adenocarcinoma", ***Briefings in Bioinformatics***, 27(1), bbaf735.

An AI-powered Multimodal Platform for Pancancer Kinase Activity Profiling and Drug Response Prediction

Yann-Jen Ho[1], Muh-Hwa Yang[2]*, Tzong-Yi Lee[1]*

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[2] *Cancer and Immunology Research Center, National Yang Ming Chiao Tung University, Taipei, Taiwan.*

Kinases are central regulators of cellular signaling and key therapeutic targets in oncology, yet their dynamic activities across diverse cancer types remain difficult to quantify and translate into actionable predictions. Here, we present an **AI-powered multi-modal platform** that integrates transcriptomic, proteomic, phosphoproteomic, and pharmacogenomic data to construct a comprehensive pan-cancer landscape of kinase activity and therapeutic response. Our framework employs advanced machine learning algorithms to infer kinase activity profiles from multi-omics signals, and to couple these profiles with large-scale drug sensitivity datasets for predictive modeling. By jointly learning across modalities, the platform achieves robust inference under noisy or incomplete data, identifies cancer-specific and shared kinase dysregulations, and uncovers predictive biomarkers of response to targeted therapies. Benchmarking against existing kinase activity profiling and drug prediction methods demonstrates improved accuracy, interpretability, and generalizability across tumor types. Case studies highlight the platform's ability to reveal novel kinase–drug associations and prioritize therapeutic opportunities in resistant cancers. Collectively, this work establishes a scalable and explainable AI approach to kinase activity profiling and drug response prediction, providing a foundation for precision oncology and rational drug development.

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Session 11：Multi-Omics in Translational Medicine

TBSB

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- 2022/08 ~ 2025/02 Assistant Professor
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- 2018/08 ~ 2022/07 Assistant Professor
  Department of Information Management, National University of Kaohsiung
- 2015/08 ~ 2018/07 Principal Engineer
  Taiwan Semiconductor Manufacturing Company Limited (TSMC), Taiwan

Research Interests

Bioinformatics
Computational and Systems Biology
Deep Learning/Machine Learning and Artificial Intelligence
Biomedical Data Science

Selected Publications

1. **T.-H. Yang**, X.-W. Li, Y.-H. Lee, S.-Y. Lu, W.-S. Wu#, and H.-C. Lee# (2025) "mirTarCLASH: a comprehensive miRNA target database based on chimeric read-based experiments," *Database*, vol. **2025**: article ID baaf023.
2. **T.-H. Yang**, Y.-Y. Chen, C.-C. Liao, H.-C. Zheng, C.-L. Hsieh, J.-S. Chen, W.-C. Tsai, Y.-Y. Tseng, W.-S. Wu# (2025) "NoAC: an automatic builder for knowledge bases and query interfaces on genomes of non-model organisms," *Journal of Molecular Biology*, vol. **437**, p. 169488.
3. **T.-H. Yang†**, Y.-H. Huang†, Y.-H. Lee, J.-N. Lai, K.-D. Chen, M.-H. Guo, Y. Pang, C.-Y. Chen, W.-S. Wu#, and H.-C. Kuo# (2025) "Identifying the risk of Kawasaki disease based solely on routine blood test features through novel construction of machine learning models," *Computational and Structural Biotechnology Journal*, vol. **27**, pp. 2832–2842. **†: co-first authors**
4. Y.-H. Yu, K.-L He, and **T.-H. Yang*** (2026) "CRM-TI: an enhanced pipeline for computationally assigning the target genes of cis-regulatory modules by considering comprehensive long-range regulation mechanisms," *BioData Mining*, vol. **19**, p. 45.
5. Y.-H. Yu, W.-C. Dai, Z.-H. Jiang, **T.-H. Yang*** (2026) "CRMIPred: Identifying the spatial interactions among cis-regulatory modules via considering their cross-attended epigenetic profiles," *Computer Methods and Programs in Biomedicine*, vol. **280**, p. 109314.
6. **T.-H. Yang†**, W.-T. Chiu†, Y.-H. Chu, H.-T. Hong, C.-W. Liu, L.-Y. Chang, H.-C. Hung, W.-C. Dai, Y.-L. Chen, S. Y. Lim, Y.-C. Chuang, W.-Y. Huang*, and T.-L. Liu* (2026) "A Fluorescence Imaging- and Deep Learning-Based Approach for Detecting Hepatitis B Virus Integration into Host Genomes," *Journal of Imaging Informatics in Medicine*. **†: co-first authors**

7. Y.-H. Yu, H.-T Hong, and **T.-H. Yang*** (2026) "Generalizable deep-learning-based mRNA-protein interaction prediction strongly depends on protein diversity," *Journal of Cheminformatics*, vol. **18**, p. 74.

Decoding Multi-layer Cis-regulatory Module Functional Mechanisms with Deep Learning

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Cis-regulatory modules (CRMs) are clustering of transcription factor binding sites in metazoan. They are the central components of transcriptional regulation, and their functions extend beyond serving as isolated DNA elements bound by transcription factors. CRMs can communicate with other regulatory regions through chromatin interactions, undergo direct transcription on them to generate non-coding RNAs (ncRNAs), and regulate distal target genes through multiple interconnected mechanisms. A unified view of these multilayers of CRM functional mechanisms across the genome is essential for studying the complicated transcriptional regulatory network.

In this talk, I will present our recent efforts to develop deep-learning approaches for systematically decoding these complementary regulatory activities from epigenomic data. By modeling interactions between regulatory regions, identifying transcriptional activity occurring directly on CRMs, and linking CRMs to their downstream target genes, these approaches collectively reconstruct a multi-layer view of CRM-mediated regulation. Our results indicate that CRM interaction, transcription on CRMs, and target-gene regulation are supported by distinct but complementary regulatory signals, and that considering these mechanisms together provides a more complete representation of CRM functions than analyzing any single regulatory layer alone. These studies demonstrate how deep learning can move toward integrative analysis of *cis*-regulatory mechanisms and provide a computational framework for investigating transcriptional regulation at the genome-wide scale.



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Education

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Experience

Postdoctoral Associate, Department of Medicine, Baylor College of Medicine, Houston, TX, USA

Research Interests

Bioinformatics
Cancer Genomics
Tumor Microenvironment
Prognostic and Predictive Biomarkers

Selected Publications

1. **Zhu, B.**; Chen, P.; Aminu, M.; **Li, J.-R.**; Fujimoto, J.; et al. Spatial and multiomics analysis of human and mouse lung adenocarcinoma precursors reveals TIM-3 as a putative target for precancer interception. *Cancer Cell* **2025**, *43* (6), 1125–1140. (**Co-first author**)
2. **Li, J.-R.**; Shaw, V.; Lin, Y.; Wang, X.; Aminu, M.; Li, Y.; Wu, J.; Zhang, J.; Amos, C. I.; Cheng, C. The prognostic effect of infiltrating immune cells is shaped by proximal M2 macrophages in lung adenocarcinoma. *Mol. Cancer* **2024**, *23*, 185.
3. **Li, J.-R.**; Cheng, C. Immune cell pair ratio captured by imaging mass cytometry has superior predictive value for prognosis of non-small cell lung cancer than cell fraction and density. *Cancer Commun.* **2024**, *44* (5), 252–257.

Deciphering Single-Cell Spatial Architecture in Lung Adenocarcinoma: Cell-Cell Interactions and TIM-3 Precancer Interception via Spatial Multiomics

Jian-Rong Li [1,2], Bo Zhu [3], Xiang Wang [4], Jianjun Zhang [3*], Chao Cheng [4*], et al.
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[2] *Molecular Medicine Research Center, Chang Gung University, Taoyuan City, Taiwan*
[3] *Department of Genomic Medicine, The University of Texas MD Anderson Cancer Center, Houston, TX, USA*
[4] *Institute for Clinical and Translational Research, Baylor College of Medicine, Houston, TX, USA*

The spatial architecture of the tumor microenvironment (TME) dictates cancer evolution and clinical outcomes in lung adenocarcinoma (LUAD). Integrating imaging mass cytometry (IMC), scRNA-seq, and high-resolution spatial transcriptomics (Visium and Xenium 5K), this study establishes a comprehensive framework bridging spatial cell-cell interaction metrics to precancer interception. We demonstrate that spatial metrics, including Cell Pair Ratios and Proximity Scores, provide superior prognostic value over conventional cell densities. Specifically, proximal M2 macrophages suppress the protective functions of Th, Tc, and B cells, whereas pre-treatment Treg-Tc spatial interactions accurately predict immunochemotherapy responses. Extending these spatial principles to early lung carcinogenesis across human precursor spectrum (AAH/AIS/MIA/IAC) and mouse models, we uncovered a coordinated transition from innate to adaptive immunity. Crucially, the immune checkpoint TIM-3 peaks specifically at the AAH precancer stage on innate immune populations. In vivo TIM-3 blockade at the precancer stage—but not at advanced invasive stages—significantly attenuates tumor burden by promoting antigen presentation in dendritic cells, reprogramming M2-to-M1 macrophage ratios, and boosting T cell activation. Together, these findings highlight single-cell spatial multiomics as a powerful tool for biomarker discovery and identify TIM-3 as a promising target for lung precancer interception.

References:

1. Li, J.-R.; Cheng, C. Immune cell pair ratio captured by imaging mass cytometry has superior predictive value for prognosis of non-small cell lung cancer than cell fraction and density. *Cancer Commun.* 2024, 44 (5), 252–257.
2. Li, J.-R.; Shaw, V.; Lin, Y.; Wang, X.; Aminu, M.; Li, Y.; Wu, J.; Zhang, J.; Amos, C. I.; Cheng, C. The prognostic effect of infiltrating immune cells is shaped by proximal M2 macrophages in lung adenocarcinoma. *Mol. Cancer* 2024, 23, 185.
3. Wang, X.; Dong, J.; Li, J.-R.; Lin, Y.; Sahoo, B.; Li, Y.; Liu, Y.; Ripley, R. T.; Wu, J.; Zhang, J.; Amos, C. I.; Cheng, C. Cytotoxic and regulatory T cell interactions calculated from image mass cytometry predict immunochemotherapy response in triple-negative breast cancer. *Cancer Commun.* 2025, 45 (4), 2523.
4. Zhu, B.; Chen, P.; Aminu, M.; Li, J.-R.; Fujimoto, J.; Tian, Y.; Hong, L.; Chen, H.; Hu, X.; Li, C.; et al. Spatial and multiomics analysis of human and mouse lung adenocarcinoma precursors reveals TIM-3 as a putative target for precancer interception. *Cancer Cell* 2025, 43 (6), 1125–1140.



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Education

PhD: **Graduate Program in Translational Agricultural Sciences, National Cheng Kung University and Academia Sinica**, Taiwan, 2017-2020
- Supervisor: Prof. Dr. Wen-Chi Chang

MSc: **Institute of Tropical Plant Sciences, National Cheng Kung University**, Tainan, Taiwan, 2013-2015
- Supervisor: Prof. Dr. Wen-Chi Chang

BA: **Department of Life Sciences, National Chung Hsing University**, Taichung, Taiwan, 2009-2013

Experience

2026 **Assistant Research Scientist.** Laboratory of Prof. Tzong-Yi Lee
Institute of Bioinformatics and Systems Biology, National Yang Ming Chiao Tung University, Hsinchu, Taiwan

2023-2025 **Assistant Research Scientist.** Laboratory of Prof. Julian Chen
School of Molecular Sciences, Arizona State University, Arizona, U.S.A.

2020-2023 **Postdoctoral Researcher.** Laboratory of Prof. Wen-Chi Chang
Institute of Tropical Plant Sciences and Microbiology, National Cheng Kung University, Tainan, Taiwan.

2015-2018 **Research Assistant.** Laboratory of Prof. Wen-Chi Chang
Institute of Tropical Plant Sciences, National Cheng Kung University, Tainan, Taiwan.

Research Interests

Bioinformatics; Systems biology
Pharmacogenomics and drug response analysis
Multi-omics data integration
Coding and non-coding gene regulatory networks construction
Biological database development and knowledgebase construction
Non-coding RNA identification

Selected Publications

1. Yang, C. W., **Chow, C. N.**, Chen, H., Tseng, K. C., Wu, N. Y., Zhuang, Y., Zhuang, W. & Chang, W. C. (2026). Peanut genome resource: a functional genomics platform for *Arachis hypogaea*. ***Plant Cell Physiol***, 67(5), 847-861. Retrieved from https://academic.oup.com/pcp/article/67/5/847/8377253. doi: 10.1093/pcp/pcaf165
2. Chou, Y. S., Logeswaran D., **Chow, C. N.**, Dunn, P. L., Podlevsky, J. D., Liu, T., Akhter, K., J. & Chen, J. J. (2025) A degenerate telomerase RNA directs telomeric DNA synthesis in Lepidopteran

insects. ***Proc Natl Acad Sci USA***, 122 (9), Retrieved from https://doi.org/10.1073/pnas.2424443122
3. Chen, Y. Y., Sun, Y., Li, C. I., Lin, S. T., Zheng, H. C., Zhang, Z. B., Lee, B. R., Hsieh, C. L., Hsiao, Y. Y., **Chow, C. N.**, Yang, C. W., Chang, W. C., Chen, H., Yang, F. X., Zhu, G. F., Zheng, Q., Zhou, C. Y., Zhao, Z., Ai, Y., Wang, L. Y., Chen, D., He, X., Huang, M. Z., Peng, D. H., Yu, H., Lan, S., Liu, Z. J., Wu, W. S., & Tsai, W. C. (2025). OrchidBase 6.0: increasing the number of Cymbidium (Orchidaceae) genomes and new bioinformatic tools for orchid genome analysis. ***BMC Plant Biol***, 25(1), 1. Retrieved from https://doi.org/10.1186/s12870-024-06024-1
4. **Chow, C. N.**, Yang, C. W., Wu, N. Y., Chiu, Y. H., & Chang, W. C. (2025). Peanut Databases: Source and Possible Applications. (1st edition, chap. 20) Retrieved from https://doi.org/10.1201/9781003360988 ISBN: 978-1-00336-098-8
5. **Chow, C. N.**, Yang, C. W., Wu, N. Y., Wang, H. T., Tseng, K. C., Chiu, Y. H., Lee, T. Y. & Chang, W. C. (2024). PlantPAN 4.0: updated database for identifying conserved non-coding sequences and exploring dynamic transcriptional regulation in plant promoters. ***Nucleic Acids Res***, 52(D1), D1569-D1578. Retrieved from https://www.ncbi.nlm.nih.gov/pubmed/37897338. doi: 10.1093/nar/gkad945
6. Cao, Y. L., Chen, Y. Y., Li, Y. L., Li, C. I., Lin, S. T., Lee, B. R., Hsieh, C. L., Hsiao, Y. Y., Fan, Y. F., Luo, Q., Zhao, J. H., Yin, Y., An, W., Shi, Z. G., **Chow, C. N.**, Chang, W. C., Huang, C. L., Chang, W. H., Liu, Z. J., Wu, W. S., & Tsai, W. C. (2024). Wolfberry genome database: integrated genomic datasets for studying molecular biology. ***Front Plant Sci***, 15, 1310346. Retrieved from https://doi.org/10.3389/fpls.2024.1310346
7. Tseng, K. C., Wu, N. Y., **Chow, C. N.**, Zheng, H. Q., Chou, C. Y., Yang, C. W., Wang, M. J., & Chang, W. C. (2023). JustRNA: a database of plant long non-coding RNA expression profiles and functional network. ***J Exp Bot***, erad186. doi:10.1093/jxb/erad186
8. **Chow, C. N.**, Yang, C. W., & Chang, W. C. (2023). Databases and prospects of dynamic gene regulation in eukaryotes: A mini review. ***Comput Struct Biotechnol J***, *21*, 2147-2159. Retrieved from https://www.ncbi.nlm.nih.gov/pubmed/37013004. doi:10.1016/j.csbj.2023.03.032
9. Chang, W. C., & **Chow, C. N.** (2023). Database for Plant Transcription Factor Binding Sites. ***Methods Mol Biol***, 2594, 173-183. Retrieved from https://www.ncbi.nlm.nih.gov/pubmed/36264496. doi:10.1007/978-1-0716-2815-7_13

AI-driven multi-omics integration for drug response prediction and clinical outcomes in precision oncology

Chi-Nga Chow

Non-coding RNAs (ncRNAs), including long non-coding RNAs (lncRNAs) and microRNAs (miRNAs), are increasingly recognized as critical regulators of gene expression, cellular signaling, and therapeutic response. Nevertheless, their systematic contribution to drug response prediction remains insufficiently characterized. The substantial heterogeneity of drug response observed across patients and cancer types presents a challenge to the identification of robust and generalizable biomarkers, given that drug sensitivity is governed by the intricate interplay among transcriptional regulation, post-transcriptional control, protein-level activity, and downstream signaling networks. Multi-omics integration thus offers a principled framework for capturing these interdependent regulatory layers and for elucidating the molecular mechanisms underlying differential drug response. Within this framework, artificial intelligence (AI) and machine learning (ML) methodologies provide powerful analytical tools for integrating heterogeneous molecular features, prioritizing informative biomarkers, and constructing interpretable predictive models of drug response.

In this study, we demonstrate the utility of ncRNAs as informative molecular features for drug response prediction, identifying lncRNA signatures through multi-omics analysis of drug response data in acute myeloid leukemia (AML). In parallel, we present dbDrugMiR (https://bioinfo.nycu.edu.tw/dbDrugMiR/), a curated database that integrates drug–miRNA–target associations with miRNA and gene expression profiles, pharmacological response data, and clinical outcome information across cancer cell lines and patient cohorts. Collectively, these complementary approaches underscore the mechanistic and translational relevance of ncRNAs in multi-omics-based drug response modeling, and illustrate how AI-driven analytical strategies, coupled with integrated data resources, can advance biomarker discovery, mechanistic investigation, and the broader goals of precision oncology.

Session 12：精準醫學臨床應用與未來治療策略

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主持人

陳相成　處長

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軍醫局醫務管理處

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Prof. Hsiang-Cheng Chen

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楊謝樂 董事長
Dr. Shieh-Yueh Yang

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CURRENT APPOINTMENTS
- 磁量生技股份有限公司董事長
- 磁量生技創辦人

EDUCATION & TRAINING
- 國立臺灣師範大學物理學系 學士（1993）
- 國立臺灣師範大學物理學系 博士（1999）

PROFESSIONAL EXPERIENCE
- 中央研究院博士後研究員，研究磁性奈米粒子特性
- 國立臺灣師範大學光電科技研究所教師，從事光電與生醫感測研究
- 2008 年創辦磁量生技，推動免疫磁減量檢測技術之商品化與臨床應用

EXPERTISE
- 免疫磁減量技術（Immunomagnetic Reduction, IMR）
- 磁性奈米粒子、超靈敏生物標記檢測
- 阿茲海默症等神經退化疾病血液生物標記

SPEAKER INTRODUCTION
楊謝樂博士為磁量生技創辦人及董事長，具有物理與奈米磁性量測研究背景，致力將免疫磁減量技術應用於血液生物標記檢測，推動研究成果走向臨床與產業應用。

量子感測精準檢驗神經退化性疾病血液蛋白質生物標記：臨床案例分享

楊謝樂
磁量生技股份有限公司

隨著人口結構高齡化，神經退化性疾病(失智症)已成為僅次於癌症的重大社會醫療挑戰。根據衛福部調查，台灣現有約 30 至 40 萬名神經退化性疾病患者，而潛在高風險族群更逾百萬，帶來沉重的照護負擔。若能落實早期診斷，將能大幅減輕社會與家庭的經濟、精神及人力負擔。然而，神經退化性疾病的臨床精準診斷極具挑戰，欲做到「極早期」的精準篩檢更是醫學界亟待突破的瓶頸。

臨床上，相同的神經退化症狀可能源自阿茲海默症、巴金森氏症、前額葉失智症或血管型失智症等不同病因，因此精準診斷的關鍵在於正確鑑別病理源頭。現行臨床主要依賴醫學影像檢查，不僅費用高昂 (整套檢查動輒十數萬台幣)、無法普及，更難以針對高風險群進行常態性的早期篩檢。相較之下，「抽血檢測」是目前最具可行性且廣受國際認同的發展趨勢。

然而，開發早期血液精準檢驗面臨兩大核心難題：第一，外周血中的神經退化病理蛋白質生物標記濃度極低(達 fg/ml 至 pg/ml 等級)，現行傳統檢驗技術的靈敏度明顯不足；第二，不同型態的神經退化性疾病對應不同的病理機制與蛋白質，必須建構完整的檢驗面板(Panel)，才能達成精準診斷。

為此，磁量生技整合學界與醫界的研發能量，創新融合量子感測、奈米科技及蛋白質體學技術，成功開發出極高靈敏度的蛋白質檢驗技術-「免疫磁減量技術 (ImmunoMagnetic Reduction, IMR)」。該技術的檢測靈敏度高達 0.01 fg/ml，且發展出能檢驗阿茲海默症、巴金森氏症、前額葉失智症、漸凍人 (肌萎縮側索硬化症) 等不同神經退化性疾病的完整蛋白質生物標記檢項，一舉克服了前述兩大難題。歷經近 20 年的商業化開發與臨床驗證，此技術已順利取得台灣衛福部第三類體外診斷醫療器材 (IVD) 上市許可及歐盟認證，將抽血檢驗落實於早期神經退化疾病診斷中，目前已成功落地台灣及亞洲各大醫療機構。

本演講將深入淺出地介紹免疫磁減量技術的核心原理與臨床驗證數據，並分享近年來的臨床實務案例，期望推動神經退化疾病於基層醫療的早期檢驗，共同守護民眾的大腦健康。



周輝政 醫師
Dr. Hei-Jen Jou

主治醫師/臺安醫院 婦產科
Visiting Staff / Department of Obstetrics and Gynecology,
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CURRENT APPOINTMENTS

- 臺安醫院婦產科主治醫師
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EDUCATION & TRAINING

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PROFESSIONAL EXPERIENCE

- 臺安醫院婦產部主任
- 臺安醫院預防醫學事業部主任
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- 國立臺北護理健康大學兼任教授
- 國立臺灣大學醫學系兼任助理教授

EXPERTISE

- 婦女健康、更年期保健與骨質疏鬆
- 循環腫瘤細胞（CTC）檢測諮詢
- 預防醫學及臨床整合照護

SPEAKER INTRODUCTION

周輝政醫師為婦產科及預防醫學領域專家，曾任臺安醫院婦產部主任、預防醫學事業部主任及資深行政副院長，現為臺灣精準醫學學會常務監事，關注循環腫瘤細胞檢測與臨床應用。

Circulating Tumor Cells in Precision Oncology: Enumeration, Phenotyping, Minimal Residual Disease, and Clinical Decision Support

Hei-Jen Jou[1, 2*]
[1] *Department of Obstetrics and Gynecology, Taiwan Adventist Hospital*
[2] *Taiwan Precision Medicine Society*

Precision medicine selects therapy according to a patient's molecular profile. However, tissue biopsy provides only a static snapshot and cannot capture tumor evolution during disease progression or treatment. Circulating tumor cells (CTCs) are intact tumor-derived cells in blood, providing morphological, phenotypic, and single-cell molecular information.

This presentation introduces the CellReveal™ automated CTC platform, which integrates the nanostructured V-BioChip with artificial intelligence-assisted image analysis. Multi-antibody capture using EpCAM, E-cadherin, and N-cadherin is designed to recover heterogeneous CTC subpopulations.

CTC enumeration has prognostic value in metastatic breast, colorectal, and prostate cancers. During serial monitoring, persistently elevated or rebounding counts may signal progression, whereas declining counts are generally associated with a more favorable course. Beyond enumeration, CTC profiling adds clinically relevant context. Lineage markers, including GATA3, SATB2, and PAX8, may support assessment of the likely tumor origin. Therapy-related markers such as ER, HER2, AR-V7, and PD-L1 may reveal phenotypic shifts, emerging resistance, or potential therapeutic targets, prompting confirmatory testing and multidisciplinary treatment review.

Automated single-cell isolation allows CTCs to function as a repeatable single-cell biopsy. Single-cell sequencing can characterize the genome and transcriptome of individual CTCs, resolve intrapatient heterogeneity, and track clonal evolution or minimal residual disease. These advances may eventually improve drug selection and adaptive treatment strategies. Because analytical and clinical validation remain incomplete, CTC findings should be integrated with imaging, pathology, and validated biomarkers as complementary decision support rather than used as a stand-alone treatment directive.



葉富強 醫師
Dr. Fu-Chiang Yeh

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CURRENT APPOINTMENTS

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- 國防醫學大學內科學科助理教授

EDUCATION & TRAINING

- 國防醫學院醫學系　醫學士（MD）
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PROFESSIONAL EXPERIENCE

- 三軍總醫院內科部及風濕免疫科住院醫師、總醫師
- 陸軍第三區支援指揮部醫務所醫官、憲兵司令部醫務所主任
- 內科、風濕科與免疫科專科醫師及指導醫師；精準醫學專科醫師

EXPERTISE

- 肺動脈高壓；結締組織疾病相關肺動脈高壓（CTD-PAH）
- 自體免疫疾病：全身性紅斑狼瘡、類風濕性關節炎、硬皮症及乾燥症
- 風濕免疫疾病臨床診療、一般內科與整合醫學

SPEAKER INTRODUCTION

葉富強醫師現任三軍總醫院風濕免疫過敏科主治醫師及實驗室主任，並任國防醫學大學內科學科助理教授。畢業於國防醫學院醫學系，取得英國倫敦帝國理工大學醫學博士學位，研究及臨床專長包括肺動脈高壓、自體免疫與風濕免疫疾病。

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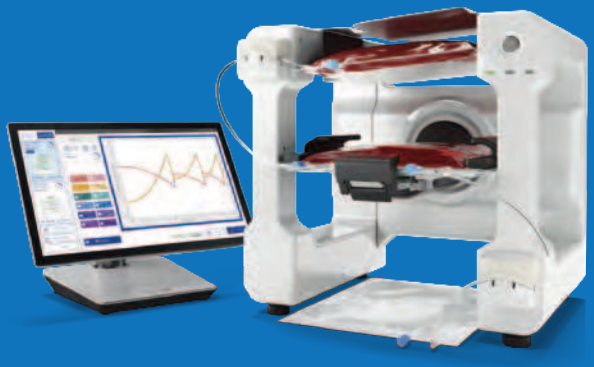


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